



Fermify GmbH
Richard Neutra Gasse 5
1210 Vienna
Austria

October 14, 2024

Dr. Susan Carlson
Director, Division of Biotechnology and GRAS Notice Review
Office of Food Additive Safety (HFS-200)
Center for Food Safety and Applied Nutrition
Food and Drug Administration
5001 Campus Drive
College Park, MD 20740-3835

RE: GRAS Determination for Fermify GmbH for the use of β -casein produced by *E. coli* stF_24_139 in non-alcoholic beverages, cheeses, cheese analogs, dairy product analogues, frozen dairy desserts, milk products and plant protein products

Dear Dr. Carlson,

Please find enclosed the Generally Recognized as Safe (GRAS) determination for the use of β -casein produced by *E. coli* stF_24_139. β -casein produced by *E. coli* stF_24_139. β -casein may be incorporated into non-alcoholic beverages, cheeses, cheese analogs, dairy product analogs, frozen dairy desserts, milk products and plant protein products as a source of dietary protein, providing nutritional value and functionality due to its properties as an emulsifier, stabilizer, formulation aid, surface-active agent, nutrient supplement, synergist and texturizer. Information regarding the product identification, intended uses, the manufacturing process and safety is presented in the attached GRAS notice.

Fermify GmbH has determined that its β -casein produced by *E. coli* stF_24_139 is GRAS for its intended uses based on scientific procedures in accordance with 21 CFR § 170.30(b) and in conformance with the guidance issued by the Food and Drug Administration (FDA) under 21 CFR § 170.36, 81 Fed. Reg. 54960 (Aug. 17, 2016). Therefore, the use of β -casein produced by *E. coli* stF_24_139, as described in this GRAS Determination, is exempt from the requirement of premarket approval set forth in the Federal Food, Drug, and Cosmetic Act.

Analytical data, published studies and information that has been used as the basis for this GRAS determination are available for FDA's review upon request.

We appreciate the FDA's consideration of this submission and are available to offer any further details required to support the assessment process. Thank you for your attention to this matter.

Sincerely,



Eva Sommer
Co-Founder & CEO
Fermify GmbH
+43 6 642 30 59 15
eva@fermify.org

**GENERALLY RECOGNISED AS SAFE DETERMINATION FOR BETA-CASEIN
FROM *E. coli* stF_24_139 IN NON-ALCOHOLIC BEVERAGES, CHEESES,
CHEESE ANALOGS, DAIRY PRODUCT ANALOGS, FROZEN DAIRY DESSERTS,
MILK PRODUCTS AND PLANT PROTEIN PRODUCTS**

Prepared by: Fermify GmbH

Richard Neutra Gasse 5
1210 Vienna
Austria

Date: October 14, 2024

Table of contents

PART 1. §170.225 SIGNED STATEMENT OF THE CONCLUSION OF GENERALLY RECOGNIZED AS SAFE (GRAS) AND CERTIFICATION OF CONFORMITY TO 21 CFR §170.205-170.260	8
1.1. Submission of GRAS notice	8
1.2. Name and address of the sponsor	8
1.3. Common or usual name of substance	8
1.4. Intended use	8
1.5. Basis for GRAS determination	8
1.6. Exemption from Premarket approval	8
1.7. Availability of information	9
1.8. Freedom of Information Act (FOIA), 5 U.S.C. 552	9
1.9. Certification of Information included in the GRAS notification	10
PART 2. §170.230 IDENTITY, METHOD OF MANUFACTURE, SPECIFICATIONS, AND PHYSICAL OR TECHNICAL EFFECT OF THE NOTIFIED SUBSTANCE	11
2.1. Product identity	11
2.1.1. β -casein	11
Figure 2. Amino acid sequence of Fermify's recombinant β -casein	12
2.2. Source organism and parental strain	13
2.2.1. Production strain	14
2.3. Production process	15
2.3.1. Description of the method of manufacture	15
2.3.2. Raw materials and processing aids	16
2.3.3. Quality control	17
2.4. Specifications	17
2.5. Batch analysis	19
2.6. Viable cells of the production strain	22
2.7. Physical or technical effect	22
PART 3. §170.235 DIETARY EXPOSURE	23
3.1. Intended use	23
3.2. Estimated daily intake	23
3.2.1. Methods	23
3.2.2. Results	25
3.3. Exposure to undesirable substances	27
PART 4. §170.240 SELF-LIMITING LEVELS OF USE	29
PART 5. §170.245 COMMON USE IN FOOD BEFORE 1958	30
PART 6. §170.250 NARRATIVE ON THE CONCLUSION OF GRAS STATUS	31
6.1. Introduction	31
6.2. Absorption, digestion, metabolism, and excretion of β -casein	31
6.3. Safety of β -casein	32
6.3.1. Bioactivity of β -casein	33
6.4. Safety of the parental strain	34
6.5. Safety of the production strain	35
6.6. Safety of β -casein produced by <i>E. coli</i> stF_24_139	35
6.7. Allergenicity of β -casein produced by <i>E. coli</i> stF_24_139	38
6.8. Summary and conclusion of the Basis for GRAS Determination	40

6.9. Data and information that is inconsistent with the GRAS conclusion	40
PART 7. §170.255 LIST OF SUPPORTING DATA	41
7.1. List of references	41
Appendix I – FDA September 11, 2003, letter to ETA as acceptable for use in enzyme manufacturing	43
Appendix II – Certificates of analysis (CoAs) of the presented batches	44
Appendix III – NHANES food codes and associated proposed uses used in exposure assessment.	45

List of tables

Table 1. Comparison of <i>in vitro</i> digestibility of native bovine β -casein and β -casein produced by <i>E. coli</i> stF_24_139	13
Table 2. Specifications for β -casein produced by <i>E. coli</i> stF_24_139.....	18
Table 3. Batch analysis of β -casein produced by <i>E. coli</i> stF_24_139.....	19
Table 4. Amino acid analysis on a wet basis of β -casein produced by <i>E. coli</i> stF_24_139 compared to values for native bovine β -casein A2 variant	20
Table 5. Analysis of contaminants in β -casein produced by <i>E. coli</i> stF_24_139	21
Table 6. Analysis of minerals and trace elements in β -casein produced by <i>E. coli</i> stF_24_139	22
Table 7. Proposed food uses, maximum use levels of β -casein produced by <i>E. coli</i> stF_24_139 by food use and description of representative foods selected for the analysis	24
Table 8. Two-day average estimated daily intake (EDI) of β -casein produced by <i>E. coli</i> stF_24_139 (g/day) from all proposed food uses among the total US population two years and older (2+ year) & subpopulations.....	26
Table 9. Two-day average estimated daily intake (EDI) of β -casein produced by <i>E. coli</i> stF_24_139 (g/kg bw/day) from all proposed food uses among the total US population two years and older (2+ year) & subpopulations.	26
Table 10. Overview of applications receiving GRAS status with no objection from the FDA	33
Table 11. Overview of the 20 most abundant host cell proteins in <i>E. coli</i> BLR(DE3) ¹	36
Table 12. Endotoxin levels of known foodstuffs obtained from the literature (LAL assay)	38

List of figures

Figure 1. Amino acid sequence of native bovine β -casein	12
Figure 2. Amino acid sequence of Fermify's recombinant β -casein.....	12
Figure 3. SDS-PAGE profile of native bovine β -casein and β -casein produced by <i>E. coli</i> stF_24_139	13
Figure 4. Flow-chart of the manufacturing process to produce β -casein produced by <i>E. coli</i> stF_24_139	16
Figure 5. β -casein produced by <i>E. coli</i> stF_24_139 sliding 80-mer window comparison to AllergenOnline.org.	39

List of abbreviations

ACS	American Chemical Society
ADP	Adenosine Diphosphate
AFNOR	Association Française de Normalisation
AI	Adequate Intake
ANSES	French Agency for Food, Environmental and Occupational Health & Safety
AROI	Acceptable Range of Oral Intake
ATCC	American Type Culture Collection
ATP	Adenosine Triphosphate
BCM-7	β -casomorphin-7
BSL	Biosafety Level
bw	body weight
CAS	Chemical Abstract Service
CFR	Code of Federal Regulation
CFU	Colony Forming Unit
cGMP	Current Good Manufacturing Practice
CoA	Certificate of Analysis
DM	Dry Matter
DNA	Deoxyribonucleic acid
DSMZ	German Collection of Microorganism
dsDNA	Double-stranded deoxyribonucleic acid
EDI	Estimated Daily Intake
EFSA	European Food Safety Authority
EN	European Standard
ETA	Enzyme Technical Association
EU	Endotoxin Units
FALCPA	Food Allergen Labeling and Consumer Protection Act
FAO	Food and Agriculture Organization of the United Nations
FCC	Food Chemical Codex
FD&C	Federal Food, Drug and Cosmetics Act
FDA	U.S. Food and Drug Administration
FOIA	Freedom of Information Act
FSIS	USDA Food Safety Inspection Service
GRAS	Generally Recognized as Safe
GRN	GRAS Notice
IB	Inclusion Bodies
ICP-OES	Inductively Coupled Plasma – Optical Emission Spectrometry
ILs	Interleukins
IPTG	Isopropyl- β -D-thiogalactopyranoside
ISO	International Organization for Standardization
ISTISAN	Italian National Institute of Health
kDa	Kilo Daltons
LAL	Limulus Amebocyte Lysate
LOD	Limit of Detection
LPS	Lipopolysaccharide

MPC	Milk Protein Concentrate
MPI	Milk Protein Isolate
MPO	Myeloperoxidase
MS	Mass Spectrometry
MS/MS	Tandem Mass Spectrometry
NCHS	National Center for Health Statistics
ND	Not Detected
NF	National Formulary
NHANES	National Health and Nutrition Examination Surveys
NIH	National Institute for Health
NOAEL	No Observed Adverse Effect Level
nLC	Nano-Liquid Chromatography
NS	Not Specified
OECD	Organisation for Economic Co-operation and Development
OOS	Out Of Specification
PBS	Phosphate Buffered Saline
PCBs	Polychlorinated biphenyls
PCDDs	Polychlorinated dibenzo-p-dioxins
PCDFs	Polychlorinated dibenzofurans
PCR	Polymerase Chain Reaction
PMNs	Polymorphonuclear neutrophils
PPM	Parts Per Million
RP-HPLC/DAD	Reverse Phase High-Performance Liquid Chromatography with Diode Array Detection
SCOGS	Select Committee on GRAS Substances
SDS-PAGE	Sodium Dodecyl Sulphate–Polyacrylamide Gel Electrophoresis
SOP	Standard Operating Procedure
TCD	Thermal Conductivity Detector
TDI	Tolerable Daily Intake
TEFs	Toxic Equivalency Factors
TEQs	Toxic Equivalents
USDA	United States Department of Agriculture
USP	United States Pharmacopeia
WGS	Whole Genome Sequencing
WHO	World Health Organization
WWEIA	What We Eat in America

PART 1. §170.225 SIGNED STATEMENT OF THE CONCLUSION OF GENERALLY RECOGNIZED AS SAFE (GRAS) AND CERTIFICATION OF CONFORMITY TO 21 CFR §170.205-170.260

In accordance with Title 21 of the Code of Federal Regulation (CFR) § 170 Subpart E, Fermify GmbH, hereby informs the U.S. Food and Drug Administration (FDA) that β -casein produced by *E. coli* stF_24_139 is not subject to the premarket approval requirements outlined in the Federal Food, Drug and Cosmetics (FD&C) Act based on Fermify's conclusion that the notified substance is Generally Recognized as Safe (GRAS) under the proposed conditions of use described herein.

1.1. Submission of GRAS notice

Fermify is hereby submitting a GRAS notice in accordance with Subpart E of Part 170.

1.2. Name and address of the sponsor

Fermify GmbH
Richard Neutra Gasse 5
1210 Vienna
Austria

1.3. Common or usual name of substance

β -casein produced by *E. coli* stF_24_139
Non-animal β -casein produced by fermentation

1.4. Intended use

β -casein produced by *E. coli* stF_24_139 is intended to be used as a food ingredient in non-alcoholic beverages, cheeses, cheese analogs, dairy product analogs, frozen dairy desserts, milk products and plant protein products at a maximum level of up to 15 g of β -casein per 100 g in cheese and cheese analogs and 3.8 g of β -casein per 100 g in non-alcoholic beverages, dairy product analogs, frozen dairy desserts, milk products and plant protein products.

The β -casein is manufactured according to current good manufacturing practice (cGMP) as specified in 21 CFR § 117 and can be used as an alternative source of casein. β -casein may be incorporated into food products as a source of dietary protein, providing nutritional value and functionality due to its properties as an emulsifier, stabilizer, formulation aid, surface-active agent, nutrient supplement, synergist and texturizer. These uses will not increase the overall protein consumption of the general population.

β -casein produced by *E. coli* stF_24_139 is not intended for use in foods where standards of identity would preclude such use, infant formula, meat, poultry, egg products, catfish, or any products that would require additional regulatory review by the United States Department of Agriculture (USDA) and the USDA Food Safety Inspection Service (FSIS).

1.5. Basis for GRAS determination

This GRAS conclusion is based on scientific procedures in accordance with 21 CFR 170.30(a)(1) and 170.30(b) and has followed the guidance issued by the FDA under 21 CFR 170.36, 81 Federal Register 54960 (Aug. 17, 2016).

The GRAS status of β -casein produced by *E. coli* stF_24_139 is based on data generally available in the public domain.

1.6. Exemption from Premarket approval

The notified substance is not subject to the premarket approval requirements of the FD&C Act based on our conclusion that the substance is GRAS under the conditions of intended use and when meeting

the specifications described herein, when used as a food ingredient and source of protein in selected conventional food products.

1.7. Availability of information

The data and information that serve as a basis for this GRAS conclusion will be available for review and copying during customary business hours at the offices of Fermify, Richard Neutra Gasse 5, 1210 Vienna, Austria; telephone: +43 6 642 30 59 15; email: eva@fermify.org or will be sent to FDA upon request.

1.8. Freedom of Information Act (FOIA), 5 U.S.C. 552

It is Fermify's view that all data and information presented in Parts 2 through 7 of this notice do not contain any trade secret, commercial, or financial information that is privileged or confidential, and therefore all data and information presented herein are not exempt from the Freedom of Information Act (FOIA), 5 U.S.C. 552.

1.9. Certification of Information included in the GRAS notification

As an authorized representative of Fermify GmbH, the undersigned hereby certifies that to the best of our knowledge, the data and information provided in this GRAS notice constitutes a complete, representative, and balanced submission. This includes all relevant information, both favorable and unfavorable, known to Fermify GmbH at the time of notification, and is pertinent to assessing the safety and GRAS status of β -casein produced by *E. coli* stF_24_139 intended for use as a food ingredient, as detailed in this notice.



Eva Sommer
Co-Founder & CEO
Fermify GmbH
Notifier

October 14, 2024

PART 2. §170.230 IDENTITY, METHOD OF MANUFACTURE, SPECIFICATIONS, AND PHYSICAL OR TECHNICAL EFFECT OF THE NOTIFIED SUBSTANCE

2.1. Product identity

β -casein produced by *E. coli* stF_24_139 is a recombinant version of the naturally occurring protein β -casein, which is present in bovine milk. β -casein is one of four primary casein proteins found in milk, alongside α S1-casein, α S2-casein, and κ -casein (Fox and Mulvihill, 1990). Fermify's β -casein is based on the sequence of one of the two major subvariants of this protein, the A2 genetic variant. This variant differs from the other major subvariant (A1) by one amino acid at position 67. The A1 genetic variant has a histidine amino acid at this position, while the A2 genetic variant contains proline at position 67 (Poulsen et al., 2017).

The common name of the notified substance is β -casein produced by *E. coli* stF_24_139. The notified substance is also referred to as β -casein produced by *E. coli* DSM 35048, in which the strain name refers to the deposit number in the Leibniz Institute DSMZ-German Collection of Microorganisms (**Section 2.2.1**).

The product is produced by fermentation using the bioengineered strain *E. coli* stF_24_139. The product is a liquid product, containing min. 2% protein in buffer solution. The protein fraction of the product consists of min. 75% β -casein, the remaining proteins present in the product originate from the host strain. The recombinant β -casein is dissolved in a sodium phosphate buffer and has a pH between 6.5 and 7.5.

Throughout this GRAS Determination, the product is referred to as " β -casein produced by *E. coli* stF_24_139" (which consists of the recombinant β -casein dissolved in the sodium phosphate buffer), and "Fermify's recombinant β -casein" refers solely to recombinant β -casein protein.

2.1.1. β -casein

β -casein is a protein that is naturally present in bovine milk and is the subject of this GRAS Determination. The amino acid sequence of Fermify's recombinant β -casein is based on the sequence of the A2 genetic variant of bovine β -casein. The amino acid sequence is listed in the UniProt database under Accession No. P02666.

Native bovine β -casein consists of 209 amino acids and has a molecular mass of 23.59 kDa for the primary structure prior to post-translational phosphorylation, and a molecular mass of 23.98 kDa following phosphorylation of 5 serine residues, i.e., Ser15, Ser17, Ser18, Ser19, and Ser35 (Farrell et al., 2004) (**Figure 1**). The A2 variant of native bovine β -casein does not contain any other post-translational modifications (Farrell et al., 2004).

1	10	20
Arg- Glu- Leu- Glu- Glu- Leu-Asn- Val- Pro- Gly- Glu- Ile- Val- Glu- SerP- Leu- SerP-SerP-SerP- Glu-		
21	30	40
Glu- Ser- Ile- Thr- Arg- Ile- Asn- Lys- Lys- Ile- Glu- Lys- Phe- Gln- SerP- Glu- Glu- Gln- Gln- Gln-		
41	50	60
Thr- Glu-Asp- Glu- Leu- Gln- Asp- Lys- Ile- His- Pro- Phe- Ala- Gln- Thr- Gln- Ser- Leu- Val- Tyr-		
61	70	80
Pro- Phe- Pro- Gly- Pro- Ile- Pro- Asn- Ser- Leu- Pro- Gln- Asn- Ile- Pro- Pro- Leu- Thr- Gln- Thr-		
81	90	100
Pro- Val- Val- Val- Pro- Pro- Phe- Leu- Gln- Pro- Glu- Val- Met- Gly- Val- Ser- Lys- Val- Lys- Glu-		
101	110	120
Ala- Met- Ala- Pro-Lys- His- Lys- Glu- Met- Pro- Phe- Pro- Lys- Tyr- Pro- Val- Glu- Pro- Phe- Thr-		
121	130	140
Glu- Ser- Gln- Ser-Leu- Thr- Leu- Thr- Asp- Val- Glu- Asn- Leu- His- Leu- Pro- Leu- Pro- Leu- Leu-		
141	150	160
Gln- Ser- Trp-Met- His- Gln- Pro- His- Gln- Pro- Leu- Pro- Pro- Thr- Val- Met- Phe- Pro- Pro- Gln-		
161	170	180
Ser- Val- Leu- Ser-Leu- Ser- Gln- Ser- Lys- Val- Leu- Pro- Val- Pro- Gln- Lys- Ala- Val- Pro- Tyr-		
181	190	200
Pro- Gln- Arg- Asp- Met- Pro- Ile- Gln- Ala- Phe- Leu- Leu- Tyr- Gln- Glu- Pro- Val- Leu- Gly- Pro-		
201		
Val- Arg- Gly- Pro- Phe- Pro- Ile- Ile- Val		

Figure 1. Amino acid sequence of native bovine β -casein. The phosphorylation sites are indicated at Ser15, Ser17, Ser18, Ser19, and Ser35. Figure based on Farrell et al. (2004).

In addition to the native sequence of bovine β -casein, Fermify's recombinant β -casein contains an additional methionine at the N-terminal end of the amino acid sequence, as a consequence of expressing the protein in *E. coli*. In bacterial systems, protein expression typically begins with methionine due to the start codon (AUG), and this methionine is typically not removed from the N-terminus, which leads to Fermify's recombinant β -casein consisting of 210 amino acids.

Substantial equivalence to native bovine β -casein

Fermify conducted a comprehensive assessment of its recombinant β -casein and concluded that it is substantially equivalent to the native bovine β -casein, based on:

- Amino acid sequence
- Molecular weight characteristics
- *In vitro* digestibility

The amino acid sequence of Fermify's recombinant β -casein was experimentally confirmed by *de novo* sequencing, submitting Fermify's recombinant β -casein to Trypsin, Chymotrypsin, Asp-N, Elastase, Lys-C, Glu-C and α LP proteolysis followed by nano-liquid chromatography (nLC)-Orbitrap Fusion Lumos Tandem Mass Spectrometry (MS/MS) analysis. The derived amino acid sequence was confirmed to be identical to the predicted sequence, consisting of 210 amino acids (N-terminal methionine + amino acid sequence of A2 variant of native bovine β -casein consisting of 209 amino acids). The derived amino acid sequence is presented below (**Figure 2**).

MRELEELNVPGEIVESLSSEESITRINKKIEKFQSEEQQQTEDELQDKIHHPFAQTQSLVYPFGPIPN
 LPQNIPLTQTTPVVPPFLQPEVMGVSKVKEAMAPKHKEMPFPKYPVEPFTESQSLTLTDVENLHLPL
 PLLQSWMHQPHQPLPPTVMFPQSVLSLSQSKVLPVPQKAVPYPQRDMPIQAFLLYQEPVLGPVRG
 PFPPIIV

Figure 2. Amino acid sequence of Fermify's recombinant β -casein

Based on the amino acid sequence (primary structure), the molecular mass of Fermify's recombinant β -casein is predicted to be 23.715 kDa. The molecular weight of Fermify's β -casein produced by *E. coli* stF_24_139 was verified by Sodium Dodecyl Sulfate–Polyacrylamide Gel Electrophoresis (SDS-PAGE) and compared against native bovine β -casein (Sigma Aldrich, C6905) (**Figure 3**). Native bovine β -casein (23.98 kDa) migrated slower and appears to have a higher molecular mass because it is phosphorylated at 5 serine residues. Phosphorylation introduces negative charges to the protein, altering its electrophoretic mobility, which causes it to migrate more slowly on the gel, making it appear heavier than its actual molecular mass.

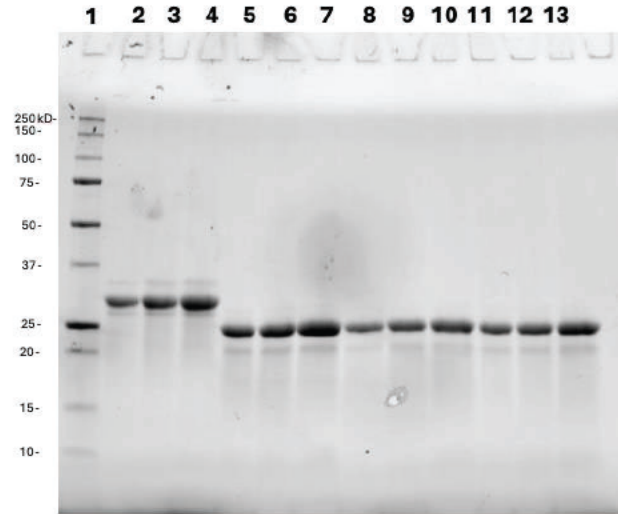


Figure 3. SDS-PAGE profile of native bovine β -casein and β -casein produced by *E. coli* stF_24_139. Lane 1: ladder (5 μ L). Lane 2 – 4: native bovine β -casein (5 μ L, 7 μ L and 10 μ L). Lane 5 – 7: β -casein produced by *E. coli* stF_24_139 (Batch Fermify008) (5 μ L, 7 μ L and 10 μ L). Lane 8 – 10: β -casein produced by *E. coli* stF_24_139 (Batch Fermify019) (5 μ L, 7 μ L and 10 μ L). Lane 11 – 13: β -casein produced by *E. coli* stF_24_139 (Batch Fermify021) (5 μ L, 7 μ L and 10 μ L).

E. coli lacks the post-translational modification machinery necessary to obtain post-translational modification patterns identical to those observed in eukaryote organisms (Hochkoeppler, 2013). Eukaryotic proteins often undergo complex post-translational modifications, such as ubiquitination, prenylation, phosphorylation, glycosylation, methylation, among others. However, the post-translational modifications of native bovine β -casein are limited to phosphorylation at 5 serine residues. Fermify's recombinant β -casein is likely to lack phosphorylation at those 5 sites.

The *in vitro* digestibility of Fermify's recombinant β -casein was measured and compared with native (bovine) β -casein (Sigma Aldrich, C6905, using the Protein Digestibility Assay procedure (Megazyme, Medallion Labs). The analysis demonstrated that β -casein produced by *E. coli* stF_24_139 is highly digestible and showed a comparable *in vitro* digestibility profile to native bovine β -casein (Table 1).

Table 1. Comparison of <i>in vitro</i> digestibility of native bovine β -casein and β -casein produced by <i>E. coli</i> stF_24_139	
Product	<i>In vitro</i> digestibility
Native bovine β -casein (Sigma Aldrich, C6905)	117.58 $\pm$ 4.01 %
Recombinant β -casein (Batch Fermify008)	123.40 $\pm$ 0.91 %
Recombinant β -casein (Batch Fermify019)	130.75 $\pm$ 7.91%
Recombinant β -casein (Batch Fermify021)	130.36 $\pm$ 1.80%

2.2. Source organism and parental strain

The β -casein is produced via fermentation using a genetically modified strain of *Escherichia coli* BL21(DE3).

Kingdom: Bacteria
 Phylum: Pseudomonadota
 Class: Gammaproteobacteria
 Order: Enterobacterales
 Family: Enterobacteriaceae
 Genus: *Escherichia*
 Species: *E. coli*
 Strain: BL21(DE3)

The *de novo* Whole Genome Sequencing (WGS) of the production strain was performed using long-read sequencing technology from Oxford Nanopore Technologies. The taxonomic identification was based on alignment and non-alignment tools using the WGS data. Based on whole genome-based analyses it was concluded that the test strain belongs to the species *Escherichia coli* and is identical to the strain *E. coli* BL21(DE3). *E. coli* BL21(DE3) is a non-pathogenic strain of *E. coli* and does not contain any genes of concern contributing to virulence, pathogenicity and/or toxigenicity (**Section 6.4**).

The parental strain that was used to construct the production strain for the β -casein was *Escherichia coli* BL21(DE3). *E. coli* BL21(DE3) is a commonly used strain in molecular biology and biotechnology for the expression of recombinant proteins.

E. coli BL21(DE3) is a non-pathogenic strain of *E. coli*, which has been classified as a Biosafety Level 1 (BSL-1) microorganism by the American Type Culture Collection (ATCC). Moreover, *E. coli* BL21(DE3) also meets the criteria for a safe production microorganism (Pariza and Foster, 1983, Pariza and Johnson, 2001).

2.2.1. Production strain

Fermify have introduced genetic modifications to the parental strain *E. coli* BL21(DE3), to create their derivative strain *E. coli* stF_24_139, which produces recombinant β -casein. *E. coli* stF_24_139 contains the following genetic modifications in comparison to the parental strain:

- Insertion of *ccdB* gene (complementation factor B) into the genome
- Transformation of the production strain with a plasmid based on pET29b(+), from which the kanamycin-resistance gene was deleted and substituted with *ccdA* gene (complementation factor A), as well as the codon optimized nucleotide sequence of the A2 variant of bovine β -casein, under control of inducible T7lac promoter.

Fermify has deposited its production strain *E. coli* stF_24_139 in the Leibniz Institute DSMZ-German Collection of Microorganisms and Cell Cultures, under its reference number DSM 35048.

Construction of the production strain

DNA constructs (plasmids) were created using standard molecular biology methods, mainly via modified Gibson assembly (Zhang et al., 2012). DNA insertion into the bacterial genome was performed using the Red recombination system of bacteriophage λ (Jensen et al., 2015).

E. coli stF_24_139 was built based on the BL21(DE3)-pET expression system. Briefly, *E. coli* BL21(DE3) expresses a high throughput T7 RNA polymerase which can transcribe genes downstream of the T7 promoter. Additionally, the T7 promoter contains a binding site for lac operon inhibitor lacI, which makes the system inducible by lactose or its synthetic analog – isopropyl- β -D-thiogalactopyranoside (IPTG). The coding sequence for β -casein was *de novo* synthesized prior to insertion downstream of the T7 promoter in the pET29b(+) plasmid. The nucleotide sequence of the β -casein gene was specifically optimized for expression in *E. coli*, however, it encodes for an amino acid sequence identical to the bovine β -casein protein (A2 variant). The expression of the protein can be controlled by the addition of IPTG. A specific transcription terminator sequence was also included downstream of the coding sequence to avoid the transcription of any unwanted DNA.

The expression system of Fermify's recombinant β -casein is based on the plasmid pET29b(+) (EMD Biosciences), which originally contained a kanamycin-resistance marker gene. The antimicrobial resistance marker was removed by substituting it using the *ccdB/ccdA* complementation system (Peubez et al., 2010). This system is naturally occurring and stabilizes *E. coli* plasmid F (Tam and Kline, 1989). Briefly, the *ccdB* protein is specifically targeting bacterial gyrase, leading to the death of the cell, unless the *ccdA* protein is present in the cell. To ensure the retention of the plasmid, the *ccdB* gene was integrated into the genome of *E. coli* stF_24_139, whereas *ccdA* was integrated into the expression plasmid.

All genetic modifications to the strain and plasmid were confirmed via polymerase chain reaction (PCR) and Sanger sequencing. Additionally, whole genome sequencing (WGS) was performed to confirm the identity of the production strain and to ensure the absence of toxins, virulence factors, antibiotic resistance genes and any unintentional changes or mutations in the genome of *E. coli* stF_24_139. The

strain is genetically stable, and the plasmid stability was assessed by re-streaking the strain on solid media repeatedly over time and followed by colony PCR to detect the presence of the plasmid. The strain stability is also demonstrated by the comparison of productivity, the recombinant protein titer remains consistent over multiple production runs.

2.3. Production process

2.3.1. Description of the method of manufacture

Upstream process

A vial of *E. coli* stF_24_139 is taken from a working cell bank and used for an inoculation/seeding stage. The seed is transferred to the production fermenter to produce biomass, followed by protein expression. The expressed β -casein protein is produced intracellularly.

Downstream process

1. The cells harvested by centrifugation at the end of the fermentation are disrupted by high pressure homogenization and subsequently washed by resuspension in fresh buffer.
2. Centrifugation is then performed to separate and remove cell debris and obtain inclusion bodies (IBs).
3. The IBs separated from cell debris are solubilized to yield crude, soluble β -casein.
4. The solution is subjected to sequential precipitation and flocculation steps for the removal of impurities, which are subsequently separated from the soluble product by centrifugation and filtration.
5. Tangential flow filtration is performed to adjust protein titer and for final product formulation. The product is retained by the ultrafiltration membrane and volumetrically concentrated to the target final concentration. An in-house developed and validated reverse-phase high-performance liquid chromatography (RP-HPLC) method is used to determine product concentration and purity during the process and for the final product. Commercially available plate-based fluorescence assays are used for determination of DNA and endotoxins on in-process samples and final products: Quant-iT™ 1X dsDNA Assay Kits, Thermofisher Scientific and respectively.
6. Finally, the product is aliquoted and stored at maximum temperature of 4 °C or 39.2 °F.

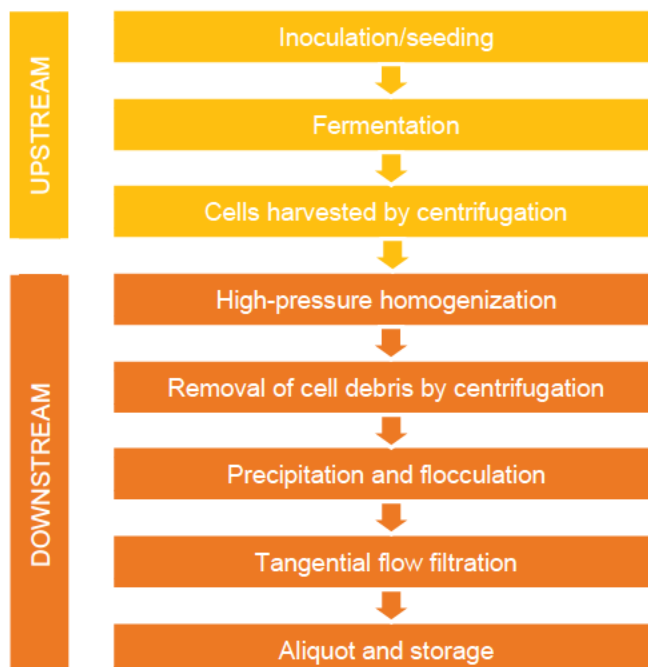


Figure 4. Flow-chart of the manufacturing process to produce β -casein produced by *E. coli* stF_24_139

2.3.2. Raw materials and processing aids

All raw materials, including the fermentation nutrients, and processing aids used to manufacture β -casein produced by *E. coli* stF_24_139 are food-grade, obtained from approved suppliers and are safe and suitable for use, conforming to either the specifications set out in the *Food Chemicals Codex* (14th edition) or to other applicable regulatory standards. The raw materials are of high purity and quality, and suitable for their intended use. They may be food-grade and GRAS or high-quality chemical or pharmaceutical grades (*United States Pharmacopeia* [USP], *National Formulary* [NF], or American Chemical Society [ACS] grades).

In the fermentation medium, soy peptone and yeast extract are used as carbon sources. The Food Allergen Labeling and Consumer Protection Act (FALCPA) labels soybean as one of the eight major food allergens. The considerations for the allergenicity profile of β -casein produced by *E. coli* stF_24_139 are addressed in **Section 6.7**. No solvents are used in the production process.

During the upstream process, Fermify adds IPTG during fermentation as an inducer of β -casein expression. The use of IPTG has been discussed in other GRAS notices such phospholipase C from *Pseudomonas fluorescens* (GRN 574, page 15), a lipase preparation from *Pseudomonas fluorescens* Biovar I (GRN 462, page 8), and an alpha amylase preparation from *Pseudomonas fluorescens* Biovar I (GRN 126), all of which have previously been reviewed by the FDA. Since IPTG is not covered by the CFR, Fermify has included a safety narrative on the substance covering its use in **Section 6.6**.

In the downstream processing, Fermify uses a specific flocculant, dimethylamine-epichlorohydrin copolymer, which is acceptable for use in the manufacturing of enzymes, as per the FDA September 11, 2003, letter to the Enzyme Technical Association (ETA) (**Appendix I**). Since Fermify's use of dimethylamine-epichlorohydrin copolymer is not fully aligned with the approved use in the Code of Federal Regulation (21 CFR § 173.60), Fermify has included the safety considerations of the substance covering its use in Fermify's production process in **Section 6.6**.

2.3.3. Quality control

β -casein produced by *E. coli* stF_24_139 is manufactured following current Good Manufacturing Practices (cGMP) with the preparation of a food safety plan in compliance with 21 CFR §117, Subparts B and C.

The batches of β -casein produced by *E. coli* stF_24_139 are evaluated against the specifications and if acceptable, transferred to packages for downstream processing, together with the Certificate of Analysis (CoA), for shipping and storage.

Fermify has established raw material specifications as part of its quality assurance program and supplier assurance program to ensure that the final product is of a suitable and safe quality. Standard operating procedures are followed to ensure a consistent production process and final product.

In the event of an out of specification (OOS) event, including the potential for an OOS production lot, steps will be taken to identify the issue(s) and appropriate corrective actions will be taken. If a specific lot cannot be corrected to meet the specifications, the lot will be destroyed in accordance with internal SOPs and cGMP.

2.4. Specifications

The specifications for β -casein produced by *E. coli* stF_24_139, along with the specification methods, are listed in **Table 2**. All methods used in the specifications are validated for their intended purpose. All tests are conducted in an ISO-17025-accredited testing facility.

Food-grade product specification limits have been established for the compositional parameters (moisture, total protein, fat, ash, and carbohydrates) and pH, as well as for β -casein as a percentage of the total protein, heavy metals (arsenic, cadmium, lead, mercury), microbiological contaminants, and endotoxins. The FCC Monograph for casein and caseinate salts (CAS RN®: 9000-71-9) is not considered appropriate for β -casein produced by *E. coli* stF_24_139 because the FCC specifications are not specific to β -casein A2 variant, and the production process is significantly different. Accordingly, additional specifications have been included for Fermify's β -casein produced by *E. coli* stF_24_139. The FCC monograph lists lead, fat and lactose as specifications. Fermify's β -casein produced by *E. coli* stF_24_139 contains trace amounts of fat and carbohydrates as the main fraction of the product is moisture and protein. Hence, fat and lactose are not appropriate specifications for Fermify's β -casein. Moreover, Fermify's β -casein is tested for arsenic, cadmium and mercury in addition to lead.

Table 2. Specifications for β -casein produced by <i>E. coli</i> stF_24_139		
Test item	Specification	Method
Color	Colorless, slightly translucent liquid	Sensory
Texture	Thin fluid	
Odor	Odorless; no off-odor	
Composition and pH		
pH	6.5 – 7.5	pH meter
Moisture	$\leq 97\%$	Gravimetry, internal method
Crude protein	$\geq 2\%$	UNI EN ISO 14891:2002, Dumas (TCD) Protein (N x 6.25)
β -casein	$\geq 75\%$ of total protein fraction	RP-HPLC/DAD (validated internal method)
Total fat	$\leq 0.1\%$	Gravimetry, internal method
Carbohydrates	$\leq 0.2\%$	Calculation
Ash	$\leq 1\%$	Rapporti ISTISAN 1996/34 pag 77, Gravimetry
Heavy metals		
Arsenic	< 0.5 mg/kg	ICP-OES, internal method
Cadmium	< 0.5 mg/kg	
Lead	< 0.5 mg/kg	
Mercury	< 0.5 mg/kg	
Microbiological contaminants		
<i>E. coli</i>	< 10 CFU/g	Internal Method SOP Mb 055 (24.05.2022), E-Cultural technique (media film) <i>Escherichia coli</i> B-Glucuronidase+
<i>Listeria monocytogenes</i> in 25 g	Not detected	Internal Method SOP Mb 010 (03.02.2021), Real-time PCR
<i>Salmonella</i> spp. in 25 g	Not detected	AFNOR 3M 01/16-11/16, Bioluminescence
Other contaminants		
Endotoxins	$\leq 20,000$ EU/mL	Ph. Eur. 10th Ed. 2.6.14, LAL turbidimetric assay
AFNOR: Association Française de Normalization; CFU: Colony Forming Units; EN: European Standard; EU: Endotoxin Units; ICP-OES: Inductively Coupled Plasma – Optical Emission Spectrometry; ISO: International Organization for Standardization; ISTISAN: Italian National Institute of Health; LAL: Limulus Amebocyte Lysate; RP-HPLC/DAD: Reverse Phase High-Performance Liquid Chromatography with Diode Array Detection; TCD: Thermal Conductivity Detector.		

2.5. Batch analysis

The results from testing three non-consecutive batches of β -casein produced by *E. coli* stF_24_139 are presented in **Table 3 – Table 6**, verifying that the product meets or exceeds the specifications. The data are presented on a wet basis. All methods used to analyze the batches were validated for their intended purpose. This data confirms that the production process is consistent among different batches. The certificates of analysis (CoAs) of the presented batches are provided in **Appendix II**.

Table 3. Batch analysis of β -casein produced by <i>E. coli</i> stF_24_139					
Test item	Unit	Specification	Batch 1 (Fermify008)	Batch 2 (Fermify019)	Batch 3 (Fermify021)
Organoleptic characteristics					
Color	-	Colorless, slightly translucent liquid	Complies	Complies	Complies
Texture	-	Thin fluid	Complies	Complies	Complies
Odor	-	Odorless; no off-odor	Complies	Complies	Complies
Proximate analysis (wet basis) and pH					
pH	-	6.5 – 7.5	7.0	7.0	7.1
Moisture	g/100 g	≤ 97	96.87	96.71	95.59
Crude Protein	g/100g	≥ 2	2.42	2.46	3.60
β -casein	% total protein	≥ 75	80.4	78.8	80.9
Total fat	g/100g	≤ 0.1	< 0.1	< 0.1	< 0.1
Carbohydrates	g/100g	≤ 0.2	0.11	0.06	0.02
Total dietary fiber	g/100g	-	< 1	< 1	< 1
Ash	g/100g	≤ 1	0.60	0.77	0.79
Proximate analysis (dry matter basis)¹					
Crude Protein	g/100g DM	-	77.32	74.77	81.63
Total fat	g/100g DM	-	$< \text{LOD}$	$< \text{LOD}$	$< \text{LOD}$
Carbohydrates	g/100g DM	-	3.51	1.82	0.45
Total dietary fiber	g/100g DM	-	$< \text{LOD}$	$< \text{LOD}$	$< \text{LOD}$
Ash	g/100g DM	-	19.17	23.40	17.91
Residual DNA (wet basis)					
DNA	$\mu\text{g/g}$	-	3.1	5.1	4.9
¹ The proximate analysis on a dry matter basis is calculated by adjusting the analyzed values to exclude moisture, representing the fraction of the nutrients in the dry matter (i.e., 100% minus the moisture content)					
DM: dry matter; CFU: colony forming units; LOD: limit of detection					

The batch analysis indicates that β -casein produced by *E. coli* stF_24_139 contains 2.42 – 3.60% of protein on a wet basis among the presented batches, which consists of 78.8% - 80.9% of Fermify's recombinant β -casein. The remaining proteins present in the product are from the production strain.

An overview of the composition of the individual amino acids in the product is presented in **Table 4**. The average total amino acid content in β -casein produced by *E. coli* stF_24_139 is 2.98% on a wet basis (**Table 4**), and, as indicated in **Table 3**, the mean crude protein content is 2.83% on wet basis. The crude protein content was analyzed using the Dumas method based on the nitrogen content (N-content $\times 6.25$) which accounts for both the protein and other nitrogen-containing substances present in the product. Additionally, the conversion factor of 6.25 assumes that proteins contain 16% nitrogen (i.e., $1/0.16 = 6.25$). However, this is an average, and the nitrogen content of proteins can vary. A more accurate analysis of protein content can be estimated based on the total amino acid content. However, considering the similar results, it can be concluded that the Dumas method (N-content $\times 6.25$) is adequate to measure the protein content in β -casein produced by *E. coli* stF_24_139.

The mean of the measured amino acid composition of the batches of β -casein produced by *E. coli* stF_24_139 (in molar % of the total protein fraction) was compared against the theoretical values of native bovine β -casein (A2 variant), which indicates a similar amino acid composition. The slight deviation from the theoretical values is due to the purity of β -casein in the product and the presence of host strain protein.

Table 4. Amino acid analysis on a wet basis of β -casein produced by <i>E. coli</i> stF_24_139 compared to values for native bovine β -casein A2 variant					
Test item	Batch 1 (Fermify008) (g/100g)	Batch 2 (Fermify019) (g/100g)	Batch 3 (Fermify021) (g/100g)	Mean values (mol %) ¹	Theoretical values for bovine β - casein (A2 variant) (mol %) ¹
Alanine	0.0469	0.0431	0.0679	1.77%	2.39%
Arginine	0.0802	0.0802	0.0813	2.70%	1.91%
Asparagine + aspartic acid	0.132	0.121	0.160	4.62%	4.31%
Cystine + cysteine	< 0.006	< 0.006	< 0.006	-	0.00%
Glutamine + glutamic acid	0.569	0.549	0.787	21.30%	18.66%
Glycine	0.0373	0.0375	0.0551	1.45%	2.39%
Histidine	0.0716	0.0658	0.100	2.65%	2.39%
Hydroxyproline	< 0.2	< 0.2	< 0.2	-	0.00%
Isoleucine	0.110	0.117	0.173	4.47%	4.78%
Leucine	0.282	0.278	0.419	10.94%	10.53%
Lysine	0.169	0.188	0.268	6.99%	5.26%
Methionine	0.100	0.0960	0.136	3.71%	2.87%
Ornithine	< 0.05	< 0.05	< 0.05	-	0.00%
Phenylalanine	0.148	0.132	0.197	5.33%	4.31%
Proline	0.371	0.282	0.575	13.73%	16.75%
Serine	0.149	0.140	0.201	5.48%	7.66%
Threonine	0.0983	0.104	0.137	3.79%	4.31%
Tryptophane	0.0177	0.0188	0.0285	0.73%	0.48%
Tyrosine	0.0695	0.0614	0.0880	2.45%	1.91%
Valine	0.205	0.204	0.297	7.89%	9.09%
Sum of amino acids	2.6565	2.5178	3.7708	100%	100%

¹The mean of the measured amino acid composition of the batches (in mol% of total protein fraction) was compared with the theoretical values of native bovine β -casein (A2 variant). The amino acid sequence of native bovine β -casein (A2 variant) was derived from the UniProt database under Accession No. P02666.

Table 5 gives an overview of potential contaminants in β -casein produced by *E. coli* stF_24_139. The results indicate that heavy metals, microbiological contaminants, and mycotoxins were below the limit of detection. Endotoxins were present at a level ranging from 6,400 to 14,000 endotoxin units per mL (EU/mL). Total WHO-PCDD/F and PCB refers to the sum of toxic equivalents (TEQs) of polychlorinated dibenzo-p-dioxins (PCDDs), polychlorinated dibenzofurans (PCDFs), and polychlorinated biphenyls (PCBs), according to toxic equivalency factors (TEFs) developed by the World Health Organization (WHO). The upper bound level reported in **Table 5** assumes that concentrations below the detection limit are equal to the detection limit. The upper bound level of total WHO-PCDD/F and PCBs ranged from 0.0694 to 0.111 pg/g or ppb among the presented batches, which is far below the tolerance value for residues of PCBs of 1.5 ppm for manufactured dairy products referenced under 21 CFR §109.30.

Table 5. Analysis of contaminants in β -casein produced by <i>E. coli</i> stF_24_139					
Test item	Unit	Specification	Batch 1 (Fermify008)	Batch 2 (Fermify019)	Batch 3 (Fermify021)
Heavy metals					
Arsenic (As)	mg/kg	< 0.5	< 0.5	< 0.5	< 0.5
Cadmium (Cd)	mg/kg	< 0.5	< 0.1	< 0.1	< 0.1
Lead (Pb)	mg/kg	< 0.5	< 0.5	< 0.5	< 0.5
Mercury (Hg)	mg/kg	< 0.5	< 0.005	< 0.005	< 0.005
Microbiological contaminants					
Total aerobic count	CFU/g	-	< 10	< 10	< 10
Enterobacteriaceae	CFU/g	-	< 10	< 10	< 10
<i>E. coli</i>	CFU/g	< 10	< 10	< 10	< 10
<i>Salmonella</i> spp.	CFU/g	ND	ND	ND	ND
<i>Listeria monocytogenes</i>	CFU/g	ND	ND	ND	ND
<i>Bacillus cereus</i>	CFU/g	-	< 10	< 10	< 10
Yeasts	CFU/g	-	< 10	< 10	< 10
Molds	CFU/g	-	< 10	< 10	< 10
<i>Cronobacter</i> spp.	CFU/g	-	ND	ND	ND
Coagulase-positive <i>Staphylococcus aureus</i>	CFU/g	-	< 10	< 10	< 10
<i>Clostridium perfringens</i>	CFU/g	-	< 10	< 10	< 10
Mycotoxins					
Aflatoxin B1	$\mu\text{g/kg}$	-	< 1.0	< 1.0	< 1.0
Aflatoxin B2	$\mu\text{g/kg}$	-	< 1.0	< 1.0	< 1.0
Aflatoxin G1	$\mu\text{g/kg}$	-	< 1.0	< 1.0	< 1.0
Aflatoxin G2	$\mu\text{g/kg}$	-	< 1.0	< 1.0	< 1.0
Deoxynivalenol	$\mu\text{g/kg}$	-	< 20	< 20	< 20
Zearalenone	$\mu\text{g/kg}$	-	< 10	< 10	< 10
Ochratoxin A	$\mu\text{g/kg}$	-	< 2.0	< 2.0	< 2.0
T2-toxin	$\mu\text{g/kg}$	-	< 10	< 10	< 10
HT2-toxin	$\mu\text{g/kg}$	-	< 10	< 10	< 10
Fumonisin B1	$\mu\text{g/kg}$	-	< 20	< 20	< 20
Fumonisin B2	$\mu\text{g/kg}$	-	< 20	< 20	< 20
Dioxins/furans and PCBs					
Total WHO(2005)- PCDD/F+PCB TEQ (lower-bound)	pg/g	-	ND	0.000609	ND
Total WHO(2005)- PCDD/F+PCB TEQ (medium-bound)	pg/g	-	0.0347	0.0351	0.0554
Total WHO(2005)- PCDD/F+PCB TEQ (upper-bound)	pg/g	-	0.0694	0.0697	0.111
Other contaminants					
Endotoxins	EU/mL	$\leq 20,000$	14,000	6,400	9,700

CFU: colony forming units; ND: not detected; WHO: World Health Organization; PCDD/F: polychlorinated dibenzop-dioxins (PCDDs), polychlorinated dibenzofurans (PCDFs); PCB: polychlorinated biphenyls. TEQ; toxic equivalents.

Table 6 provides analytical data on all minerals and trace elements that could potentially be carried over from the fermentation medium during the upstream process of the production process. The levels of elemental phosphorus ranged from 1,500 to 1,700 mg/kg among the presented batches of β -casein produced by *E. coli* stF_24_139 and originate from the sodium phosphate buffer in which the product is dissolved. The levels of sodium ranged from 2,300 to 2,500 mg/kg among the presented batches, which also originate from the buffer. The levels of elemental sulphur ranged from 210 to 320 mg/kg among the presented batches and originated from the protein fraction of the product through the presence of sulphur-containing amino acid methionine. Boron and copper were detected slightly above the limit of detection in two batches (Fermify019 and Fermify021), at levels of 0.3 mg/kg and 0.1 mg/kg, respectively, and magnesium was detected at levels ranging from 0.8 to 9.5 mg/kg, among all presented batches. All other minerals and trace elements were below the limit of detection.

Table 6. Analysis of minerals and trace elements in β -casein produced by <i>E. coli</i> stF_24_139				
Test item	Unit	Batch 1 (Fermify008)	Batch 2 (Fermify019)	Batch 3 (Fermify021)
Antimony (Sb)	mg/kg	< 0.50	< 0.50	< 0.50
Boron (B)	mg/kg	< 0.2	0.3	0.3
Cobalt (Co)	mg/kg	< 0.2	< 0.2	< 0.2
Copper (Cu)	mg/kg	< 0.1	0.1	0.1
Iron (Fe)	mg/kg	< 0.5	< 0.5	< 0.5
Magnesium (Mg)	mg/kg	9.5	1.1	0.8
Manganese (Mn)	mg/kg	< 0.1	< 0.1	< 0.1
Molybdenum (Mo)	mg/kg	< 0.2	< 0.2	< 0.2
Nickel (Ni)	mg/kg	< 0.1	< 0.1	< 0.1
Phosphorus (P)	mg/kg	1,500	1,700	1,700
Potassium (K)	mg/kg	< 3	6	< 3
Selenium (Se)	mg/kg	< 0.5	< 0.5	< 0.5
Sodium (Na)	mg/kg	2,300	2,500	2,500
Sulphur (S)	mg/kg	210	220	320
Tin (Sn)	mg/kg	< 0.5	< 0.5	< 0.5
Zinc (Zn)	mg/kg	< 0.5	< 0.5	< 0.5

2.6. Viable cells of the production strain

No viable *E. coli* cells were detected in the product using a validated internal method which detects β -glucuronidase enzyme activity, a biomarker for the presence of *E. coli*. Based on the batch results indicated in Table 5, it can be concluded that no *E. coli* cells are present in the product, neither contamination, nor as remaining viable cells of the production strain.

2.7. Physical or technical effect

β -casein produced by *E. coli* stF_24_139 is intended for use as an emulsifier, stabilizer, formulation aid, surface-active agent, nutrient supplement, synergist and texturizer.

PART 3. §170.235 DIETARY EXPOSURE

3.1. Intended use

β -casein produced by *E. coli* stF_24_139 is intended to be used as a food ingredient, at a maximum level of up to 15 g of dry matter of β -casein produced by *E. coli* stF_24_139 per 100 g in cheeses and cheese analogs and at a maximum level of 3.8 g of dry matter of β -casein produced by *E. coli* stF_24_139 per 100 g in non-alcoholic beverages, dairy product analogs, frozen dairy desserts, milk products and plant protein products. Users may therefore need to concentrate β -casein produced by *E. coli* stF_24_139 to reach the desired use level. The maximum use levels are based on the inclusion of β -casein produced by *E. coli* stF_24_139 on a dry matter basis in the food products.

The proposed use of β -casein produced by *E. coli* stF_24_139 is not intended for use in foods where standards of identity would preclude such use, infant formula, meat, poultry, egg products, catfish, or any products that would require additional regulatory review by USDA.

3.2. Estimated daily intake

3.2.1. Methods

The estimated daily intake (EDI) of β -casein produced by *E. coli* stF_24_139 when used up to 15 g/100g in cheeses and cheese analogs and 3.8 g/100g in non-alcoholic beverages, dairy product analogs, frozen dairy desserts, milk products and plant protein products, was calculated for the total US population 2-year-old and older (2+ year), and four subpopulations including 2-4-year-old young children, 5-12-year-old children, 13–18-year-old adolescents and 19+-year-old adults using food consumption data from the 2017 - March 2020 pre-pandemic (2017-2020 pp) National Health and Nutrition Examination Surveys (NHANES), according to the FDA Guidance for Industry: Estimating Dietary Intake of Substances in Food (FDA, 2006). The data and methods used to conduct the assessment and results are summarized below.

The list of food codes reported consumed in NHANES 2017-2020 pp were reviewed. **Table 7** represents the food types for which β -casein produced by *E. coli* stF_24_139 is proposed for use, a description of representative food categories selected for the analysis, and the maximum intended use level of β -casein produced by *E. coli* stF_24_139. The list of the NHANES food codes included in the analysis are provided in **Appendix III**.

Table 7. Proposed food uses, maximum use levels of β -casein produced by <i>E. coli</i> stF_24_139 by food use and description of representative foods selected for the analysis		
Food use	Description of NHANES Foods selected for analysis (Appendix III)	Maximum use level (g per 100 g food as consumed) ¹
Beverages (non-alcoholic)	Smoothies and grain drinks, fruit drinks, sport and energy drinks	3.8 g/100 g
Cheeses	Cheese (all types, including blue cheese, brick cheese, Camembert, Cheddar, Colby, feta, Fontina, goat cheese, Gouda, Gruyère, Monterey, mozzarella, muenster, parmesan, provolone, Swiss and American cheese), cottage/ricotta cheese, cream cheese, sour cream, whipped cream	15 g/100 g
Cheese analogs	Imitation cheese	15 g/100 g
Dairy analog products	Cream substitutes and milk substitutes (excluding coconut milk)	3.8 g/100 g
Frozen dairy desserts	Dairy-based ice cream (all types, including novelty bars, sandwiches, cones, etc.) and frozen yoghurt	3.8 g/100 g
Milk products	Dairy-based milk (all types, including flavored milk), dairy-based yogurt (all types, including regular and Greek yogurt), milk shakes	3.8 g/100 g
Plant protein products	Meat analog/vegetarian/meatless products including frankfurter, bacon strip, breakfast link, chicken, luncheon slice, meatball, vegetarian burger/patty, sandwich spread (meat substitute type), meat substitute	3.8 g/100 g
The estimates for the intake of β -casein produced by <i>E. coli</i> stF_24_139 were generated using the maximum use level for each intended food-use, with food consumption data available from the 2017-2020 pp NHANES datasets with the DaDiet software.		
¹ The maximum use levels reported in this table are based on the inclusion of Fermify's β -casein produced by <i>E. coli</i> stF_24_139 on a dry matter basis in the final product, excluding the moisture content of the product. Users may therefore need to concentrate Fermify's β -casein produced by <i>E. coli</i> stF_24_139 to reach the desired use level.		

Consumption data from individual dietary records, detailing food items ingested by each survey participant (What We Eat in America (WWEIA) dietary component of the NHANES conducted in 2017-2020 pp), were used to generate estimates for the intake of β -casein produced by *E. coli* stF_24_139 when used in non-alcoholic beverages, cheeses, cheese analogs, dairy analog products, frozen dairy desserts, milk products, and plant protein products, by the US population. Statistical analysis and data management were conducted in DaDiet Software (Version 17.04, Dazult Ltd). DaDiet Software is a web-based software tool that allows an accurate estimate of exposure to nutrients and to substances added to foods, including contaminants, food additives and novel ingredients. The main input components are concentration (use level) data and food consumption data. Datasets are combined in the software to provide accurate and efficient exposure assessments.

Estimates for the daily intake of β -casein produced by *E. coli* stF_24_139 represent projected 2-day averages for each individual from Day 1 and Day 2 of NHANES 2017-2020 pp; these average amounts comprised the distribution from which mean, and percentile intake estimates were determined. Mean and percentile estimates were generated incorporating survey weights to provide representative intakes for the entire U.S. population. "Per capita" intake refers to the estimated intake of β -casein produced by *E. coli* stF_24_139 averaged over all individuals surveyed, regardless of whether they consumed food products in which β -casein produced by *E. coli* stF_24_139 is proposed for use, and therefore includes individuals with "zero" intakes (i.e., those who reported no intake of food products containing β -casein produced by *E. coli* stF_24_139 during the 2 survey days). "Consumer-only" intake refers to the estimated intake of β -casein produced by *E. coli* stF_24_139 by those individuals who reported consuming food products in which the use of β -casein produced by *E. coli* stF_24_139 is currently under consideration. Individuals were considered "consumers" if they reported consumption of 1 or more food products in which β -casein produced by *E. coli* stF_24_139 is proposed for use on either Day 1 or Day 2 of the survey.

This approach and software to estimate daily intake has been reviewed by the FDA as part of different GRAS notices for which a "no questions letter" was issued (e.g., GRN No. 881, GRN No. 943, GRN No. 996, GRN No. 997).

3.2.2. Results

The estimated daily intake (EDI) of β -casein produced by *E. coli* stF_24_139 was calculated using DaDiet Software (Version 17.04, Dazult Ltd) and NHANES 2017-2020 pp consumption data and maximum proposed use levels in non-alcoholic beverages, cheeses, cheese analogs, dairy analog products, frozen dairy desserts, milk products, and plant protein products (**Table 7**). Intake estimates are provided on a *per capita* and *per user* basis at the mean and 90th percentile of intake and expressed in g per day (g/day) and g per kg body weight per day (g/kg body weight/day) for the total US population (+ 2-year-old) and 4 subpopulations (2–5-year-old young children, 6–12-year-old children, 13-18-year-old adolescents, and 19+ year old adults). The total EDI of β -casein produced by *E. coli* stF_24_139 from the consumption of any of the proposed food uses are presented in **Table 8** (in g/day) and **Table 9** (in g/kg body weight/day).

Among the US population 2+ y, 88.4% were identified as consumers of foods proposed to contain β -casein produced by *E. coli* stF_24_139. The EDI of β -casein produced by *E. coli* stF_24_139 among the US population two years and older at the *per user* mean and 90th percentile of intake was 13.380 g β -casein/day (0.245 g β -casein/kg bw/day) and 27.816 g β -casein/day (0.567 g β -casein/kg bw/day), respectively. *Per user* mean intake of β -casein produced by *E. coli* stF_24_139 ranged from 12.379 g β -casein/day among adults to 17.258 g β -casein/day among children 6-12 y. On body weight basis, the highest *per user* mean intake is among children 2-5 y at 0.995 g β -casein/kg bw/day. The *per user* 90th intake estimates of β -casein produced by *E. coli* stF_24_139 range from 26.355 g β -casein/day among adults, to 31.633 g β -casein/day among children 6-12y. The highest *per user* 90th percentile of intake on a bodyweight basis is among children 2-5 y at 1.791 g β -casein/kg bw/day.

It should be considered that these exposure values represent the worst-case scenario, and are considered to be an overestimation, as it considers the intake of the product at the maximum use levels in all the proposed food categories β -casein produced by *E. coli* stF_24_139 is not expected to have such a vast market range, and the real-life exposure is considered to be much lower.

β -casein produced by *E. coli* stF_24_139 is intended to replace animal-derived casein and other protein in the selected food categories and will be substitutional for bovine β -casein and other dietary protein in the diet. Given the proposed uses for β -casein produced by *E. coli* stF_24_139, the cumulative exposure to β -casein (both from dietary exposure from milk and milk protein products and from the intended uses of β -casein produced by *E. coli* stF_24_139) is not expected to increase.

Table 8. Two-day average estimated daily intake (EDI) of β-casein produced by <i>E. coli</i> stF_24_139 (g/day) from all proposed food uses among the total US population two years and older (2+ year) & subpopulations.						
Population	N*	% User	Per Capita		Per User	
			Mean	90 th percentile	Mean	90 th percentile
US population (2+ years)	8797	88.4%	11.831	26.604	13.380	27.816
Children (2-5 years)	793	99.5%	16.518	28.594	16.608	28.643
Children (6-12 years)	1339	97.0%	16.743	31.633	17.258	31.633
Adolescents (13-18 years)	1010	88.5%	13.792	28.584	15.580	30.205
Adults (19+ years)	5655	86.7%	10.728	24.691	12.379	26.355
*Unweighted number of users, % users, and <i>per capita</i> and <i>per user</i> estimates based on NHANES 2017-2020 pp were derived using the statistical weights provided by the National Center for Health Statistics (NCHS).						

Table 9. Two-day average estimated daily intake (EDI) of β-casein produced by <i>E. coli</i> stF_24_139 (g/kg bw/day) from all proposed food uses among the total US population two years and older (2+ year) & subpopulations.						
Population	N*	% User	Per Capita		Per User	
			Mean	90 th percentile	Mean	90 th percentile
US population (2+ years)	8705	88.4%	0.216	0.528	0.245	0.567
Children (2-5 years)	776	99.4%	0.989	1.791	0.995	1.791
Children (6-12 years)	1331	97.0%	0.491	0.965	0.507	0.965
Adolescents (13-18 years)	1002	88.4%	0.212	0.483	0.240	0.495
Adults (19+ years)	5596	86.7%	0.134	0.315	0.154	0.336
*Unweighted number of users, % users, and <i>per capita</i> and <i>per user</i> estimates based on NHANES 2017-2020 pp were derived using the statistical weights provided by the National Center for Health Statistics (NCHS).						

3.3. Exposure to undesirable substances

Sodium

The levels of sodium in β -casein produced by *E. coli* stF_24_139 ranged from 2,300 to 2,500 mg/kg among the presented batches on a wet basis, which originated from the sodium phosphate buffer in which the product is dissolved. On a dry matter basis, these levels translate to a range from 5.67 to 7.60 g per 100 g of product. Taking the worst-case scenario into account of 7.60 g/100 g of sodium in the product on a dry matter basis, the mean and 90th percentile exposure of sodium through consumption of β -casein produced by *E. coli* stF_24_139 accounts for 1.017 g/day (or 0.019 g/kg bw/day) and 2.114 g/day (or 0.043 g/kg bw/day), respectively, among the US population two years and older. The World Health Organization (WHO) recommends a sodium intake of maximum 2 g/day in adults (WHO, 2012), which is slightly exceeded by the 90th percentile intake of the product. Since the exposure to the product is considered an overestimation, and real-life exposure to the product is considered to be significantly lower, the intake of sodium through the consumption of β -casein produced by *E. coli* stF_24_139 is not considered to be of concern.

Magnesium

The levels of sodium in β -casein produced by *E. coli* stF_24_139 ranged from 0.8 to 9.5 mg/kg among the presented batches on a wet basis, which equals a range from 18 to 303.5 mg/kg of the product on a dry matter basis. Taking the worst-case scenario into account of 303.5 mg/kg of product on a dry matter basis, the mean and 90th percentile exposure to magnesium through consumption of β -casein produced by *E. coli* stF_24_139 accounts for 4.06 mg/day (or 0.074 mg/kg bw/day) and 8.44 mg/day (or 0.172 mg/kg bw/day), respectively, among the US population two years and older. Magnesium is considered to be a nutrient, and the European Food Safety Authority (EFSA) sets an adequate intake (AI) for adults of magnesium at 350 mg/day for men and 300 mg/day for women (EFSA, 2015). The intake of magnesium through the consumption of β -casein produced by *E. coli* stF_24_139 is not considered to be of concern.

Boron

Boron was detected in two batches of β -casein produced by *E. coli* stF_24_139 (Fermify019 and Fermify021), at levels of 0.3 mg/kg, on a wet basis. On a dry matter basis, this value equals to 9.12 mg/kg for batch Fermify019 and 6.80 mg/kg for batch Fermify021. Taking the level of 9.12 mg/kg, the mean and 90th percentile exposure to boron through consumption of β -casein produced by *E. coli* stF_24_139 accounts for 122 μ g/day (or 2.23 μ g/kg bw/day) and 254 μ g/day (or 5.17 μ g/kg bw/day), respectively, among the US population two years and older. The WHO refers to a tolerable daily intake (TDI) of boron of 0.17 mg/day (WHO, 2009). The 90th percentile intake of boron through consumption of β -casein produced by *E. coli* stF_24_139, i.e., 5.17 μ g/kg bw/day, represents 3.04% of the TDI of boron. Therefore, the intake of boron through consumption of β -casein produced by *E. coli* stF_24_139 is not considered to be of concern.

Copper

Copper was detected in two batches of β -casein produced by *E. coli* stF_24_139 (Fermify019 and Fermify021), at levels of 0.1 mg/kg, on a wet basis. On a dry matter basis, this value equals to 3.04 mg/kg for batch Fermify019 and 2.27 mg/kg for batch Fermify021. Taking the level of 3.04 mg/kg into account, the mean and 90th percentile exposure to copper through consumption of β -casein produced by *E. coli* stF_24_139 accounts for 40.67 μ g/day (or 0.75 μ g/kg bw/day) and 84.56 μ g/day (or 1.72 μ g/kg bw/day), respectively, among the US population two years and older. The WHO reports that the upper limit of the acceptable range of oral intake (AROI) in adults is uncertain but is most likely in the range of 2 to 3 mg/day in adults (WHO, 2004). The 90th percentile intake of copper through consumption of β -casein produced by *E. coli* stF_24_139, i.e., 84.56 μ g/day bw/day, represents 4.23% of the AROI of copper. Therefore, the intake of copper through consumption of β -casein produced by *E. coli* stF_24_139 is not considered to be of concern.

Residual DNA

DNA was detected in β -casein produced by *E. coli* stF_24_139 at levels ranging from 3.1 $\mu\text{g/g}$ to 5.1 $\mu\text{g/g}$, on a wet basis. These values equal a range from 99 $\mu\text{g/g}$ to 155 $\mu\text{g/g}$ on dry matter basis. Taking the level of 155 $\mu\text{g/g}$ into account, the mean and 90th percentile exposure to DNA through consumption of β -casein produced by *E. coli* stF_24_139 accounts for 2.07 mg/day (or 0.037 mg/kg bw/day) and 4.31 mg/day (or 0.088 mg/kg bw/day), respectively, among the US population two years and older. The FAO/WHO/UNICEF Protein Advisory Group recommends limiting the dietary exposure to nucleic acids from single cell protein novel foods to no more than 2 g/person/day (WHO, 2007). Taking the worst-case intake scenario into account, considering the 90th percentile intake, i.e., 4.31 mg/day, the intake of DNA through consumption of β -casein produced by *E. coli* stF_24_139, would only represent 0.22% of the accepted intake of 2 g/person/day of nucleic acids.

Endotoxins

Endotoxins were present in β -casein produced by *E. coli* stF_24_139 at a level ranging from 6,400 to 14,000 endotoxin units per mL (EU/mL) on a wet basis. Converting these values on a dry matter basis, they range from 195,000 EU/g to 450,000 EU/g. Considering a level of 450,000 EU/g, the mean and 90th percentile exposure to endotoxins through consumption of β -casein produced by *E. coli* stF_24_139 accounts for 6.02×10^6 EU/day (or 110,000 EU/kg bw/day) and 1.25×10^7 EU/day (or 255,000 EU/kg bw/day), respectively. The specification for endotoxins in β -casein produced by *E. coli* stF_24_139 was set at 20,000 EU/mL, which translates to 660,000 EU/g on a dry matter basis, considering a moisture level of 97%. This value can be considered the worst-case scenario representing the highest potential endotoxin level in the product. Taking the value of 660,000 EU/g on dry matter basis of the product into account, the mean and 90th percentile exposure to endotoxins through consumption of β -casein produced by *E. coli* stF_24_139 accounts for 8.83×10^6 EU/day (or 162,000 EU/kg bw/day) and 1.84×10^7 EU/day (or 374,000 EU/kg bw/day), respectively. The safety considerations resulting from the intake of endotoxins through consumption of β -casein produced by *E. coli* stF_24_139 are considered in **Section 6.6**.

PART 4. §170.240 SELF-LIMITING LEVELS OF USE

There are no known inherent self-limiting levels of use.

PART 5. §170.245 COMMON USE IN FOOD BEFORE 1958

The GRAS conclusion for β -casein produced by *E. coli* stF_24_139 is based on scientific procedures, and thus, experience based on common use in food prior to 1958 is not considered pivotal information. Whilst β -casein has a long history of safe consumption in the human diet, β -casein produced by *E. coli* stF_24_139 has not been commonly used in foods prior to 1958.

PART 6. §170.250 NARRATIVE ON THE CONCLUSION OF GRAS STATUS

6.1. Introduction

β -casein produced by *E. coli* stF_24_139 is a liquid product containing Fermify's recombinant β -casein at a concentration of 1.5 to 4% on a wet basis and is intended as a food ingredient in non-alcoholic beverages, cheeses, cheese analogs, dairy analog products, frozen dairy desserts, milk products, and plant protein products.

β -casein is a protein that is naturally present in bovine milk. Bovine milk contains on average about 3.5% protein. The proteins present in milk can be categorized into two principal groups: caseins and whey (or non-casein) proteins. Caseins are phosphoproteins insoluble at pH 4.6 (the isoelectric pH) at 20 °C or 68 °F. Whey proteins are soluble under these conditions. On average, bovine milk consists of 80% casein and 20% whey protein (Yalçin, 2006). Isoelectric casein is known to consist of four primary proteins (gene products), which are α S1-casein, α S2-casein, β -casein, and κ -casein, present in the approximate ratio of 40:10:35:12 of the total protein fraction in milk (Fox and Mulvihill, 1990). Caseins in milk form complex aggregates or micelles of proteins and mineral calcium phosphate. Caseins utilize a calcium-sequestering mechanism to regulate the calcium phosphate concentration in their environment (Fox & Brodtkorp, 2008).

β -casein produced by *E. coli* stF_24_139 contains min. 1.5% of Fermify's recombinant β -casein, which is comparable to the total casein concentration naturally present in bovine milk (ca. 2.8%).

The structure of β -casein in bovine milk can vary depending on the breed and genetic properties of different cows. The two major subvariants for this protein are the A1 and A2 types, which differ by an amino acid at position 67. The A1 genetic variant has a histidine amino acid at this position, while the A2 genetic variant contains proline at position 67 (Poulsen et al., 2017). While the A1 and A2 genetic variants are the most common variants, also other subvariants exist. Most commercial milks contain a mixture of A1 and A2 β -casein, which can arrive from either heterozygous A1/A2 cows or from mixing milk from A1/A1 and A2/A2 cows (Jianqin et al., 2016).

Fermify's β -casein has an amino acid sequence that is based on the A2 genetic variant of β -casein and contains a proline at position 67. In addition, Fermify's β -casein contains an additional methionine at the N-terminal end of the amino acid sequence, as a consequence of expressing the protein in *E. coli*. In bacterial systems, protein expression typically begins with methionine due to the start codon (AUG), and this methionine is typically not removed from the N-terminus. Based on the amino acid sequence, amino acid composition and molecular weight characteristics of Fermify's β -casein and native bovine β -casein presented in **Section 2.1.1**, it can be concluded that both β -caseins are substantially equivalent, and the safety of Fermify's β -casein can be based on the safety of native bovine β -casein.

6.2. Absorption, digestion, metabolism, and excretion of β -casein

β -casein is a protein and is digested and metabolized in the same way as other dietary proteins. An overview of the absorption, digestion, metabolism, and excretion (ADME) profile of dietary protein is provided below.

Upon ingestion, β -casein is first exposed to the acidic environment of the stomach, where it undergoes partial hydrolysis by gastric enzymes, primarily pepsin. This enzymatic action breaks down β -casein into larger polypeptides. These polypeptides then move into the small intestine, where they encounter pancreatic enzymes such as chymotrypsin, elastase, and carboxypeptidases. These pancreatic proteases further hydrolyze the peptides into smaller units, consisting of 6 to 8 amino acids. Free amino acids are also formed to a small extent by the action of the carboxypeptidases. The resulting peptides are further hydrolyzed by peptidases associated with the brush border membrane of enterocytes, which have a specificity of oligopeptides consisting of 6 to 8 amino acids and break them down into small peptides containing 2 to 3 amino acids. The dipeptides and tripeptides and free amino acids are absorbed by the enterocytes through active transport mechanisms (Bhutia and Ganapathy, 2018).

Once absorbed into the bloodstream, free amino acids, di- and tripeptides are transported to the liver via the portal vein. In the liver, they may be utilized for protein synthesis, converted into other biologically

active molecules, or used as an energy source through gluconeogenesis or the citric acid cycle (Chandel, 2021).

The ADME profile of Fermify's recombinant β -casein is expected to follow the ADME profile of dietary protein and is expected to be equal to milk derived β -casein. An *in vitro* digestibility of β -casein produced by *E. coli* stF_24_139 (Megazyme, Medallion Labs; **Section 2.1.1 – Table 1**) demonstrated that the product is highly digestible and is comparable to the *in vitro* digestibility profile of native bovine β -casein.

6.3. Safety of β -casein

Due to the equivalence between native bovine β -casein and Fermify's recombinant β -casein, the safety of Fermify's recombinant β -casein can be based on the long history of safe consumption of bovine milk and milk-derived products, both in liquid form, dried form (e.g., milk powder), or as milk-derived ingredients.

Casein and caseinates (including sodium and calcium salts), which contain β -casein, have an extensive history of safe consumption as they are naturally occurring proteins in bovine milk. Sodium caseinate is GRAS for multiple purposes when used in accordance with Good Manufacturing Practice (GMP) under 21 CFR §182.1748. The safety of casein and sodium caseinate and calcium caseinate was reviewed by the FDA Select Committee on GRAS Substances (SCOGS) who concluded the following:

- *It is essential that food grade specifications for casein, sodium caseinate, and calcium caseinate be established including provisions for acceptable levels of lysinoalanine, nitrite, and nitrosamines. Assuming that acceptable levels of lysinoalanine, nitrite, and nitrosamine are established, there is no evidence in the available information on casein, sodium caseinate, or calcium caseinate that demonstrates or suggests reasonable grounds to suspect a hazard when they are used at levels that are now current or that may reasonably be expected in the future".*
- *There is no evidence in the available information on casein that demonstrates or suggests reasonable grounds to suspect a hazard when it is used in paper and paperboard products for food packaging at levels that are now current or that might reasonably be expected in the future.*

The FDA reviewed several applications containing casein or casein-derived substances such as hydrolysates. An overview of these GRAS affirmations and notices is provided in **Table 10**. These precedents demonstrate the FDA's confidence in the safety of casein and casein-derived substances. The safety of concentrated milk proteins including caseinates was the subject of GRN No. 504, GRN No. 633 and GRN No. 959.

Table 10. Overview of applications receiving GRAS status with no objection from the FDA				
GRN No.	Substance	Casein specifications	Use	FDA's letter
11	Calcium casein peptone-calcium phosphate	Hydrolyzed casein	Texturizer in chewing gum not to exceed 5%	FDA has no questions
199	Concentrated hydrolyzed milk protein	Hydrolyzed milk protein, manufactured either through milk fermentation or enzymatic hydrolysis of casein	Ingredient in orange juice, yogurt (including yogurt drinks), nutrition bars and margarine spread; ingredient in hard and soft candies, cereal, cheese, chewing gum, chocolate, cookies, crackers, cranberry juice, energy drinks, frozen dinners, grape juice, ketchup, meal replacements, milk and dairy-based beverages, tea beverages and tomato juice	FDA has no questions
504	Milk protein concentrate (MPC) and milk protein isolate (MPI)	- MPC contains < 40% milk protein, although several forms of MPC are commercially available. - MPI contains < 90% milk protein. - Casein-to-whey ratio in the finished product is equivalent to that of the original milk, generally 80:20.	Ingredient in meal replacements and meal supplements, milk products including milk drinks, yogurt, fermented milks, spreads, dips, non-standardized cheese products, dairy product analogs, frozen dairy desserts and mixes, desserts and mousses, confection and frostings, snack foods, coatings and fillings, salad dressings, soups and soup mixes, and sauces.	FDA has no questions
633	Concentrated milk protein with a $\geq$ 60:40 whey:casein ratio	Concentrated milk protein, contains > 79.5% protein with casein-to-whey ratio of 60:40	For use as an emulsifier, flavoring agent, formulation aid, humectants, stabilizer, thickener, texturizers, and protein source in the following food categories at varying use levels depending on technical effect: meal replacements and meal supplements; powdered nutritional beverages; nutritional bars; acidified sports beverages; milk products; yogurt and fermented milk products; non-standardized cheese products; spreads, dips and cream substitutes; frozen dairy desserts and mixes; desserts and mousses; confections; snack foods; coatings and fillings; salad dressings; soups, soup mixes, and sauces	FDA has no questions
959	Iron milk proteinate	Ferric iron linked to sodium caseinate, contains > 50% protein	Dietary source of iron in the same food categories as other iron salts at levels consistent with current good manufacturing practice	FDA has no questions

The safety of casein was also evaluated in a dose-repeated toxicity study following OECD guideline 408, in which Wistar rats were orally administered casein nanoparticles at a dose of 50, 150 or 500 mg/kg bw/day over 90 days. No evidence of significant alterations in the treated animals were reported. The nanoparticles were not absorbed and remained within the gut as observed by radiolabeling in the biodistribution study. After 28 days, there was a generalized hyperchloremia in the males and females treated with the highest dose of 500 mg/kg bw, that was coupled with hypernatremia in the females. These effects were related to the presence of mannitol which was used as excipient in the formulation of casein nanoparticles (Gil et al., 2017). It should be noted that β -casein produced by *E. coli* stF_24_139 does not contain casein nanoparticles and will be present in dissolved form in food when consumed and is therefore not expected to remain in the gut. β -casein produced by *E. coli* stF_24_139 is expected to be absorbed as described in Section 6.2.

6.3.1. Bioactivity of β -casein

As mentioned in Section 6.1, two subvariants of β -casein can be found in bovine milk, A1 β -casein and A2 β -casein, which differ in one amino acid at position 67. The A1 polymorphism is more likely to undergo enzymatic cleavage resulting in the product peptide β -casomorphin-7 (BCM-7). The

production of BCM-7 is primarily linked with A1 β -casein, which is connected to the development of several adverse effects (Kay et al., 2021).

In general, BCM-7 released from A1 β -casein binds to μ -opioid receptors found throughout the gastrointestinal tract. This interaction contributes to various adverse gastrointestinal effects, such as lactose intolerance–like symptoms (borborygmus, flatulence, bloating, abdominal pain, stool frequency and stool consistency changes), altered gut microbiota (decreased total stool mass, decreased gastrointestinal motility, decreased bacterial mass, increased inflammatory response), and increased mucin production (Kay et al., 2021). In addition to gastrointestinal effects, BCM-7 has also been shown to exert oxidative stress and has been associated with adverse effects in the cardiovascular, neurological, and endocrine systems (Kay et al., 2021).

A preclinical study in a mouse model, in which mice were fed with a basal diet and β -casein variants (A1A1, A2A2 and A1A2) at a dose of 85 mg/animal/day suspended in 200 μ L phosphate-buffered saline (PBS), suggested that the A1 variant can trigger elevated inflammation when compared to the A2 variant, mediated by the Th₂-pathway in the gut, that was measured by increased inflammatory markers, gut permeability, and IL-4, suggesting the differentiation of naïve T cells to Th₂ cells (UI Haq et al., 2014). This study showed that feeding of A1A1 and A1A2 β -casein increased myeloperoxidase (MPO) activity (MPO is a marker for inflammation, which is released from stimulated polymorphonuclear neutrophils (PMNs) that produce the powerful oxidant hypochlorous acid) in the murine intestine by 204.20% and 43.54%, respectively, compared to control group mice. IL-4 levels on feeding A1A1 and A1A2 β -casein variants increased by 272.13% and 282.29 %, respectively, compared to A2A2 β -casein consumption. No changes were observed in feeding A2A2 β -casein variant compared to the control group mice. The increase in IL-4 levels on feeding A1 variants of β -casein in this study suggests a mechanism that may lead to the differentiation of naïve helper T cells to Th₂ cells (UI Haq et al., 2014).

Since the sequence of β -casein produced by *E. coli* stF_24_139 is based on the A2 variant, and therefore the adverse effects associated with BCM-7, including increased inflammation and lactose intolerance–like symptoms, are not expected to occur.

6.4. Safety of the parental strain

The subject of this GRAS notification is recombinant β -casein, which is produced by the bioengineered strain *E. coli* stF_24_139, derived from *E. coli* BL21(DE3).

Escherichia coli, which naturally occurs in the intestinal microflora of humans and many animals, are part of the *Enterobacteriaceae* family known for their Gram-negative, non-spore forming, and facultatively anaerobic nature. Taxonomically, these *E. coli* strains fall into five ECOR phylogroups (A, B1, B2, D, and E) based on housekeeping gene sequence similarities (Archer et al., 2011). Human commensal strains are predominantly found in phylogroups A and B1, while groups B2, D, and E contain non-related pathogenic strains. Among the numerous *E. coli* isolates, laboratory strains such as K-12, B, and C, along with their derivatives, are identified as Risk Group 1 organisms according to biological safety guidelines (Daegelen et al., 2009, Archer et al., 2011). The National Institute for Health (NIH) guidelines classify Risk Group 1 organisms as “*Agents not associated with disease in healthy adult humans*” (CDC/NIH, 2009). Among these strains, *E. coli* K-12 and its B derivatives (such as REL606 and BL21) are some of the most utilized for manufacturing biotechnology products, including those for industrial, pharmaceutical, and food biotechnology applications.

The safety of the parental strain *E. coli* stF_24_139 has been evaluated in several GRAS notices, including GRN 1015 (page 20), GRN 921 (page 24-25), GRN 571 (Appendix K, page 282-300), the GRN 571 Supplement, and GRN 485 (page 15-18), all of which received “no questions” letters from the FDA. In these GRAS notices, several studies are referenced to confirm the safety of the parental strain *E. coli* BL21(DE3). *E. coli* BL21(DE3) lacks the pathogenic elements commonly found in *E. coli* strains responsible for most gastrointestinal infections. Consequently, *E. coli* BL21(DE3) is regarded as non-pathogenic, with a low likelihood of surviving in host tissues or causing disease (Chart et al., 2000). This strain was among the first to have its entire genome sequenced, showing only minor differences from the widely used production strain *E. coli* K-12 (Studier et al., 2009). The sequencing revealed that it lacks genes for virulence factors such as invasion proteins, adhesion molecules, and enterotoxins (Jeong et al., 2009). Additionally, an acute oral toxicity test found that the endotoxin from *E. coli*

BL21(DE3) did not produce toxicity in mice, even at the maximum dose of 1,000,000 EU (3.3 mg/kg body weight) (Harper et al., 2011).

6.5. Safety of the production strain

Overall, the safety of the production strain can be based on the safety of the parental strain. During the creation of the production strain, no genes of concern were introduced (Section 2.2.2), since all genes included in the genome of the parental strain express naturally occurring proteins or have been minimally modified. No additional antimicrobial resistance genes are present in the production strain *E. coli* stF_24_139. Additionally, no viable cells of the production strain are present in the final product.

6.6. Safety of β -casein produced by *E. coli* stF_24_139

The batch analysis of β -casein produced by *E. coli* stF_24_139 (Section 2.5) shows that the product is sufficiently characterized and meets the set specifications (Section 2.4). β -casein produced by *E. coli* stF_24_139 is demonstrated to be absent from chemical (e.g., heavy metals) and microbiological impurities that would raise any safety concerns. β -casein produced by *E. coli* stF_24_139 has a protein content of at least 2%, and out of the protein fraction, min. 75% of the total protein is Fermify's recombinant β -casein and max. 25% are proteins originating from the host strain. The safety of β -casein is discussed in Section 6.3, and the safety of the host strain proteins is discussed below. Considering the production process and nature of the product, three potential impurities IPTG, dimethylamine-epichlorohydrin copolymer, and lipopolysaccharides (LPS; endotoxins) are discussed below. Considering the manufacturing process and the measures taken to control these impurities, and the exposure to the ingredient, Fermify does not expect these to be a safety concern when β -casein produced by *E. coli* stF_24_139 is consumed through its proposed uses.

Residual host strain protein

Disela et al. (2023) characterized the host cell proteome of two *E. coli* strains, *E. coli* HMS174 and *E. coli* BLR(DE3) using mass-spectrometry-based proteomics. *E. coli* BLR(DE3) is a *recA*-deficient-derivative of *E. coli* BL21(DE3) (Goffin and Dehottay, 2017), and therefore also closely related to the production organism of Fermify's β -casein, *E. coli* stF_24_139. Considering the close taxonomic relationship between *E. coli* BLR(DE3) and *E. coli* stF_24_139, the proteome of *E. coli* BLR(DE3) is considered to be representative of the proteome of *E. coli* stF_24_139. **Table 11** provides an overview of the 20 most abundant host cell proteins identified in *E. coli* BLR(DE3). The most abundant protein appeared to be adenosine triphosphate (ATP) synthase F1 subunit epsilon. This protein generates ATP from adenosine diphosphate (ADP) in the presence of a proton gradient across the membranes and accounted for 73.03 molar % (mol %) of the total host protein fraction, making it the most abundant protein by a high margin. This protein was also found to be the most abundant protein in *E. coli* HMS174. ATP synthase F1 subunit epsilon, and the 19 other most abundant proteins that were identified in *E. coli* BLR(DE3) are considered housekeeping proteins, which are responsible for the maintenance of basic cellular functions required for survival. None of these proteins are considered to contribute to virulence, pathogenicity and/or toxigenicity. Moreover, none of the major food allergens were identified.

Table 11. Overview of the 20 most abundant host cell proteins in <i>E. coli</i> BLR(DE3) ¹			
Protein name	Avg. mass (Da)	Protein content (mol %)	Accession closest <i>E. coli</i> K12 analog
F1 sector of membrane-bound ATP synthase epsilon subunit	15,068	73.03	P0A6E6
Phosphohistidinoprotein-hexose phosphotransferase component of PTS system (Hpr)	9,119	7.30	P0AA04
50S ribosomal subunit protein L22	12,226	1.30	P61175
50S ribosomal subunit protein L33	6,372	0.73	P0A7N9
CopC family protein	23,410	0.73	P0AA57
Putative uncharacterized protein YciS	11,351	0.73	P0ACV4
50S ribosomal subunit protein L7/L12	12,295	0.41	P0A7K2
50S ribosomal subunit protein L24	11,316	0.41	P60624
Glutamate/aspartate ABC transporter substrate-binding protein	33,375	0.39	P37902
Glycine betaine ABC transporter substrate-binding protein	36,046	0.27	P0AFM2
50S ribosomal subunit protein L13	16,019	0.23	P0AA10
Fructose-bisphosphate aldolase class II	39,147	0.23	P0AB71
Methylglyoxal synthase	16,919	0.23	P0A731
30S ribosomal subunit protein	1,173	0.18	P02358
30S ribosomal protein S1	61,158	0.16	P0AG67
Malate dehydrogenase NAD(P)-binding	32,337	0.16	P61889
30S ribosomal subunit protein S15	10,269	0.16	P0ADZ4
Galactitol-specific enzyme IIB component of PTS	10,270	0.16	P37188
Aspartate-semialdehyde dehydrogenase NAD(P)-binding	40,034	0.16	P0A9Q9
50S ribosomal subunit protein L18	12,770	0.13	P0C018

¹The information presented in this table is a simplified version of the information on the 20 most abundant proteins in *E. coli* BLR(DE3) presented in Disela et al. (2023).

IPTG

IPTG is added during the fermentation process of the production strain as an inducer of β -casein production by the production strain. The FDA previously reviewed the safety of IPTG as a part of the manufacturing process of phospholipase C from *Pseudomonas fluorescens* (GRN 574, page 15), a lipase preparation from *Pseudomonas fluorescens* Biovar I (GRN 462, page 8), alpha amylase preparation from *Pseudomonas fluorescens* Biovar I (GRN 126).

Fermify is adding IPTG at the fermentation stage at a concentration of 0.2383 g/L. By applying a dilution factor of 571,552, which is expected by the downstream processing of the ingredient, a theoretical maximum concentration of 0.417 μ g/L or 0.417 ppb IPTG on a wet basis may be present in β -casein produced by *E. coli* stF_24_139. This is considered the “worst-case” estimate, as IPTG is likely to be removed during the downstream processing and the actual residue level in the final product is likely to be below 0.417 ppb on a wet basis.

Dimethylamine-epichlorohydrin copolymer

Dimethylamine-epichlorohydrin copolymer is added in the downstream processing of the manufacturing process of the ingredient as a flocculant. The substance is a polycation and its primary role in the production process of β -casein produced by *E. coli* stF_24_139 is to remove undesirable residues from the ingredient. Dimethylamine-epichlorohydrin is a common processing aid used in the enzyme industry, and is listed in the FDA September 11, 2003, letter to ETA as acceptable for use in enzyme manufacturing (Appendix I). Fermify is adding 150 ppm of the substance at two separate stages of the downstream processing.

The French Agency for Food, Environmental and Occupational Health & Safety (ANSES) reported a no-observed adverse effect level (NOAEL) of the substance of 1,000 mg/kg body weight/day according to the OECD 408 guideline, reported in a safety evaluation of the substance in the context of the use of the substance as a processing aid (flocculant) in sugar confectionery and sugar refinery (ANSES, 2021). If a safety factor for intra- and interspecies variation is applied, an upper limit of dimethylamine-

epichlorohydrin copolymer of 10 mg/kg body weight/day can be considered safe for humans. Taking the 90th percentile exposure of β -casein produced by *E. coli* stF_24_139 (worst-case scenario) among the general US population of 2 years and older into account, i.e., 567 mg/kg body weight/day, the maximum concentration of dimethylamine-epichlorohydrin copolymer that would be considered safe for consumption, accounts for 1.76% of the product on dry matter basis, which equals to 529 ppm on wet basis, considering that the product contains 97% of moisture. Since dimethylamine-epichlorohydrin copolymer is added at levels of 150 ppm at two separate stages of the downstream processing, the remaining level in the final product is expected to be much lower than 529 ppm on a wet basis and is likely to be removed during the downstream processing.

Endotoxins

The batch variation data of β -casein produced by *E. coli* stF_24_139 have shown the presence of bacterial endotoxins at a level of ranging from 6,400 to 14,000 EU/mL on wet basis. The specification for endotoxins in β -casein produced by *E. coli* stF_24_139 was set at 20,000 EU/mL, which would represent the worst-case scenario as highest tolerated level of endotoxins in the product. This value translates to 660,000 EU/g on dry matter basis, considering a moisture level of 97%. Considering this value, the mean and 90th percentile exposure to endotoxins through consumption of β -casein produced by *E. coli* stF_24_139 accounts for 8.83×10^6 EU/day (or 162,000 EU/kg bw/day) and 1.84×10^7 EU/day (or 374,000 EU/kg bw/day), respectively.

Bacterial endotoxins, found in the outer membrane of Gram-negative bacteria are members of a class of phospholipids called lipopolysaccharides (LPS). LPS consists of long chains of sugar moieties covalently linked to lipids. These chains create a dense network that forms a gelatinous layer on the bacterial surface which keep the bacterial surface moist and slightly negatively charged to protect against harmful compounds and allowing nutrients uptake (Wassenaar and Zimmermann, 2018). The actual physical barrier separating the inside of a bacterial cell from the external environment is the membrane, a double lipid layer with interspersed proteins, to which LPS is connected via lipid A, a phosphorylated lipid. The toxicity of LPS primarily stems from lipid A, whereas the polysaccharides are less toxic. In Gram-negative bacteria, LPS is anchored to the outer membrane through lipid A (Wassenaar and Zimmermann, 2018). In the gut, intestinal phosphatase is produced, which detoxifies lipid A by the removal of phosphate (Beumer et al., 2003). LPS is synthesized by enzymes, meaning there are no genes directly coding for LPS itself, but there are genes for the enzymes that produce LPS (Wassenaar and Zimmermann, 2018).

LPS are known to act as endotoxins when introduced into the bloodstream, eliciting strong immune responses and potentially causing fever and inflammation. However, the route of exposure is critical in determining the potential health risk posed by LPS. LPS is regularly consumed via the diet through various foods without adverse effects. For instance, LPS is present in meat, dairy, fermented foods and infant formula (**Table 12**).

If a cheese analog was to be produced containing the maximum tolerated level of 660,000 EU/g of β -casein produced by *E. coli* stF_24_139 on dry matter basis (equal to the maximum specification of 20,000 EU/mL on a wet basis of the product with a moisture content of 97%), at the maximum inclusion rate of 15 g/100 g of cheese analog (worst-case scenario), the product will contain endotoxins at a level of 99,000 EU/g. This level is in line with some products already on the market, such as ground beef and certain probiotic products (**Table 12**).

Table 12. Endotoxin levels of known foodstuffs obtained from the literature (LAL assay)		
Food stuff	LAL test results	Reference
Ground beef	500 – 75,000 EU/g	Jay et al. (1979)
Powdered instant formula milk	40 – 55,000 EU/g; median 3000 EU/g	Townsend et al. (2007)
Milk	1 – 1,000,000 EU/mL; median 100 EU/mL	Gehring et al. (2008)
Whole raw milk (fresh)	3 – 6,144 EU/mL	Sipka et al. (2015)
Processed shop milk (fresh)	60 – 240 EU/mL	Sipka et al. (2015)
Probiotic products ¹	ND – 400,000 EU/mL	Wassenaar and Zimmermann (2018)
Commercial infant cereals	1,400 – 24,200 EU/g	Oscarsson et al. (2020)
¹ Probiotic products containing a suspension of viable Gram-positive bacteria, and viable and lysed <i>E. coli</i> cells LAL; Limulus Amebocyte Lysate		

A further source of orally ingested LPS is the saliva. Human saliva contains approximately 1 mg of endotoxins per mL (Leenstra et al., 1996), which is equivalent to 1×10^6 EU. Saliva is produced at levels exceeding 500 mL/day (Iorgulescu, 2009), which amounts to 5×10^8 EU/day. The 90th percentile intake of endotoxins through consumption of β -casein produced by *E. coli* stF_24_139 containing 660,000 EU/g on dry matter basis (i.e., 2.02×10^7 EU/day) would account for an intake of 4.04% of the overall daily intake of endotoxins through saliva.

6.7. Allergenicity of β -casein produced by *E. coli* stF_24_139

Casein is a natural component of milk which is considered a major food allergen subject to labeling under the Food Allergen Labeling and Consumer Protection Act (FALCPA) as regulated by the FDA. As explained above, Fermify's recombinant β -casein is substantially equivalent to the β -casein found in bovine milk and isolated milk proteins. Therefore, the notified substance may produce a milk protein allergic response when consumed. All products containing the notified substance will indicate that the product contains an allergen (e.g., a protein also found in milk) to inform those consumers who are allergic to milk and comply with food allergen labeling requirements.

To predict potential allergy or allergenic cross-reactivity, the CODEX standard of more than 35% identity in the amino acid sequence over 80 amino acid window was used (FAO, 2001). In the assessment below, the amino acid sequence of Fermify's β -casein and the 20 most abundant residual host proteins identified in *E. coli* BLR(DE3) (amino acid sequences derived from UniProt) that are predicted to be present in the final product were screened for sequence homology to known allergens.

80mer Sliding Window Search Results

Database	AllergenOnline Database v22 (May 25, 2023)
Input Query	>query MRELEELNVPGEIVESLSSEESITRINKKIEKFQSEEQQQTEDELQDKIHHPAQTQSLV YFPFGPIPNLQNPPLTQTPVVVPPFLQPEVMGVSKVKEAMAPKHKEMPPFKYPVEPF TESQSLTLTDVENLHLPLQLQSWMHQPHQLPPTVMFPQSVLSLSQSKVLPVPQKAVP YPQDMPIQAFLLYQEPVLGPFVRGPFPIIV
Length	210
Number of 80 mers	131
Number of Sequences with hits	8

Hit #	Define	Species	Best % ID	# Hits > 35%	Full Alignment			Links	
					E-val	% ID	length	NCBI	Details
1	gil162805 gid1374 beta-casein	Bos taurus	100.00%	131 of 131	1.7e-074	99.50%	209	gil162805	GO!
2	gil459292 gid1374 beta-casein A3 [Bos taurus]	Bos taurus	100.00%	131 of 131	1.2e-074	99.50%	209	gil459292	GO!
3	gil942073448 gid1374 PREDICTED: beta-casein isoform	Bos grunniens mutus	100.00%	131 of 131	4.2e-075	100.00%	209	gil942073448	GO!
4	gil162797 gid1374 beta-casein precursor	Bos taurus	100.00%	131 of 131	2e-073	98.10%	209	gil162797	GO!
5	gil886967 gid1154 low molecular weight glutenin [T	Triticum aestivum	36.60%	5 of 131	0.026	28.40%	194	gil886967	GO!
6	gil170702 gid1152 gamma gliadin precursor	Triticum aestivum	36.27%	4 of 131	5.7e-005	29.60%	159	gil170702	GO!
7	gil208605344 gid1150 ID-type LMW glutenin subunit [	Triticum aestivum	35.40%	4 of 131	4.3e-007	27.40%	179	gil208605344	GO!
8	gil508732623 gid1150 omega-gliadin, partial [Triti	Triticum aestivum	35.40%	1 of 131	9.2e-009	29.20%	185	gil508732623	GO!

Figure 5. β -casein produced by *E. coli* stF_24_139 sliding 80-mer window comparison to AllergenOnline.org. Three bovine β -casein protein alignments were identified.

As indicated in **Figure 5**, the sequence of Fermify's recombinant β -casein was a 100% match to bovine milk protein allergens. The sequence also showed a > 35% match in the amino acid sequence over an 80 amino acid window to wheat (*Triticum aestivum*) allergens. Although 35% identity in the amino acid sequence over 80 amino acid window is a standard criterion, it is also considered to be conservative. This is because sequence identity < 50% is thought to rarely lead to cross-reactivity with known allergens, with 70% identity over most of the sequence considered to be a more realistic threshold (Aalberse, 2000, Ladics et al., 2011).

Only one of the 20 most abundant residual host proteins identified in *E. coli* BLR(DE3), malate dehydrogenase (UniProt accession: P61889), showed a 67.55% match with an allergen of *Malassezia furfur*, which is a known skin allergen, and therefore it is not considered of concern as a food allergen. The protein is also predicted to be present in very low concentrations, as it is present at a level of 0.15 mol % of the total protein fraction of *E. coli* BLR(DE3). Based on this assessment, it is concluded that the predicted residual host strain proteins present in β -casein produced by *E. coli* stF_24_139 do not pose a risk of food allergy.

In the fermentation medium, soy peptone and yeast extract are used as carbon sources. FALCPA labels soybean as one of the eight major food allergens. Considering that during the downstream processing, a dilution factor of 571,552 is applied, the presence of traces of soybean in the product are not considered of concern.

Based on the fact the allergenicity assessment, it is concluded that β -casein produced by *E. coli* stF_24_139 is considered to have the same allergenic profile as other products such as milk and milk protein products containing native bovine β -casein.

6.8. Summary and conclusion of the Basis for GRAS Determination

Fermify has determined that β -casein produced by fermentation of *E. coli* stF_24_139 is GRAS for the intended use in food based on the following:

- The intended uses and the estimated consumption of β -casein.
- Fermify's recombinant β -casein is substantially equivalent to native bovine β -casein. The safety of Fermify's recombinant β -casein is based on the safety of native bovine β -casein and its long history of safe use in milk and milk protein as food.
- β -casein produced by *E. coli* stF_24_139 is manufactured under cGMP for food (21 CFR Part 117)
- Potential contaminants, such as heavy metals and pathogenic microorganisms, are either not detected or below toxicological and regulatory limits.
- β -casein produced by *E. coli* stF_24_139 is considered to have the same allergenic profile as other products such as milk and milk protein products containing native bovine β -casein

6.9. Data and information that is inconsistent with the GRAS conclusion

We have reviewed the available data and information and are not aware of any data and information that are, or may appear to be, inconsistent with our conclusion of GRAS status.

PART 7. §170.255 LIST OF SUPPORTING DATA

7.1. List of references

- AALBERSE, R. C. 2000. Structural biology of allergens. *J Allergy Clin Immunol*, 106, 228-38.
- ANSES 2021. AVIS de l'Agence nationale de sécurité sanitaire de l'alimentation, de l'environnement et du travail relatif aux demandes d'autorisation d'emploi du 1,2-Ethdiamine, polymère with 2-(chlorométhyl)oxirane and N-méthylmethanamine [polycondensat d'épichlorhydrine et de méthylamine], en tant qu'auxiliaire technologique, comme flocculant en sucrerie et en raffinerie de sucre. *Saisines n°2019-SA-0130 et 2019-SA-0131*.
- ARCHER, C. T., KIM, J. F., JEONG, H., PARK, J. H., VICKERS, C. E., LEE, S. Y. & NIELSEN, L. K. 2011. The genome sequence of *E. coli* W (ATCC 9637): comparative genome analysis and an improved genome-scale reconstruction of *E. coli*. *BMC Genomics*, 12, 9.
- BEUMER, C., WULFERINK, M., RAABEN, W., FIECHTER, D., BRANDS, R. & SEINEN, W. 2003. Calf intestinal alkaline phosphatase, a novel therapeutic drug for lipopolysaccharide (LPS)-mediated diseases, attenuates LPS toxicity in mice and piglets. *J Pharmacol Exp Ther*, 307, 737-44.
- BHUTIA, Y. & GANAPATHY, V. 2018. Protein Digestion and Absorption.
- CDC/NIH 2009. *Biosafety in Microbiological and Biomedical Laboratories*, 5th Edition.
- CHANDEL, N. S. 2021. Amino Acid Metabolism. *Cold Spring Harb Perspect Biol*, 13.
- CHART, H., SMITH, H. R., LA RAGIONE, R. M. & WOODWARD, M. J. 2000. An investigation into the pathogenic properties of *Escherichia coli* strains BLR, BL21, DH5alpha and EQ1. *J Appl Microbiol*, 89, 1048-58.
- DAEGELEN, P., STUDIER, F. W., LENSKI, R. E., CURE, S. & KIM, J. F. 2009. Tracing ancestors and relatives of *Escherichia coli* B, and the derivation of B strains REL606 and BL21(DE3). *J Mol Biol*, 394, 634-43.
- DISELA, R., BUSSY, O. L., GELDHOF, G., PABST, M. & OTTENS, M. 2023. Characterisation of the *E. coli* HMS174 and BLR host cell proteome to guide purification process development. *Biotechnol J*, 18, e2300068.
- EFSA 2015. Scientific Opinion on Dietary Reference Values for magnesium. *EFSA Journal*, 13, 4186.
- FAO 2001. Evaluation of Allergenicity of Genetically Modified Foods. *Report of a Joint FAO/WHO Expert Consultation on Allergenicity of Foods Derived from Biotechnology*, 22-25 (2001).
- FARRELL, H. M., JIMENEZ-FLORES, R., BLECK, G. T., BROWN, E. M., BUTLER, J. E., CREAMER, L. K., HICKS, C. L., HOLLAR, C. M., NG-KWAI-HANG, K. F. & SWAISGOOD, H. E. 2004. Nomenclature of the Proteins of Cows' Milk—Sixth Revision. *Journal of Dairy Science*, 87, 1641-1674.
- FDA 2006. Guidance for Industry: Estimating Dietary Intake of Substances in Food.
- FOX, P. F. & MULVIHILL, D. M. 1990. Casein. In: HARRIS, P. (ed.) *Food Gels*. Dordrecht: Springer Netherlands.
- GEHRING, U., SPITHOVEN, J., SCHMID, S., BITTER, S., BRAUN, C., DALPHIN, J. C., A., PEKKANEN, J., RIEDLER, J., WEILAND, S. K., G., VON MUTIUS, E., VUITTON, D. A. & BRUNEKREEF, B. 2008. Endotoxin levels in cow's milk samples from farming and non-farming families - the PASTURE study. *Environ Int*, 34, 1132-6.
- GIL, A. G., IRACHE, J. M., I., NAVARRO, C. J. DE CERAIN, A. 2017. Toxicity and biodistribution of orally administered casein nanoparticles. *Food and Chemical Toxicology*, 106, 477-486.
- GOFFIN, P. & DEHOTTAY, P. 2017. Complete Genome Sequence of *Escherichia coli* BLR(DE3), a recA-Deficient Derivative of *E. coli* BL21(DE3). *Genome Announc*, 5.
- HARPER, M. S., CARPENTER, C., KLOCKE, D. J., CARLSON, G., DAVIS, T. & DELANEY, B. 2011. *E. coli* Lipopolysaccharide: acute oral toxicity study in mice. *Food Chem Toxicol*, 49, 1770-2.
- HOCHKOEPLER, A. 2013. Expanding the landscape of recombinant protein production in *Escherichia coli*. *Biotechnol Lett*, 35, 1971-81.
- IORGULESCU, G. 2009. Saliva between normal and pathological. Important factors in determining systemic and oral health. *J Med Life*, 2, 303-7.
- JAY, J. M., MARGITIC, S., SHEREDA, A. L. & COVINGTON, H. V. 1979. Determining endotoxin content of ground beef by the *Limulus* amoebocyte lysate test as a rapid indicator of microbial quality. *Appl Environ Microbiol*, 38, 885-90.
- JEONG, H., BARBE, V., LEE, C. H., VALLENET, D., YU, D. S., CHOI, S. H., COULOUX, A., LEE, S. W., YOON, S. H., CATTOLICO, L., HUR, C. G., PARK, H. S., SÉGURENS, B., KIM, S. C., OH, T. K., LENSKI, R. E., STUDIER, F. W., DAEGELEN, P. & KIM, J. F. 2009. Genome sequences of *Escherichia coli* B strains REL606 and BL21(DE3). *J Mol Biol*, 394, 644-52.

- JIANQIN, S., LEIMING, X., LU, X., YELLAND, G. W., NI, J. & CLARKE, A. J. 2016. Effects of milk containing only A2 beta casein versus milk containing both A1 and A2 beta casein proteins on gastrointestinal physiology, symptoms of discomfort, and cognitive behavior of people with self-reported intolerance to traditional cows' milk. *Nutrition Journal*, 15, 35.
- KAY, S. S., DELGADO, S., MITTAL, J., ESHRAGHI, R. S., MITTAL, R. & ESHRAGHI, A. A. 2021. Beneficial Effects of Milk Having A2 β -Casein Protein: Myth or Reality? *J Nutr*, 151, 1061-1072.
- LADICS, G. S., CRESSMAN, R. F., HEROUET-GUICHENEY, C., HERMAN, R. A., PRIVALLE, L., SONG, P., WARD, J. M. & MCCLAIN, S. 2011. Bioinformatics and the allergy assessment of agricultural biotechnology products: industry practices and recommendations. *Regul Toxicol Pharmacol*, 60, 46-53.
- LEENSTRA, T. S., VAN SAENE, J. J., VAN SAENE, H. K. & MARTIN, M. V. 1996. Oral endotoxin in healthy adults. *Oral Surg Oral Med Oral Pathol Oral Radiol Endod*, 82, 637-43.
- OSCARSSON, E., HÅRD AF SEGERSTAD, E. M., LARSSON, C., ÖSTBRING, K., AGARDH, D. & HÅKANSSON, Å. 2020. Commercial infant cereals contain high concentrations of endotoxins and viable *Bacillus* spp. *Food Control*, 118, 107367.
- PARIZA, M. W. & FOSTER, E. M. 1983. Determining the Safety of Enzymes Used in Food Processing. *J Food Prot*, 46, 453-468.
- PARIZA, M. W. & JOHNSON, E. A. 2001. Evaluating the safety of microbial enzyme preparations used in food processing: update for a new century. *Regul Toxicol Pharmacol*, 33, 173-86.
- PEUBEZ, I., CHAUDET, N., MIGNON, C., HILD, G., HUSSON, S., COURTOIS, V., DE LUCA, K., SPECK, D. & SODOYER, R. 2010. Antibiotic-free selection in *E. coli*: new considerations for optimal design and improved production. *Microbial Cell Factories*, 9, 65.
- POULSEN, N. A., GLANTZ, M., ROSENGAARD, A. K., PAULSSON, M. & LARSEN, L. B. 2017. Comparison of milk protein composition and rennet coagulation properties in native Swedish dairy cow breeds and high-yielding Swedish Red cows. *J Dairy Sci*, 100, 8722-8734.
- SIPKA, S., BÉRES, A., BERTÓK, L., VARGA, T. & BRUCKNER, G. 2015. Comparison of endotoxin levels in cow's milk samples derived from farms and shops. *Innate Immun*, 21, 531-6.
- STUDIER, F. W., DAEGELEN, P., LENSKI, R. E., MASLOV, S. & KIM, J. F. 2009. Understanding the differences between genome sequences of *Escherichia coli* B strains REL606 and BL21(DE3) and comparison of the *E. coli* B and K-12 genomes. *J Mol Biol*, 394, 653-80.
- TAM, J. E. & KLINE, B. C. 1989. Control of the *ccd* operon in plasmid F. *Journal of Bacteriology*, 171, 2353-2360.
- TOWNSEND, S., CAUBILLA BARRON, J., LOC-CARRILLO, C. & FORSYTHE, S. 2007. The presence of endotoxin in powdered infant formula milk and the influence of endotoxin and *Enterobacter sakazakii* on bacterial translocation in the infant rat. *Food Microbiol*, 24, 67-74.
- UL HAQ, M. R., KAPILA, R., SHARMA, R., SALIGANTI, V. & KAPILA, S. 2014. Comparative evaluation of cow β -casein variants (A1/A2) consumption on Th2-mediated inflammatory response in mouse gut. *Eur J Nutr*, 53, 1039-49.
- WASSENAAR, T. M. & ZIMMERMANN, K. 2018. Lipopolysaccharides in Food, Food Supplements, and Probiotics: Should We be Worried? *Eur J Microbiol Immunol (Bp)*, 8, 63-69.
- WHO 2004. Copper in Drinking-water. *Geneva, World Health Organization (WHO)*, 2004.
- WHO 2007. Protein and amino acid requirements in human nutrition. *World Health Organ Tech Rep Ser*, 1-265, back cover.
- WHO 2009. Boron in Drinking-water. *Geneva, World Health Organization (WHO)*, 2009.
- WHO 2012. Guideline: Sodium intake for adults and children. *Geneva, World Health Organization (WHO)*, 2012.
- YALÇIN, A. S. 2006. Emerging therapeutic potential of whey proteins and peptides. *Curr Pharm Des*, 12, 1637-43.

Appendix I – FDA September 11, 2003, letter to ETA as acceptable for use in enzyme manufacturing



September 11, 2003

Mr. Gary Yingling
Kirkpatrick & Lockhart LLP
1800 Massachusetts Avenue, NW
Second Floor
Washington, DC 20036-1221

Dear Mr. Yingling:

You requested, on behalf of the Enzyme Technical Association, that OFAS review the use of certain defoaming and flocculating agents in the manufacture of enzyme preparations used in food. You provided information related to these compounds in your letters of December 20, 1996 (to Dr. Alan Rulis), 4-24-1998 (to Dr. Zofia Olempska-Beer), and 11-30-99 (to Dr. Zofia Olempska-Beer). You also arranged for a teleconference between ETA members and OFAS representatives, facilitated telephone contacts with technical experts from ETA member companies, and responded to numerous requests for clarification. We appreciate your and ETA's cooperation.

We reviewed the information on defoaming and flocculating agents that you submitted as well as the information provided in GRAS affirmation petitions and GRAS notices for enzyme preparations. The enclosed attachment provides a brief overview of our evaluation and itemizes the evaluated defoamers (Table 1) and flocculants (Table 2). We conclude that these compounds are used by enzyme manufacturers in accordance with the principles of good manufacturing practice (GMP).

Sincerely yours,


Laura M. Tarantino, Ph.D.
Acting Director
Office of Food Additive Safety, HFS-200
Center for Food Safety and Applied Nutrition

Defoaming and Flocculating Agents Used in the Manufacture of Enzyme Preparations Used in Food

Enzyme Preparations

Most enzymes currently used in food are derived from microorganisms. The manufacturing process of such enzymes includes three major steps: fermentation, enzyme recovery, and enzyme formulation. The formulated products are generally referred to as enzyme preparations. In addition to the enzymes of interest, enzyme preparations contain added substances such as diluents, preservatives, and stabilizers. They may also contain metabolites derived from the production microorganism and the residues of substances used in the manufacturing process, such as components of the fermentation medium or defoaming and flocculating agents used during fermentation and recovery. When FDA reviews safety data on enzyme preparations, it considers all components of the preparation.

Defoaming Agents

Defoaming agents (defoamers) are used by enzyme manufacturers to reduce or prevent foaming during fermentation and recovery. They are formulated with ancillary ingredients such as surface-active agents or carriers. Defoamers currently used in the manufacture of food enzymes are listed in Table 1. The Table includes five major defoamers that are identified by a double asterisk and several compounds that are used either as secondary defoamers or ancillary ingredients in defoamer formulations.

The major defoamers are added to the fermentation broth at levels within the range of 0.05-1% on a weight basis. Some of these defoamers, for example, polyoxyethylene-polyoxypropylene block copolymer, may contain trace levels of ethylene oxide, propylene oxide, and 1,4-dioxane which are known to cause cancer in laboratory animals. The Office of Food Additive Safety (OFAS) has evaluated the use of defoamers listed in Table 1 and determined that human exposure to the residues of these defoamers in enzyme preparations does not present human safety concern.

Flocculating Agents

Flocculating agents (flocculants) are used in the enzyme recovery step to separate microbial cells and cell debris from the fermentation broth containing the dissolved enzyme. The flocculation typically consists of two steps - primary flocculation and secondary flocculation. In the primary flocculation, inorganic salts (such as calcium chloride or aluminum sulfate) or "low molecular weight" polymers (such as polyamines) are used to agglomerate the cellular debris. The primary flocculation is usually followed by the secondary flocculation in which "high molecular weight" polymers are used to aid the formation of larger agglomerates that are subsequently removed by centrifugation or filtration. The polymers used as flocculants can be either cationic or anionic. The cationic polymers are added to the fermentation broth at levels not higher than 1% on a

weight basis. The anionic polymers are used at levels at or below 0.025%.

The flocculants used in the manufacture of food enzymes are listed in Table 2. They include inorganic salts, polyamines, and polyacrylamides. Several of these compounds are regulated in 21 CFR either as food additives or GRAS substances. Certain polyamines may contain traces of epichlorohydrin and 1,3-dichloro-2-propanol. Polyacrylamides usually contain very low levels of acrylamide. These contaminants of polyamines and polyacrylamides are known to cause cancer in laboratory animals. OFAS has evaluated all polymers included in Table 2 and determined that human exposure to the residues of these flocculants in enzyme preparations does not present human safety concern.

Sources of Information on Defoamers and Flocculants

OFAS compiled data on defoamers and flocculants listed in Tables 1 and 2 using information voluntarily submitted by the Enzyme Technical Association. OFAS also relied on the information provided in GRAS affirmation petitions and GRAS notices for enzyme preparations. Other sources of information included published articles, computer searches, and Material Safety Data Sheets issued by manufacturers of defoamers and flocculants.

Table 1. Defoamers Used in the Manufacture of Food Enzymes

Compound	CAS Reg. No.	Supplemental Information
Polypropylene glycol**	25322-69-4	Average MW: 2000
Polyglycerol polyethylene-polypropylene glycol ether oleate**	78041-14-2	
Polyoxyethylene-polyoxypropylene block copolymer**	9003-11-6	Average MW: 2000
Polypropylene glycol monobutyl ether**	9003-13-8	
Polydimethylsiloxane**	63148-62-9 68083-18-1	
Silica	7631-86-9 63231-67-4	
Stearic acid	57-11-4	
Sorbitan sesquioleate	8007-43-0	
Glycerol monostearate	123-94-4	
Polysorbates (polyoxyethylene sorbitan fatty acid esters)		Polysorbate 60 (CAS No. 9005-67-8), Polysorbate 65 (CAS No. 9005-71-4), and polysorbate 80 (CAS No. 9005-65-6) are regulated as food additives and components of defoamer formulations
Rape oil mono- and diglycerides	93763-31-6	
White mineral oil	64742-47-8	

Table 2. Flocculants Used in the Manufacture of Food Enzymes

Compound	CAS Reg. No.	Supplemental Information
Dimethylamine-epichlorohydrin copolymer	25988-97-0	Cationic polyamine
Methylamine-epichlorohydrin copolymer	31568-35-1	Cationic polyamine
Dimethylamine-epichlorohydrin-ethylenediamine terpolymer	42751-79-1	Cationic polyamine
Polyacrylamide modified by condensation with formaldehyde and dimethylamine	67953-80-4	Cationic polyacrylamide
Acrylamide-acryloxyethyl-trimethyl-ammonium chloride copolymer	69418-26-4	Cationic polyacrylamide
Acrylamide-acrylic acid copolymer	25987-30-8 9003-06-9	Anionic polyacrylamide
Aluminum sulfate	10043-01-3	
Calcium chloride	10035-04-8 10043-52-4	

Appendix II – Certificates of analysis (CoAs) of the presented batches

Fermify GmbH
[REDACTED]
Richard-Neutra-Gasse 5
1210 Wien
ÖSTERREICH

Analytical Service Manager: [REDACTED]

Issue date 02.10.2024

Page 1/8

Analytical report AR-24-FA-035082-01

Batch code EUATWI-00252207

Sample No 975-2024-00041679

Sample description	Fermify008 (Beta-casein solution)
Number	1
order type	D01-Analysis of selected parameters
Additional information	Probenahme erfolgte nicht durch die ELO. Gegebenenfalls entnehmen Sie weitere Details den Fotos.
Start analysis / End analysis	20.09.2024 / 27.09.2024
Reception date / Reception temperature	20.09.2024 2,1°C

Results

Sensory Analysis

FASCK simple descriptive analysis (Ak: N)

Method DIN 10964:2014, Organoleptic evaluation

sensory indication

Aussehen: farblose, ganz leicht trübe Flüssigkeit; in durchsichtiger Verpackung

Konsistenz: dünnflüssig

Geruch: arteigen, neutral

Appearance: colourless, slightly cloudy liquid; in transparent packaging

Consistency: thin fluid

Smell: characteristic, no deviation from expected smell

Physical-chemical Analysis

JJ0EW Aflatoxin B1, B2, G1, G2 (spices, special matrix) (Ak: F)

Method	DIN EN 14123 (2008-03), mod., IAC-LC-FLD		
Sub: S1			
Aflatoxin B1	< 1.0		µg/kg
Aflatoxin B2	< 1.0		µg/kg
Aflatoxin G1	< 1.0		µg/kg
Aflatoxin G2	< 1.0		µg/kg
Sum of all positive Aflatoxins	nicht berechenbar		µg/kg
JJ0BG Fumonisin (Ak: F)			
Method	Internal Method, LC-MS/MS		
Sub: S1			
Fumonisin B1 (FB1)	< 20		µg/kg
Fumonisin B2 (FB2)	< 20		µg/kg
Fumonisin sum (B1+B2)	nicht berechenbar		µg/kg
JJ62B Deoxynivalenol, Zearalenon (DON, ZON) (Ak: F)			
Method	Internal Method, LC-MS/MS		
Sub: S1			
Deoxynivalenol (Vomitoxin)	< 20		µg/kg
Zearalenone (ZON)	< 10		µg/kg
JJ0G5 Ochratoxin A (spices, special matrix) (Ak: F)			
Method	DIN EN 14132 (2009-09), mod., IAC-LC-FLD		
Sub: S1			
Ochratoxin A	< 2.0		µg/kg
JJ0FE Fusarium toxins (DON, ZON, T2, HT2) (Ak: F)			
Method	Internal Method, LC-MS/MS		
Sub: S1			
Deoxynivalenol (Vomitoxin)	< 20		µg/kg
Zearalenone (ZON)	< 10		µg/kg
T-2 Toxin	< 10		µg/kg
HT-2 Toxin	< 10		µg/kg
sum T-2 HT-2 toxin	nicht berechenbar		µg/kg
ID04R Ash (Ak: F)			
Method	Rapporti ISTISAN 1996/34 pag 77, Gravimetry		
Sub: S2			
Ash	0.60	± 0.08	g/100 g
ID0HV Total dietary fiber (High molecular weight) (Ak: F)			
Method	AOAC 991.43 2000, Enzymatic-gravimetry		
Sub: S2			
Total dietary fibre	< 1		g/100 g
GFL05 Dioxins and Furans (17 PCDD/F) (Ak: F)			
Method	Internal, GC-MS/MS		
Sub: S3			
2,3,7,8-TetraCDD	< 0.0130		pg/g
1,2,3,7,8-PentaCDD	< 0.0170		pg/g
1,2,3,4,7,8-HexaCDD	< 0.0250		pg/g
1,2,3,6,7,8-HexaCDD	< 0.0350		pg/g
1,2,3,7,8,9-HexaCDD	< 0.0330		pg/g
1,2,3,4,6,7,8-HeptaCDD	< 0.0530		pg/g
OctaCDD	< 0.387		pg/g

2,3,7,8-TetraCDF	< 0.0350		pg/g
1,2,3,7,8-PentaCDF	< 0.0240		pg/g
2,3,4,7,8-PentaCDF	< 0.0370		pg/g
1,2,3,4,7,8-HexaCDF	< 0.0390		pg/g
1,2,3,6,7,8-HexaCDF	< 0.0360		pg/g
1,2,3,7,8,9-HexaCDF	< 0.0270		pg/g
2,3,4,6,7,8-HexaCDF	< 0.0330		pg/g
1,2,3,4,6,7,8-HeptaCDF	< 0.0370		pg/g
1,2,3,4,7,8,9-HeptaCDF	< 0.0260		pg/g
OctaCDF	< 0.0800		pg/g
WHO(2005)-PCDD/F TEQ (lower-bound)	ND		pg/g
WHO(2005)-PCDD/F TEQ (medium-bound)	0.0347	± 0,00868	pg/g
WHO(2005)-PCDD/F TEQ (upper-bound)	0.0694	± 0,0174	pg/g
ID16B	Total fat (Ak: F)		
Method	Internal method, Gravimetry		
Sub: S2			
Total fat	< 0.1		g/100 g
ID04U	Carbohydrates (calculated as difference) (Ak: N)		
Method	Internal method, Calculation		
Sub: S2			
Carbohydrate calculated (difference)	< 0.5		g/100 g
GFL11	polychlorinated biphenyls (12 WHO PCB + 6 ICES PCB) (Ak: F)		
Method	Internal, GC-MS/MS		
Sub: S3			
PCB 77	< 1.20		pg/g
PCB 81	< 0.180		pg/g
PCB 105	< 2.60		pg/g
PCB 114	< 0.353		pg/g
PCB 118	< 9.33		pg/g
PCB 123	< 0.267		pg/g
PCB 126	< 0.167		pg/g
PCB 156	< 1.47		pg/g
PCB 157	< 0.273		pg/g
PCB 167	< 0.733		pg/g
J1054	Sulphur (S) (Ak: F)		
Method	Internal method, ICP-OES		
Sub: S1			
Sulphur total (S)	210	± 42	mg/kg
J1055	Selenium (Se) (Ak: F)		
Method	Internal method, ICP-OES		
Sub: S1			
Selenium (Se)	< 0.5		mg/kg
J2014	Antimony (Sb) (Ak: F)		
Method	Internal method, ICP-OES		
Sub: S1			
Antimony (Sb)	< 0.50		mg/kg
J1076	Molybdenum (Mo) (Ak: F)		
Method	Internal method, ICP-OES		

Sub: S1			
Molybdenum (Mo)	< 0.2		mg/kg
J1040	Cobalt (Co) (Ak: F)		
Method	Internal method, ICP-OES		
Sub: S1			
Cobalt (Co)	< 0.2		mg/kg
JJ0C4	Boron (B) (Ak: F)		
Method	Internal method, ICP-OES		
Sub: S1			
Boron (B)	< 0.2		mg/kg
GFTE1	TEQ-Totals WHO-PCDD/F and PCB (Ak: F)		
Method	Internal, Calculation		
Sub: S3			
WHO(2005)-PCDD/F+PCB TEQ (lower-bound)	ND		pg/g
WHO(2005)-PCDD/F+PCB TEQ (medium-bound)	0.0554	± 0,0138	pg/g
WHO(2005)-PCDD/F+PCB TEQ (upper-bound)	0.111	± 0,0277	pg/g

Microbiological Analysis

ZMLQR	Salmonella species D (Food & Feed) [AT Food] Abs Pres /25 g AFNOR 3M 01/16-11/16 (Ak: A)		
Method	AFNOR 3M 01/16-11/16, Bioluminescence		
Salmonella	Not Detected		/25 g
ZM42U	Clostridium perfringens [g] (Ak: A)		
Method	ISO 7937:2004, E-Cultural technique (non-chromogenic media)		
Clostridium perfringens	< 10		cfu/g
ZMBDT	Cronobacter species D (Food & Feed) [AT Food] Abs Pres /100 g ISO 22964 (Ak: A)		
Method	EN ISO 22964:2017, D-Cultural techn. (chrom. + non-chromogenic media)		
Cronobacter spp	Not Detected		/100 g
ZMH3G	Escherichia coli B-Glucuronidase+ E (Food & Feed) [AT Food] <10 >1 500 /g (1) PSEC-PF Internal Metho (Ak: A)		
Method	Internal Method SOP Mb 055 (24.05.2022), E-Cultural technique (media film)		
Escherichia coli B-Glucuronidase+	< 10		cfu/g
ZMH3B	Enterobacteriaceae E (Food & Feed) [AT Food] <10 >1 000 /g (1) PEB-PF Internal Method SOP Mb 052 (22 (Ak: A)		
Method	Internal Method SOP Mb 052 (24.05.2022), E-Cultural technique (media film)		
Enterobacteriaceae	< 10		cfu/g
ZMH43	Aerobic Plate Count 30°C E (Food & Feed) [AT Food] <10 >300 000 /g (1-3) PRAC-PF Internal Method SOP (Ak: A)		
Method	Internal Method SOP Mb 051 (24.05.2022), E-Cultural technique (media film)		
Aerobic Plate Count 30°C	< 10		cfu/g
ZMH48	Mould-Yeast E (Food & Feed) [AT Food] <10 >150 000 /g (1-3) PRYM-PF Internal Method SOP Mb 058 (22.1 (Ak: A)		
Method	SOP Mb 058, E-Cultural technique (media film)		
Moulds	< 10		cfu/g
Yeast	< 10		cfu/g
ZMH3C	Coagulase positive staphylococci E (Food & Feed) [AT Food] <10 >1 500 /g (1) PSTX-PF Internal Method (Ak: A)		
Method	Internal Method SOP Mb 056 (24.05.2022), E-Cultural technique (media film)		
Coagulase positive staphylococci	< 10		cfu/g
ZM4S1	Listeria monocytogenes [25 g] (Ak: A)		
Method	Internal Method SOP Mb 010 (03.02.2021), Real-time PCR		

Listeria monocytogenes	Not Detected		/25 g
ZM4JS	Presumptive Bacillus cereus [g] (Ak: A)		
Method	ISO 7932:2004, E-Cultural technique (non-chromogenic media)		
Presumptive Bacillus cereus	< 10		cfu/g
ID0A1	Nitrogen, proteins (by calculation) (Ak: F)		
Method	UNI EN ISO 14891:2002, Dumas (TCD)		
Sub: S2			
Protein (Nx6,25)	2.42	± 0.25	g/100 g
Nitrogen	0.388	± 0.040	g/100 g
ID957	Moisture (Ak: F)		
Method	Internal method, Gravimetry		
Sub: S2			
Moisture	96.87	± 3.06	g/100 g
Dry matter	3.13	± 3.06	g/100 g

Physical-chemical Analysis

GFL11	polychlorinated biphenyls (12 WHO PCB + 6 ICES PCB) (Ak: F)		
Method	Internal, GC-MS/MS		
Sub: S3			
PCB 169	< 0.800		pg/g
PCB 189	< 0.267		pg/g
WHO(2005)-PCB TEQ (lower-bound)	ND		pg/g
WHO(2005)-PCB TEQ (medium-bound)	0.0207	± 0,00517	pg/g
WHO(2005)-PCB TEQ (upper-bound)	0.0413	± 0,0103	pg/g
PCB 28	< 0.0670		ng/g
PCB 52	< 0.0670		ng/g
PCB 101	< 0.0670		ng/g
PCB 138	< 0.0670		ng/g
PCB 153	< 0.0670		ng/g
PCB 180	< 0.0670		ng/g
Total 6 ndl-PCB (lower-bound)	ND		ng/g
Total 6 ndl-PCB (medium-bound)	0.201	± 0,0503	ng/g
Total 6 ndl-PCB (upper-bound)	0.402	± 0,101	ng/g
DJ009	Tryptophane (Ak: F)		
Method	EU 152/2009, LC-FLD		
Sub: S4			
Tryptophan (Total)	0.0177	± 0.002	g/100 g
DJ011	Cysteine+Cystine & methionine (ox-amino acid) (Ak: F)		
Method	ISO 13903:2005, IC-UV		
Sub: S4			
Cysteine +Cystine	<0,006 (LOQ)		g/100 g
Methionine	0.100	± 0.014	g/100 g
DI004	Amino acids (acid hydrolysis) (Ak: F)		
Method	ISO 13903:2005, IC-UV		
Sub: S4			
Alanine	0.0469	± 0.007	g/100 g
Aspartic acid	0.132	± 0.018	g/100 g
Arginine	0.0802	± 0.011	g/100 g
Glutamic acid	0.569	± 0.080	g/100 g

Glycine	0.0373	± 0.005	g/100 g
Histidine	0.0716	± 0.010	g/100 g
Isoleucine	0.110	± 0.015	g/100 g

LP01H Quantification of Bacterial Endotoxins according to Eu.Pharmacopoeia 2.6.14 (Ak: F)
Method Ph. Eur. 10th Ed. 2.6.14, LAL turbidimetric assay

Sub: S5

Bacterial Endotoxins	14000	± 50%	EU/ml
----------------------	-------	-------	-------

Physical-chemical Analysis
DI004 Amino acids (acid hydrolysis) (Ak: F)
Method ISO 13903:2005, IC-UV

Sub: S4

Leucine	0.282	± 0.039	g/100 g
Lysine	0.169	± 0.024	g/100 g
Phenylalanine	0.148	± 0.021	g/100 g
Proline	0.371	± 0.052	g/100 g
Serine	0.149	± 0.021	g/100 g
Tyrosine	0.0695	± 0.010	g/100 g
Valine	0.205	± 0.029	g/100 g
Threonine	0.0983	± 0.014	g/100 g
Hydroxyproline	<0,2 (LOQ)		g/100 g
Ornithine	<0,05 (LOQ)		g/100 g

J1051 Lead (Pb) (Ak: F)
Method Internal method, ICP-OES

Sub: S1

Lead (Pb)	< 0.5	mg/kg
-----------	-------	-------

J1039 Cadmium (Cd) (Ak: F)
Method Internal method, ICP-OES

Sub: S1

Cadmium (Cd)	< 0.1	mg/kg
--------------	-------	-------

JCHG2 Mercury (Hg) (Ak: F)
Method DIN EN 15763:2010 (2010-04), mod., ICP-MS

Sub: S1

Mercury (Hg)	< 0.005	mg/kg
--------------	---------	-------

J1033 Arsenic (As) (Ak: F)
Method Internal method, ICP-OES

Sub: S1

Arsenic (As)	< 0.5	mg/kg
--------------	-------	-------

J1042 Copper (Cu) (Ak: F)
Method Internal method, ICP-OES

Sub: S1

Copper [Cu]	< 0.1	mg/kg
-------------	-------	-------

J1043 Iron (Fe) (Ak: F)
Method Internal method, ICP-OES

Sub: S1

Iron (Fe)	< 0.5	mg/kg
-----------	-------	-------

J1049 Nickel (Ni) (Ak: F)
Method Internal method, ICP-OES

Sub: S1

Nickel (Ni)	< 0.1	mg/kg
-------------	-------	-------

J1061	Zinc (Zn) (Ak: F)			
Method	Internal method, ICP-OES			
Sub: S1				
Zinc [Zn]		< 0.5		mg/kg
J1057	Tin (Sn) (Ak: F)			
Method	Internal method, ICP-OES			
Sub: S1				
Tin (Sn)		< 0.5		mg/kg
J1047	Manganese (Mn) (Ak: F)			
Method	Internal method, ICP-OES			
Sub: S1				
Manganese (Mn)		< 0.1		mg/kg
J1048	Sodium (Na) (Ak: F)			
Method	Internal method, ICP-OES			
Sub: S1				
Sodium (Na)		2300	± 460	mg/kg
J1045	Potassium (K) (Ak: F)			
Method	Internal method, ICP-OES			
Sub: S1				
Potassium [K]		< 3		mg/kg
J1046	Magnesium (Mg) (Ak: F)			
Method	Internal method, ICP-OES			
Sub: S1				
Magnesium (Mg)		9.5	± 1.9	mg/kg
J1050	Phosphorus (P) (Ak: F)			
Method	Internal method, ICP-OES			
Sub: S1				
Phosphorus (P)		1500	± 300	mg/kg

Ak: Accredited Status

A Accredited
N Not accredited
F Accredited by partner lab

Sub: Subcontractors

S1:Eurofins WEJ Contaminants GmbH (HH), Neuländer Kamp 1, Hamburg, GERMANY
S2:Eurofins Chemical Control S.r.l. (Cuneo), VIA CELDIT 2, Madonna dell'Olmo (Cuneo), ITALY
S3:Eurofins GfA Lab Service GmbH (Hamburg), Neuländer Kamp 1a, Hamburg, GERMANY
S4:Eurofins Vitamin Testing Denmark, Ladelundvej 85, Vejle, DENMARK
S5:Eurofins Food & Feed Testing Sweden (Lidköping), Sockerbruksg 3, port 2, Lidköping, SWEDEN

V1.4

The tests were carried out in the respective specialist department within the scope of competence of the authorized signatory in accordance with the QM manual. The analyses, unless otherwise indicated, are carried out by the company Eurofins Lebensmittelanalytik Österreich GmbH, Palmersstraße 2 in 2351 Wiener Neudorf.

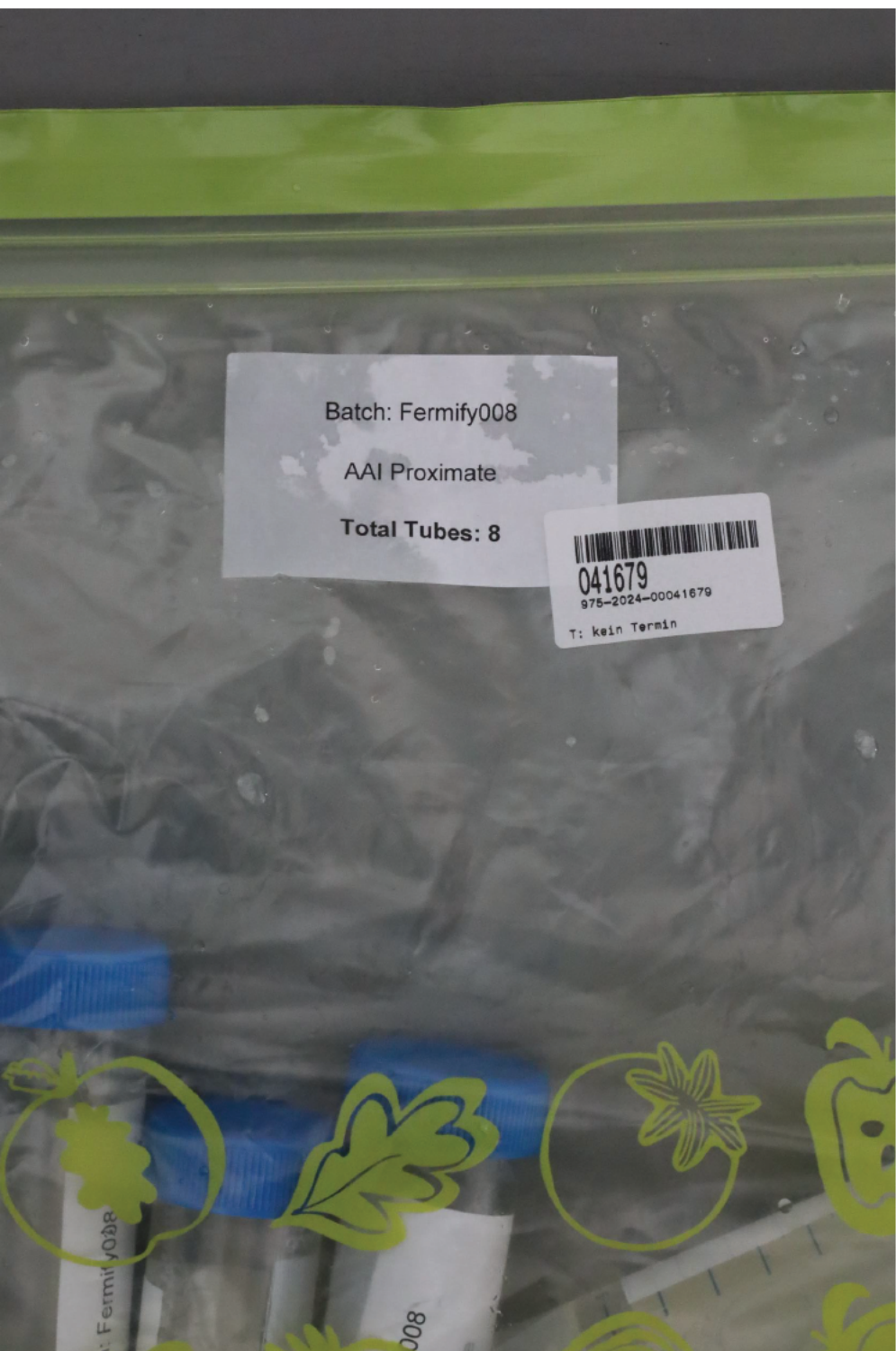
All tests are subject to a quality assurance program according to ÖVE/ÖNORM EN ISO/IEC 17025. Explanations of the decision rules in conformity assessments according to ÖVE/ÖNORM EN ISO/IEC 17025:2018 are set out in the General Terms & Conditions (GT&C), deviations from which are set out in separate agreements with the client. The test results relate exclusively to the test object and the scope of the tests performed. The samples were stored according to the manufacturer's instructions. Deviating storage conditions are documented in the analytical report. All information regarding the sample(s), with the exception of data collected by Eurofins Lebensmittelanalytik Österreich GmbH upon receipt of the sample or on site, has been provided by the customer. Data provided by the customer could have an influence on the validity of the results. Tests marked with "HB" were analyzed at the end of shelf life. Results expressed in % indicate mass ratios unless otherwise stated. cfu: colony forming units; u.B.: below limit of quantification. The GT&C of Eurofins Lebensmittelanalytik Österreich GmbH, which are available on the internet at <https://www.eurofins.at/geschaeftsbedingungen/> apply. Publication of the report requires written permission. Publication and reproduction of extracts is not permitted.

Signature



Zeichnungsberechtigt für die Prüfstelle

*** End of Report ***





Fermify GmbH
[REDACTED]
Richard-Neutra-Gasse 5
1210 Wien
ÖSTERREICH

Analytical Service Manager: [REDACTED]

Issue date 10.10.2024

Page 1/7

Analytical report AR-24-FA-03{ 2{ 7-01

Batch code EUATWI-002524{ 3

Sample No 975-2024-00042{ { 5

Sample description	Fermify019 (beta-casein solution)
Number	4
order type	D01-Analysis of selected parameters
Additional information	Probenahme erfolgte nicht durch die ELO. Gegebenenfalls entnehmen Sie weitere Details den Fotos.
Start analysis / End analysis	2{ .09.2024 / 10.10.2024
Reception date / Reception temperature	2{ .09.2024 2,2°C

Results

Sensory Analysis

FASCK simple descriptive analysis (Ak: N)

Method DIN 109{ 4:2014, Organoleptic evaluation

sensory indication

Aussehen: farblose, ganz leicht trübe Flüssigkeit; in durchsichtiger Verpackung

Konsistenz: dünnflüssig

Geruch: arteigen, neutral

Appearance: colourless, slightly cloudy liquid; in transparent packaging

Consistency: thin fluid

Smell: characteristic, no deviation from expected smell

Physical-chemical Analysis

JJ0EW Aflatoxin B1, B2, G1, G2 (spices, special matrix) (Ak: F)

Method DIN EN 14123 (2008-03), mod., IAC-LC-FLD

Sub: S1			
Aflatoxin B1	< 1.0		µg/kg
Aflatoxin B2	< 1.0		µg/kg
J2016 Antimony (Sb) (Ak: F)			
Method	Internal method, ICP-OES		
Sub: S1			
Antimony (Sb)	< 0.50		mg/kg
J10DZ Molybdenum (Mo) (Ak: F)			
Method	Internal method, ICP-OES		
Sub: S1			
Molybdenum (Mo)	< 0.2		mg/kg
J1060 Cobalt (Co) (Ak: F)			
Method	Internal method, ICP-OES		
Sub: S1			
Cobalt (Co)	< 0.2		mg/kg
JJ0C6 Boron (B) (Ak: F)			
Method	Internal method, ICP-OES		
Sub: S1			
Boron (B)	0.3	Z 0.2	mg/kg
GFOE1 OE5 -Totals WTH-PCI I /F and PCB (Ak: F)			
Method	Internal, Calculation		
Sub: S2			
WHO(2005)-PCDD/F+PCB TE± (lower-bound)	0.000{ 09	Z 0,000152	pg/g
WHO(2005)-PCDD/F+PCB TE± (medium-bound)	0.0558	Z 0,0140	pg/g
WHO(2005)-PCDD/F+PCB TE± (upper-bound)	0.111	Z 0,0277	pg/g

Microbiological Analysis

4MV5 R Salmonella species I (Food w Feed) LAOFood7Abs Pres /2Ug AFNHR +M 01/1Z-11/1Z (Ak: A)			
Method	AFNOR 3M 01/1{ -11/1{ , Bioluminescence		
Salmonella	Not Detected		/25 g
4M62Q Clostridium perfringens Ig7(Ak: A)			
Method	ISO 7937:2004, E-Cultural technique (non-chromogenic media)		
Clostridium perfringens	< 10		cfu/g
4MBI O Cronobacter species I (Food w Feed) LAOFood7Abs Pres /100 g SH 22[Z6 (Ak: A)			
Method	EN ISO 229{ 4:2017, D-Cultural techn. (chrom. + non-chromogenic media)		
Cronobacter spp	Not Detected		/100 g
4MT+G Escherichia coli B-Glucuronidase] E (Food w Feed) LAOFood7310 91 U00 /g (1) PSEC-PF &ternal Metho (Ak: A)			
Method	Internal Method SOP Mb 055 (24.05.2022), E-Cultural technique (media film)		
Escherichia coli B-Glucuronidase+	< 10		cfu/g
4MT+B Enterobacteriaceae E (Food w Feed) LAOFood7310 91 000 /g (1) PEB-PF &ternal Method SHP Mb 0U2 (22 (Ak: A)			
Method	Internal Method SOP Mb 052 (24.05.2022), E-Cultural technique (media film)		
Enterobacteriaceae	< 10		cfu/g
4MT6+ Aerobic Plate Count +0<C E (Food w Feed) LAOFood7310 9+00 000 /g (1+) PRAC-PF &ternal Method SHP (Ak: A)			
Method	Internal Method SOP Mb 051 (24.05.2022), E-Cultural technique (media film)		
Aerobic Plate Count 30°C	< 10		cfu/g
4MT6> Mould-° east E (Food w Feed) LAOFood7310 91U0 000 /g (1-+) PR° M-PF &ternal Method SHP Mb 0U> (22& (Ak: A)			

Method	SOP Mb 058, E-Cultural technique (media film)			
Moulds		< 10		cfu/g
Yeast		< 10		cfu/g
4MT+C	Coagulase positive staphylococci E (Food w Feed) IAO Food 7310 91 U00 /g (1) PS0Y-PF			
	Internal Method (Ak: A)			
Method	Internal Method SOP Mb 05{ (24.05.2022), E-Cultural technique (media film)			
Coagulase positive staphylococci		< 10		cfu/g
4M6S1	Visteria monocytogenes l2Ug7 (Ak: A)			
Method	Internal Method SOP Mb 010 (03.02.2021), Real-time PCR			
Listeria monocytogenes		Not Detected		/25 g
4M6JS	Presumptive Bacillus cereus lg7 (Ak: A)			
Method	ISO 7932:2004, E-Cultural technique (non-chromogenic media)			
Presumptive Bacillus cereus		< 10		cfu/g
& 0A1	Nitrogen, proteins (by calculation) (Ak: F)			
Method	UNI EN ISO 14891:2002, Dumas (TCD)			
Sub: S3				
Protein (Nx{ ,25)		2.4{	Z 0.25	g/100 g
Nitrogen		0.393	Z 0.040	g/100 g
& [UD	Moisture (Ak: F)			
Method	Internal method, Gravimetry			
Sub: S3				
Dry matter		3.29	Z 3.05	g/100 g
Moisture		9{ .71	Z 3.05	g/100 g
VP01T	5 uantification of Bacterial Endotoxins according to Eu8Pharmacopoeia 28316 (Ak: F)			
Method	Ph. Eur. 10th Ed. 2.{ .14, LAL turbidimetric assay			
Sub: S4				
Bacterial Endotoxins		{ 400	Z 50Y	EU/ml

Physical-chemical Analysis

JJ0EW	Aflatoxin B1, B2, G1, G2 (spices, special matrix) (Ak: F)			
Method	DIN EN 14123 (2008-03), mod., IAC-LC-FLD			
Sub: S1				
Aflatoxin G1		< 1.0		µg/kg
Aflatoxin G2		< 1.0		µg/kg
Sum of all positive Aflatoxins		nicht berechenbar		µg/kg
JJ0BG	Fumonisin (Ak: F)			
Method	Internal Method, LC-MS/MS			
Sub: S1				
Fumonisin B1 (FB1)		< 20		µg/kg
Fumonisin B2 (FB2)		< 20		µg/kg
Fumonisin sum (B1+B2)		nicht berechenbar		µg/kg
JJ0GU	Ochratoxin A (spices, special matrix) (Ak: F)			
Method	DIN EN 14132 (2009-09), mod., IAC-LC-FLD			
Sub: S1				
Ochratoxin A		< 2.0		µg/kg
JJ0FE	Fusarium toxins (I HN, 4HN, O2, T O2) (Ak: F)			
Method	Internal Method, LC-MS/MS			
Sub: S1				
Deoxynivalenol (Vomitoxin)		< 20		µg/kg
%earalene (%ON)		< 10		µg/kg

T-2 Toxin	< 10		µg/kg
HT-2 Toxin	< 10		µg/kg
sum T-2 HT-2 toxin	nicht berechenbar		µg/kg
& 06R	Ash (Ak: F)		
Method	Rapporti ISTISAN 199{ /34 pag 77, Gravimetry		
Sub: S3			
Ash	0.77	Z 0.08	g/100 g
& 0T.	Total dietary fiber (Tigh molecular Xeight) (Ak: F)		
Method	AOAC 991.43 2000, Enzymatic-gravimetry		
Sub: S3			
Total dietary fibre	< 1		g/100 g
GFV0U	I ioxins and Furans (1DPCI I /F) (Ak: F)		
Method	Internal, GC-MS/MS		
Sub: S2			
2,3,7,8-TetraCDD	< 0.0130		pg/g
1,2,3,7,8-PentaCDD	< 0.0170		pg/g
1,2,3,4,7,8-HexaCDD	< 0.0250		pg/g
1,2,3,{ ,7,8-HexaCDD	< 0.0350		pg/g
1,2,3,7,8,9-HexaCDD	< 0.0330		pg/g
1,2,3,4,{ ,7,8-HeptaCDD	< 0.0530		pg/g
OctaCDD	< 0.387		pg/g
2,3,7,8-TetraCDF	< 0.0350		pg/g
1,2,3,7,8-PentaCDF	< 0.0240		pg/g
2,3,4,7,8-PentaCDF	< 0.0370		pg/g
1,2,3,4,7,8-HexaCDF	< 0.0390		pg/g
1,2,3,{ ,7,8-HexaCDF	< 0.03{ 0		pg/g
1,2,3,7,8,9-HexaCDF	< 0.0270		pg/g
2,3,4,{ ,7,8-HexaCDF	< 0.0330		pg/g
1,2,3,4,{ ,7,8-HeptaCDF	0.0{ 09	Z 0,0183	pg/g
1,2,3,4,7,8,9-HeptaCDF	< 0.02{ 0		pg/g
OctaCDF	< 0.0800		pg/g
WHO(2005)-PCDD/F TE± (lower-bound)	0.000{ 09	Z 0,000152	pg/g
WHO(2005)-PCDD/F TE± (medium-bound)	0.0351	Z 0,00878	pg/g
WHO(2005)-PCDD/F TE± (upper-bound)	0.0{ 97	Z 0,0174	pg/g
& 1ZB	Total fat (Ak: F)		
Method	Internal method, Gravimetry		
Sub: S3			
Total fat	< 0.1		g/100 g
& 06Q	Carbohydrates (calculated as difference) (Ak: N)		
Method	Internal method, Calculation		
Sub: S3			
Carbohydrate calculated (difference)	< 0.5		g/100 g
GFV11	polychlorinated biphenyls (12 WTH PCB] Z &ES PCB) (Ak: F)		
Method	Internal, GC-MS/MS		
Sub: S2			
PCB 77	< 1.20		pg/g
PCB 81	< 0.180		pg/g
PCB 105	< 2.{ 0		pg/g

PCB 114	< 0.353		pg/g
PCB 118	< 9.33		pg/g
PCB 123	< 0.2{ 7		pg/g
PCB 12{	< 0.1{ 7		pg/g
PCB 15{	< 1.47		pg/g
PCB 157	< 0.273		pg/g
PCB 1{ 7	< 0.733		pg/g
PCB 1{ 9	< 0.800		pg/g
PCB 189	< 0.2{ 7		pg/g
WHO(2005)-PCB TE± (lower-bound)	ND		pg/g
WHO(2005)-PCB TE± (medium-bound)	0.0207	Z 0,00517	pg/g
WHO(2005)-PCB TE± (upper-bound)	0.0413	Z 0,0103	pg/g
PCB 28	< 0.0{ 70		ng/g
PCB 52	< 0.0{ 70		ng/g
PCB 101	< 0.0{ 70		ng/g
PCB 138	< 0.0{ 70		ng/g
PCB 153	< 0.0{ 70		ng/g
PCB 180	< 0.0{ 70		ng/g
Total { ndl-PCB (lower-bound)	ND		ng/g
Total { ndl-PCB (medium-bound)	0.201	Z 0,0503	ng/g
Total { ndl-PCB (upper-bound)	0.402	Z 0,101	ng/g
I J00[Cryptophane (Ak: F)			
Method EU 152/2009, LC-FLD			
Sub: S5			
Tryptophan (Total)	0.0188	Z 0,00188	g/100 g
I J011 Cysteine] Cystine w methionine (ox-amino acid) (Ak: F)			
Method ISO 13903:2005, IC-UV			
Sub: S5			
Cysteine +Cystine	<0.00{		g/100 g
Methionine	0.09{ 0	Z 0,0134	g/100 g
I J06 Amino acids (acid hydrolysis) (Ak: F)			
Method ISO 13903:2005, IC-UV			
Sub: S5			
Alanine	0.0431	Z 0,00{ 03	g/100 g
Aspartic acid	0.121	Z 0,01{ 9	g/100 g
Arginine	0.0802	Z 0,0112	g/100 g
Glutamic acid	0.549	Z 0,07{ 9	g/100 g
Glycine	0.0375	Z 0,00525	g/100 g
Histidine	0.0{ 58	Z 0,00921	g/100 g
Isoleucine	0.117	Z 0,01{ 4	g/100 g
Leucine	0.278	Z 0,0389	g/100 g
Lysine	0.188	Z 0,02{ 3	g/100 g
Phenylalanine	0.132	Z 0,0185	g/100 g
Proline	0.282	Z 0,0395	g/100 g
Serine	0.140	Z 0,019{	g/100 g
Tyrosine	0.0{ 14	Z 0,008{ 0	g/100 g
Valine	0.204	Z 0,028{	g/100 g

Threonine	0.104	Z 0,014{	g/100 g
Hydroxyproline	<0.2		g/100 g
Ornithine	<0.05		g/100 g
J10U1 Vead (Pb) (Ak: F)			
Method Internal method, ICP-OES			
Sub: S1			
Lead (Pb)	< 0.5		mg/kg
J10+{ Cadmium (Cd) (Ak: F)			
Method Internal method, ICP-OES			
Sub: S1			
Cadmium (Cd)	< 0.1		mg/kg
JCT G2 Mercury (Tg) (Ak: F)			
Method DIN EN 157{ 3:2010 (2010-04), mod., ICP-MS			
Sub: S1			
Mercury (Hg)	< 0.005		mg/kg
J10++ Arsenic (As) (Ak: F)			
Method Internal method, ICP-OES			
Sub: S1			
Arsenic (As)	< 0.5		mg/kg
J1062 Copper (Cu) (Ak: F)			
Method Internal method, ICP-OES			
Sub: S1			
Copper [Cu]	0.1	Z 0.1	mg/kg
J106+ Iron (Fe) (Ak: F)			
Method Internal method, ICP-OES			
Sub: S1			
Iron (Fe)	< 0.5		mg/kg
J106{ Nickel (Ni) (Ak: F)			
Method Internal method, ICP-OES			
Sub: S1			
Nickel (Ni)	< 0.1		mg/kg
J10Z1 4inc (4n) (Ak: F)			
Method Internal method, ICP-OES			
Sub: S1			
%inc [%n]	< 0.5		mg/kg
J10UD Qin (Sn) (Ak: F)			
Method Internal method, ICP-OES			
Sub: S1			
Tin (Sn)	< 0.5		mg/kg
J106D Manganese (Mn) (Ak: F)			
Method Internal method, ICP-OES			
Sub: S1			
Manganese (Mn)	< 0.1		mg/kg
J106> Sodium (Na) (Ak: F)			
Method Internal method, ICP-OES			
Sub: S1			
Sodium (Na)	2500	Z 500	mg/kg
J106U Potassium (K) (Ak: F)			
Method Internal method, ICP-OES			
Sub: S1			

Potassium [K]	{	Z 3	mg/kg
J106Z Magnesium (Mg) (Ak: F)			
Method			Internal method, ICP-OES
Sub: S1			
Magnesium (Mg)	1.1	Z 0.5	mg/kg
J10U0 Phosphorus (P) (Ak: F)			
Method			Internal method, ICP-OES
Sub: S1			
Phosphorus (P)	1700	Z 340	mg/kg
J10U6 Sulphur (S) (Ak: F)			
Method			Internal method, ICP-OES
Sub: S1			
Sulphur total (S)	220	Z 44	mg/kg
J10UU Selenium (Se) (Ak: F)			
Method			Internal method, ICP-OES
Sub: S1			
Selenium (Se)	< 0.5		mg/kg

Ak: AccreditedStatus

A Accredited
N Not accredited
F Accredited by partner lab

Sub: Subcontractors

S1:Eurofins WEJ Contaminants GmbH (HH), Neuländer Kamp 1, Hamburg, GERMANQ
S2:Eurofins GfA Lab Service GmbH (Hamburg), Neuländer Kamp 1a, Hamburg, GERMANQ
S3:Eurofins Chemical Control S.r.l. (Cuneo), VIA CELDIT 2, Madonna dell'Olmo (Cuneo), ITALQ
S4:Eurofins Food j Feed Testing Sweden (Lidk&ping), Sockerbruksg 3, port 2, Lidk&ping, SWEDEN
S5:Eurofins Vitamin Testing Denmark, Ladelundveø85, Veøen, DENMARK

v1.4

The tests were carried out in the respective specialist department within the scope of competence of the authorized signatory in accordance with the QM manual. The analyses, unless otherwise indicated, are carried out by the company Eurofins Lebensmittelanalytik Österreich GmbH, Palmersstraße 2 in 2351 Wiener Neudorf.

All tests are subject to a quality assurance program according to ÖVE/ÖNORM EN ISO/IEC 17025. Explanations of the decision rules in conformity assessments according to ÖVE/ÖNORM EN ISO/IEC 17025:2018 are set out in the General Terms & Conditions (GT&C), deviations from which are set out in separate agreements with the client. The test results relate exclusively to the test object and the scope of the tests performed. The samples were stored according to the manufacturer's instructions. Deviating storage conditions are documented in the analytical report. All information regarding the sample(s), with the exception of data collected by Eurofins Lebensmittelanalytik Österreich GmbH upon receipt of the sample or on site, has been provided by the customer. Data provided by the customer could have an influence on the validity of the results. Tests marked with "HB" were analyzed at the end of shelf life. Results expressed in % indicate mass ratios unless otherwise stated. cfu: colony forming units; u.B.: below limit of quantification. The GT&C of Eurofins Lebensmittelanalytik Österreich GmbH, which are available on the internet at <https://www.eurofins.at/geschaeftsbedingungen/> apply. Publication of the report requires written permission. Publication and reproduction of extracts is not permitted.

Signature


%zeichnungsberechtigt für die Prüfstelle

*** End of Report ***





Fermify GmbH
[REDACTED]
Richard-Neutra-Gasse 5
1210 Wien
ÖSTERREICHAnalytical Service Manager: [REDACTED]
[REDACTED]

Issue date 10.10.2024

Page 1/7

Analytical report AR-24-FA-03{ 2{ 8-01

Batch code EUATWI-002524{ 3

Sample No 975-2024-00042{ { {

Sample description	Fermify021 (beta-casein solution)
Number	5
order type	D01-Analysis of selected parameters
Additional information	Probenahme erfolgte nicht durch die ELO. Gegebenenfalls entnehmen Sie weitere Details den Fotos.
Start analysis / End analysis	2{ .09.2024 / 10.10.2024
Reception date / Reception temperature	2{ .09.2024 2,2°C

Results

Sensory Analysis

FASCK simple descriptive analysis (Ak: N)

Method DIN 109{ 4:2014, Organoleptic evaluation

sensory indication

Aussehen: farblose, ganz leicht trübe Flüssigkeit; in durchsichtiger Verpackung

Konsistenz: dünnflüssig

Geruch: arteigen, neutral

Appearance: colourless, slightly cloudy liquid; in transparent packaging

Consistency: thin fluid

Smell: characteristic, no deviation from expected smell

Physical-chemical Analysis

JJ0EW Aflatoxin B1, B2, G1, G2 (spices, special matrix) (Ak: F)

Method DIN EN 14123 (2008-03), mod., IAC-LC-FLD

Sub: S1			
Aflatoxin B1	< 1.0		µg/kg
Aflatoxin B2	< 1.0		µg/kg
Aflatoxin G1	< 1.0		µg/kg
Aflatoxin G2	< 1.0		µg/kg
Sum of all positive Aflatoxins	nicht berechenbar		µg/kg
JJ0BG Fumonisinis (Ak: F)			
Method Internal Method, LC-MS/MS			
Sub: S1			
Fumonisin B1 (FB1)	< 20		µg/kg
Fumonisin B2 (FB2)	< 20		µg/kg
Fumonisin sum (B1+B2)	nicht berechenbar		µg/kg
JJ0G6 Dchratoxin A (spices, special matrix) (Ak: F)			
Method DIN EN 14132 (2009-09), mod., IAC-LC-FLD			
Sub: S1			
Ochratoxin A	< 2.0		µg/kg
JJ0FE Fusarium toxins (ZDN, ODN, 52, T52) (Ak: F)			
Method Internal Method, LC-MS/MS			
Sub: S1			
Deoxynivalenol (Vomitoxin)	< 20		µg/kg
Zearalenone (ZON)	< 10		µg/kg
T-2 Toxin	< 10		µg/kg
HT-2 Toxin	< 10		µg/kg
sum T-2 HT-2 toxin	nicht berechenbar		µg/kg
I201R Ash (Ak: F)			
Method Rapport ISTISAN 199{ /34 pag 77, Gravimetry			
Sub: S2			
Ash	0.79	± 0.09	g/100 g
I20T4 5total dietary fiber (Tigh molecular Veight) (Ak: F)			
Method AOAC 991.43 2000, Enzymatic-gravimetry			
Sub: S2			
Total dietary fibre	< 1		g/100 g
GFw06 Zioxins and Furans (1L PCZZ/F) (Ak: F)			
Method Internal, GC-MS/MS			
Sub: S3			
2,3,7,8-TetraCDD	< 0.0130		pg/g
1,2,3,7,8-PentaCDD	< 0.0170		pg/g
1,2,3,4,7,8-HexaCDD	< 0.0253		pg/g
1,2,3,{ , 7,8-HexaCDD	< 0.0350		pg/g
1,2,3,7,8,9-HexaCDD	< 0.0330		pg/g
1,2,3,4,{ , 7,8-HeptaCDD	< 0.0532		pg/g
OctaCDD	< 0.387		pg/g
2,3,7,8-TetraCDF	< 0.0350		pg/g
1,2,3,7,8-PentaCDF	< 0.0240		pg/g
2,3,4,7,8-PentaCDF	< 0.0373		pg/g
1,2,3,4,7,8-HexaCDF	< 0.0393		pg/g
1,2,3,{ , 7,8-HexaCDF	< 0.03{ 0		pg/g
1,2,3,7,8,9-HexaCDF	< 0.0270		pg/g

2,3,4,{ ,7,8-HexaCDF	< 0.0330		pg/g
1,2,3,4,{ ,7,8-HeptaCDF	< 0.0373		pg/g
1,2,3,4,7,8,9-HeptaCDF	< 0.02{ 0		pg/g
OctaCDF	< 0.0800		pg/g
WHO(2005)-PCDD/F TEQ (lower-bound)	ND		pg/g
WHO(2005)-PCDD/F TEQ (medium-bound)	0.0348	± 0,008{ 9	pg/g
WHO(2005)-PCDD/F TEQ (upper-bound)	0.0{ 9{	± 0,0174	pg/g
IZ17B 5total fat (Ak: F)			
Method Internal method, Gravimetry			
Sub: S2			
Total fat	< 0.1		g/100 g
IZ01 U Carbohydrates (calculated as difference) (Ak: N)			
Method Internal method, Calculation			
Sub: S2			
Carbohydrate calculated (difference)	< 0.5		g/100 g
GFw11 polychlorinated biphenyls (12 WTD PCB + 7 ICES PCB) (Ak: F)			
Method Internal, GC-MS/MS			
Sub: S3			
PCB 77	< 1.20		pg/g
PCB 81	< 0.180		pg/g
PCB 105	< 2. { 0		pg/g
PCB 114	< 0.353		pg/g
PCB 118	< 9.33		pg/g
PCB 123	< 0.2{ 7		pg/g
PCB 12{	< 0.1{ 7		pg/g
PCB 15{	< 1.47		pg/g
PCB 157	< 0.273		pg/g
PCB 1{ 7	< 0.733		pg/g
PCB 1{ 9	< 0.800		pg/g
PCB 189	< 0.2{ 7		pg/g
WHO(2005)-PCB TEQ (lower-bound)	ND		pg/g
WHO(2005)-PCB TEQ (medium-bound)	0.0207	± 0,00517	pg/g
WHO(2005)-PCB TEQ (upper-bound)	0.0413	± 0,0103	pg/g
PCB 28	< 0.0{ 70		ng/g
PCB 52	< 0.0{ 70		ng/g
PCB 101	< 0.0{ 70		ng/g
PCB 138	< 0.0{ 70		ng/g
PCB 153	< 0.0{ 70		ng/g
PCB 180	< 0.0{ 70		ng/g
Total { ndl-PCB (lower-bound)	ND		ng/g
Total { ndl-PCB (medium-bound)	0.201	± 0,0503	ng/g
Total { ndl-PCB (upper-bound)	0.402	± 0,101	ng/g
ZJ00Q 5ryptophane (Ak: F)			
Method EU 152/2009, LC-FLD			
Sub: S4			
Tryptophan (Total)	0.0285	± 0,00285	g/100 g
ZJ011 Cysteine+Cystine & methionine (ox-amino acid) (Ak: F)			
Method ISO 13903:2005, IC-UV			

Sub: S4

Cysteine +Cystine

<0.00{ g/100 g

Methionine

0.13{ ± 0,0190 g/100 g

J1001 Amino acids (acid hydrolysis) (Ak: F)
Method ISO 13903:2005, IC-UV

Sub: S4

Alanine

0.0{ 79 ± 0,00951 g/100 g

Aspartic acid

0.1{ 0 ± 0,0224 g/100 g

Arginine

0.0813 ± 0,0114 g/100 g

Glutamic acid

0.787 ± 0,110 g/100 g

Glycine

0.0551 ± 0,00771 g/100 g

Histidine

0.100 ± 0,0140 g/100 g

Isoleucine

0.173 ± 0,0242 g/100 g

Leucine

0.419 ± 0,0587 g/100 g

Lysine

0.2{ 8 ± 0,0375 g/100 g

Phenylalanine

0.197 ± 0,027{ g/100 g

Proline

0.575 ± 0,0805 g/100 g

Serine

0.201 ± 0,0281 g/100 g

Tyrosine

0.0880 ± 0,0123 g/100 g

Valine

0.297 ± 0,041{ g/100 g

Threonine

0.137 ± 0,0192 g/100 g

Hydroxyproline

<0.2 g/100 g

Ornithine

<0.05 g/100 g

J1061 wead (Pb) (Ak: F)
Method Internal method, ICP-OES

Sub: S1

Lead (Pb)

< 0.5 mg/kg

J10{ Q Cadmium (Cd) (Ak: F)
Method Internal method, ICP-OES

Sub: S1

Cadmium (Cd)

< 0.1 mg/kg

JCTG2 Mercury (Tg) (Ak: F)
Method DIN EN 157{ 3:2010 (2010-04), mod., ICP-MS

Sub: S1

Mercury (Hg)

< 0.005 mg/kg

J10{ [Arsenic (As) (Ak: F)
Method Internal method, ICP-OES

Sub: S1

Arsenic (As)

< 0.5 mg/kg

J10I 2 Copper (Cu) (Ak: F)
Method Internal method, ICP-OES

Sub: S1

Copper %Cu%

0.1 ± 0.1 mg/kg

J10I [Iron (Fe) (Ak: F)
Method Internal method, ICP-OES

Sub: S1

Iron (Fe)

< 0.5 mg/kg

J10I Q Nickel (Ni) (Ak: F)
Method Internal method, ICP-OES

Sub: S1			
Nickel (Ni)	< 0.1		mg/kg
J1071 Zinc (Zn) (Ak: F)			
Method Internal method, ICP-OES			
Sub: S1			
Zinc Zn%	< 0.5		mg/kg
J106L Tin (Sn) (Ak: F)			
Method Internal method, ICP-OES			
Sub: S1			
Tin (Sn)	< 0.5		mg/kg
J101 L Manganese (Mn) (Ak: F)			
Method Internal method, ICP-OES			
Sub: S1			
Manganese (Mn)	< 0.1		mg/kg
J101 J Sodium (Na) (Ak: F)			
Method Internal method, ICP-OES			
Sub: S1			
Sodium (Na)	2500	± 500	mg/kg
J101 6 Potassium (K) (Ak: F)			
Method Internal method, ICP-OES			
Sub: S1			
Potassium K%	< 3		mg/kg
J101 7 Magnesium (Mg) (Ak: F)			
Method Internal method, ICP-OES			
Sub: S1			
Magnesium (Mg)	0.8	± 0.4	mg/kg
J1060 Phosphorus (P) (Ak: F)			
Method Internal method, ICP-OES			
Sub: S1			
Phosphorus (P)	1700	± 340	mg/kg
J106I Sulphur (S) (Ak: F)			
Method Internal method, ICP-OES			
Sub: S1			
Sulphur total (S)	320	± { 4	mg/kg
J1066 Selenium (Se) (Ak: F)			
Method Internal method, ICP-OES			
Sub: S1			
Selenium (Se)	< 0.5		mg/kg
J201I Antimony (Sb) (Ak: F)			
Method Internal method, ICP-OES			
Sub: S1			
Antimony (Sb)	< 0.50		mg/kg
J10L7 Molybdenum (Mo) (Ak: F)			
Method Internal method, ICP-OES			
Sub: S1			
Molybdenum (Mo)	< 0.2		mg/kg
J101 0 Cobalt (Co) (Ak: F)			
Method Internal method, ICP-OES			
Sub: S1			
Cobalt (Co)	< 0.2		mg/kg

JJ0CI	Boron (B) (Ak: F)			
Method	Internal method, ICP-OES			
Sub: S1				
Boron (B)		0.3	± 0.2	mg/kg
GF5E1	5E3 -5otals WTD-PCZZ/F and PCB (Ak: F)			
Method	Internal, Calculation			
Sub: S3				
WHO(2005)-PCDD/F+PCB TEQ (lower-bound)		ND		pg/g
WHO(2005)-PCDD/F+PCB TEQ (medium-bound)		0.0554	± 0,0139	pg/g
WHO(2005)-PCDD/F+PCB TEQ (upper-bound)		0.111	± 0,0277	pg/g

Microbiological Analysis

OMw3 R	Salmonella species Z (Food & Feed) 9A5 Food<Abs Pres /26 g AFNDR [M 01/17-11/17 (Ak: A)			
Method	AFNOR 3M 01/1{ -11/1{ , Bioluminescence			
Salmonella	Not Detected			/25 g
OMI 2U	Clostridium perfringens 9g<(Ak: A)			
Method	ISO 7937:2004, E-Cultural technique (non-chromogenic media)			
Clostridium perfringens	< 10			cfu/g
OMBZ 5	Cronobacter species Z (Food & Feed) 9A5 Food<Abs Pres /100 g HSD 22Q7I (Ak: A)			
Method	EN ISO 229{ 4:2017, D-Cultural techn. (chrom. + non-chromogenic media)			
Cronobacter spp	Not Detected			/100 g
OMT [G	Escherichia coli B-Glucuronidase+ E (Food & Feed) 9A5 Food<>10 ° 1 600 /g (1) PSEC-PF Internal Metho (Ak: A)			
Method	Internal Method SOP Mb 055 (24.05.2022), E-Cultural technique (media film)			
Escherichia coli B-Glucuronidase+	< 10			cfu/g
OMT [B	Enterobacteriaceae E (Food & Feed) 9A5 Food<>10 ° 1 000 /g (1) PEB-PF Internal Method SDP Mb 062 (22 (Ak: A)			
Method	Internal Method SOP Mb 052 (24.05.2022), E-Cultural technique (media film)			
Enterobacteriaceae	< 10			cfu/g
OMTI [	Aerobic Plate Count [08C E (Food & Feed) 9A5 Food<>10 ° [00 000 /g (1- [) PRAC-PF Internal Method SDP (Ak: A)			
Method	Internal Method SOP Mb 051 (24.05.2022), E-Cultural technique (media film)			
Aerobic Plate Count 30°C	< 10			cfu/g
OMTI [	Mould-Yeast E (Food & Feed) 9A5 Food<>10 ° 160 000 /g (1- [) PRYM-PF Internal Method SDP Mb 06] (22.1 (Ak: A)			
Method	SOP Mb 058, E-Cultural technique (media film)			
Moulds	< 10			cfu/g
[east	< 10			cfu/g
OMT [C	Coagulase positive staphylococci E (Food & Feed) 9A5 Food<>10 ° 1 600 /g (1) PS5X-PF Internal Method (Ak: A)			
Method	Internal Method SOP Mb 05{ (24.05.2022), E-Cultural technique (media film)			
Coagulase positive staphylococci	< 10			cfu/g
OMI S1	Listeria monocytogenes 926 g<(Ak: A)			
Method	Internal Method SOP Mb 010 (03.02.2021), Real-time PCR			
Listeria monocytogenes	Not Detected			/25 g
OMI JS	Presumptive Bacillus cereus 9g<(Ak: A)			
Method	ISO 7932:2004, E-Cultural technique (non-chromogenic media)			
Presumptive Bacillus cereus	< 10			cfu/g
IZ0A1	Nitrogen, proteins (by calculation) (Ak: F)			
Method	UNI EN ISO 14891:2002, Dumas (TCD)			

Sub: S2			
Protein (Nx{ ,25)	3.{ 0	± 0.35	g/100 g
Nitrogen	0.57{	± 0.057	g/100 g
IZ Q6L	Moisture (Ak: F)		
Method	Internal method, Gravimetry		
Sub: S2			
Dry matter	4.41	± 3.02	g/100 g
Moisture	95.59	± 3.02	g/100 g
wP01T	3 uantification of Bacterial Endotoxins according to Eu.Pharmacopoeia 2.7.1I (Ak: F)		
Method	Ph. Eur. 10th Ed. 2.{ .14, LAL turbidimetric assay		
Sub: S5			
Bacterial Endotoxins	9700	± 50]	EU/ml

Ak: Accredit. Status

A Accredited
N Not accredited
F Accredited by partner lab

Sub: Subcontractors

S1:Eurofins WEJ Contaminants GmbH (HH), Neuländer Kamp 1, Hamburg, GERMANY
S2:Eurofins Chemical Control S.r.l. (Cuneo), VIA CELDIT 2, Madonna dell'Olmo (Cuneo), ITALY
S3:Eurofins GfA Lab Service GmbH (Hamburg), Neuländer Kamp 1a, Hamburg, GERMANY
S4:Eurofins Vitamin Testing Denmark, Ladelundvej 85, Vejle, DENMARK
S5:Eurofins Food & Feed Testing Sweden (Lidköping), Sockerbruksg 3, port 2, Lidköping, SWEDEN

V1.4

The tests were carried out in the respective specialist department within the scope of competence of the authorized signatory in accordance with the QM manual. The analyses, unless otherwise indicated, are carried out by the company Eurofins Lebensmittelanalytik Österreich GmbH, Palmersstraße 2 in 2351 Wiener Neudorf.

All tests are subject to a quality assurance program according to ÖVE/ÖNORM EN ISO/IEC 17025. Explanations of the decision rules in conformity assessments according to ÖVE/ÖNORM EN ISO/IEC 17025:2018 are set out in the General Terms & Conditions (GT&C), deviations from which are set out in separate agreements with the client. The test results relate exclusively to the test object and the scope of the tests performed. The samples were stored according to the manufacturer's instructions. Deviating storage conditions are documented in the analytical report. All information regarding the sample(s), with the exception of data collected by Eurofins Lebensmittelanalytik Österreich GmbH upon receipt of the sample or on site, has been provided by the customer. Data provided by the customer could have an influence on the validity of the results. Tests marked with "HB" were analyzed at the end of shelf life. Results expressed in % indicate mass ratios unless otherwise stated. cfu: colony forming units; u.B.: below limit of quantification. The GT&C of Eurofins Lebensmittelanalytik Österreich GmbH, which are available on the internet at <https://www.eurofins.at/geschaeftsbedingungen/> apply. Publication of the report requires written permission. Publication and reproduction of extracts is not permitted.

Signature



Zeichnungsberechtigt für die Prüfstelle

*** End of Report ***



Batch: Family021
AAB Mycotoxins
Total Tubes: 5
042666
879-2024-000268
1. Batch 02101

Appendix III – NHANES food codes and associated proposed uses used in exposure assessment.

Proposed use	Food code	Food description
Beverages (non-alcoholic)	11551050	Licudo or Batido
	11553100	Fruit smoothie, NFS
	11553110	Fruit smoothie, with whole fruit and dairy
	11553120	Fruit smoothie, with whole fruit and dairy, added protein
	11553130	Fruit smoothie juice drink, with dairy
	42403010	Coconut water, unsweetened
	42404010	Coconut water, sweetened
	64134015	Fruit smoothie, with whole fruit, no dairy
	64134020	Fruit smoothie, with whole fruit, no dairy, added protein
	64134025	Fruit smoothie, with whole fruit, non-dairy
	64134030	Fruit smoothie juice drink, no dairy
	64134100	Fruit smoothie, light
	64134200	Fruit smoothie, bottled
	64200100	Fruit nectar, NFS
	64201010	Apricot nectar
	64201500	Banana nectar
	64202010	Cantaloupe nectar
	64203020	Guava nectar
	64204010	Mango nectar
	64205010	Peach nectar
	64210010	Papaya nectar
	64213010	Passion fruit nectar
	64215010	Pear nectar
	64221010	Soursop, nectar
	78101100	Fruit and vegetable smoothie, with dairy
	78101110	Fruit and vegetable smoothie, added protein
	78101115	Fruit and vegetable smoothie, non-dairy
	78101118	Fruit and vegetable smoothie, non-dairy, added protein
	78101120	Fruit and vegetable smoothie, bottled
	78101125	Fruit and vegetable smoothie, no dairy
	78101130	Vegetable smoothie
	92306100	Corn beverage
	92432000	Fruit juice drink, citrus, carbonated
	92433000	Fruit juice drink, noncitrus, carbonated
	92510610	Fruit juice drink
	92510650	Tamarind drink
	92510720	Fruit punch, made with fruit juice and soda
	92510730	Fruit punch, made with soda, fruit juice, and sherbet or ice cream
	92510955	Lemonade, fruit juice drink
	92510960	Lemonade, fruit flavored drink
	92511000	Lemonade, frozen concentrate, not reconstituted
	92511015	Fruit flavored drink
	92511250	Fruit juice beverage, 40-50% juice, citrus
	92512040	Frozen daiquiri mix, frozen concentrate, not reconstituted
	92512050	Frozen daiquiri mix, from frozen concentrate, reconstituted
	92512090	Pina Colada, nonalcoholic
	92512110	Margarita mix, nonalcoholic
	92513000	Slush frozen drink
	92513010	Slush frozen drink, no sugar added
	92530410	Fruit flavored drink, with high vitamin C
	92530510	Cranberry juice drink, with high vitamin C
	92530610	Fruit juice drink, with high vitamin C
	92530950	Vegetable and fruit juice drink, with high vitamin C
	92531030	Fruit juice drink (Sunny D)
	92541010	Fruit flavored drink, powdered, reconstituted
	92542000	Fruit flavored drink, with high vitamin C, powdered, reconstituted
	92550030	Fruit juice drink, with high vitamin C, light
	92550035	Fruit juice drink, light
	92550040	Fruit juice drink, diet
	92550110	Cranberry juice drink, with high vitamin C, light
	92550200	Grape juice drink, light
	92550350	Orange juice beverage, 40-50% juice, light
	92550360	Apple juice beverage, 40-50% juice, light
	92550370	Lemonade, fruit juice drink, light
	92550380	Pomegranate juice beverage, 40-50% juice, light
	92550400	Vegetable and fruit juice drink, with high vitamin C, diet
	92550405	Vegetable and fruit juice drink, with high vitamin C, light
	92550610	Fruit flavored drink, with high vitamin C, diet
	92550620	Fruit flavored drink, diet
	92552000	Fruit flavored drink, with high vitamin C, powdered, reconstituted, diet
	92552010	Fruit flavored drink, powdered, reconstituted, diet
	92552020	Fruit juice drink, reduced sugar (Sunny D)
	92552030	Fruit juice drink (Capri Sun)
	92582100	Fruit juice drink, with high vitamin C, plus added calcium
	92582110	Fruit juice drink, added calcium (Sunny D)
	92610020	Horchata beverage, made with water
	92610030	Horchata beverage, made with milk
	92611010	Oatmeal beverage with water
	92611100	Oatmeal beverage with milk
	92612010	Sugar cane beverage
	92613010	Cornmeal beverage
	92613510	Cornmeal beverage with chocolate milk
	92801000	Wine, nonalcoholic
	92802000	Wine, light, nonalcoholic
	92803000	Nonalcoholic malt beverage
	92804000	Shirley Temple
	95310200	Energy drink (Full Throttle)
	95310400	Energy drink (Monster)
	95310500	Energy drink (Mountain Dew AMP)
	95310550	Energy drink (No Fear)
	95310555	Energy drink (No Fear Motherload)
	95310560	Energy drink (NOS)

GRAS Determination for the use of β -casein produced by *E. coli* stF_24_139

Notifier: Fermify GmbH

Date: October 14, 2024

Cheeses

95310600	Energy drink (Red Bull)
95310700	Energy drink (Rockstar)
95310750	Energy drink (SoBe Energize Energy Juice Drink)
95310800	Energy drink (Vault)
95311000	Energy Drink
95312400	Energy drink, low calorie (Monster)
95312410	Energy drink, sugar free (Monster)
95312500	Energy drink, sugar free (Mountain Dew AMP)
95312550	Energy drink, sugar free (No Fear)
95312555	Energy drink, sugar-free (NOS)
95312560	Energy drink (Ocean Spray Cran-Energy Juice Drink)
95312600	Energy drink, sugar-free (Red Bull)
95312700	Energy drink, sugar free (Rockstar)
95312800	Energy drink, sugar free (Vault)
95312900	Energy drink (XS)
95312905	Energy drink (XS Gold Plus)
95313200	Energy drink, sugar free
95320200	Sports drink (Gatorade G)
95320500	Sports drink (Powerade)
95321000	Sports drink, NFS
95322200	Sports drink, low calorie (Gatorade G2)
95322500	Sports drink, low calorie (Powerade Zero)
95323000	Sports drink, low calorie
12140000	Cream, whipped
12220200	Whipped topping
12220270	Whipped topping, fat free
12220280	Whipped topping, sugar free
12310100	Sour cream, regular
12310350	Sour cream, light
12310370	Sour cream, fat free
14010000	Cheese, NFS
14101010	Cheese, Blue or Roquefort
14102010	Cheese, Brick
14103010	Cheese, Camembert
14103020	Cheese, Brie
14104100	Cheese, Cheddar
14104110	Cheese, Cheddar, reduced fat
14104115	Cheese, Cheddar, nonfat or fat free
14104200	Cheese, Colby
14104250	Cheese, Colby Jack
14104400	Cheese, Feta
14104600	Cheese, Fontina
14104700	Cheese, goat
14105010	Cheese, Gouda or Edam
14105200	Cheese, Gruyere
14106010	Cheese, Limburger
14106200	Cheese, Monterey
14106500	Cheese, Monterey, reduced fat
14107010	Cheese, Mozzarella, NFS
14107030	Cheese, Mozzarella, part skim
14107040	Cheese, Mozzarella, reduced sodium
14107060	Cheese, Mozzarella, nonfat or fat free
14107200	Cheese, Muenster
14107250	Cheese, Muenster, reduced fat
14108010	Cheese, Parmesan, dry grated
14108015	Cheese, Parmesan, dry grated, reduced fat
14108020	Cheese, Parmesan, hard
14108060	Cheese, Parmesan, dry grated, fat free
14108200	Cheese, Port du Salut
14108400	Cheese, Provolone
14108420	Cheese, provolone, reduced fat
14109010	Cheese, Swiss
14109020	Cheese, Swiss, reduced sodium
14109030	Cheese, Swiss, reduced fat
14109040	Cheese, Swiss, nonfat or fat free
14110010	Cheese, Cheddar, reduced sodium
14120010	Cheese, Mexican blend
14120020	Cheese, Mexican blend, reduced fat
14131000	Queso Anejo, aged Mexican cheese
14131500	Queso Asadero
14133000	Queso Fresco
14134000	Queso cotija
14200100	Cheese, cottage, NFS
14201010	Cheese, cottage, creamed, large or small curd
14201200	Cottage cheese, farmer's
14201500	Cheese, Ricotta
14202010	Cheese, cottage, with fruit
14202020	Cheese, cottage, with vegetables
14203010	Cheese, cottage, dry curd
14203020	Cheese, cottage, salted, dry curd
14203510	Puerto Rican white cheese
14204010	Cheese, cottage, low fat
14204020	Cheese, cottage, lowfat, with fruit
14206010	Cheese, cottage, lowfat, low sodium
14207010	Cheese, cottage, lowfat, lactose reduced
14301010	Cream cheese, regular, plain
14301100	Cream cheese, regular, flavored
14303010	Cream cheese, light
14410100	Cheese, American and Swiss blends
14410110	Cheese, American
14410120	Cheese, American, reduced fat
14410130	Cheese, American, nonfat or fat free
14410210	Cheese, American, reduced sodium
14410330	Cheese spread, American or Cheddar cheese base, reduced fat
14410380	Cream cheese spread, fat free
14410500	Cheese, processed cheese food
14410600	Cheese, processed, with vegetables
14410620	Cheese, with wine
14420100	Cheese spread, American or Cheddar cheese base
14420160	Cheese spread, Swiss cheese base

GRAS Determination for the use of β -casein produced by *E. coli* stF_24_139

Notifier: Fermify GmbH

Date: October 14, 2024

	14420200	Cheese spread, cream cheese, regular
	14420210	Cheese spread, cream cheese, light
	14420300	Cheese spread, pressurized can
	14610200	Cheese, cottage cheese, with gelatin dessert
	14610210	Cheese, cottage cheese, with gelatin dessert and fruit
	14610250	Cheese, cottage cheese, with gelatin dessert and vegetables
	14610520	Cheese ball
	14660200	Mozzarella sticks, breaded, baked, or fried
Dairy product analogs	11300100	Non-dairy milk, NFS
	11512030	Hot chocolate / Cocoa, ready to drink, made with non-dairy milk
	11512120	Hot chocolate / Cocoa, ready to drink, made with non-dairy milk and whipped cream
	11513310	Chocolate milk, made from dry mix with non-dairy milk
	11513375	Chocolate milk, made from reduced sugar mix with non-dairy milk
	11513385	Chocolate milk, made from dry mix with non-dairy milk (Nesquik)
	11513395	Chocolate milk, made from no sugar added dry mix with non-dairy milk (Nesquik)
	11513750	Chocolate milk, made from syrup with non-dairy milk
	11513805	Chocolate milk, made from light syrup with non-dairy milk
	11513855	Chocolate milk, made from sugar free syrup with non-dairy milk
	11514150	Hot chocolate / Cocoa, made with dry mix and non-dairy milk
	11514360	Hot chocolate / Cocoa, made with no sugar added dry mix and non-dairy milk
	11519215	Strawberry milk, non-dairy
Cheese analogs	12320100	Sour cream, imitation
	14502000	Imitation cheese
Frozen dairy desserts and mixes	11459990	Frozen yogurt, NFS
	11460000	Frozen yogurt, vanilla
	11460100	Frozen yogurt, chocolate
	11460500	Frozen yogurt, soft serve, vanilla
	11460510	Frozen yogurt, soft serve, chocolate
	11461200	Frozen yogurt sandwich
	11461210	Frozen yogurt bar, vanilla
	11461220	Frozen yogurt bar, chocolate
	11461250	Frozen yogurt cone, chocolate
	11461260	Frozen yogurt cone, vanilla
	11461300	Frozen yogurt cone, vanilla, waffle cone
	11461320	Frozen yogurt cone, chocolate, waffle cone
	13110000	Ice cream, NFS
	13110100	Ice cream, vanilla
	13110102	Ice cream, vanilla, with additional ingredients
	13110110	Ice cream, chocolate
	13110112	Ice cream, chocolate, with additional ingredients
	13110200	Ice cream, soft serve, vanilla
	13110210	Ice cream, soft serve, chocolate
	13110460	Gelato, vanilla
	13110470	Gelato, chocolate
	13120050	Ice cream bar, vanilla
	13120100	Ice cream bar, vanilla, chocolate coated
	13120110	Ice cream candy bar
	13120140	Ice cream bar, chocolate
	13120500	Ice cream sandwich, vanilla
	13120510	Ice cream sandwich, chocolate
	13120550	Ice cream cookie sandwich
	13120730	Ice cream cone, scooped, vanilla
	13120735	Ice cream cone, scooped, vanilla, waffle cone
	13120740	Ice cream cone, NFS
	13120770	Ice cream cone, scooped, chocolate
	13120775	Ice cream cone, scooped, chocolate, waffle cone
	13120782	Ice cream cone, soft serve, vanilla
	13120784	Ice cream cone, soft serve, chocolate
	13120786	Ice cream cone, soft serve, vanilla, waffle cone
	13120788	Ice cream cone, soft serve, chocolate, waffle cone
	13120790	Ice cream cone, vanilla, prepackaged
	13120792	Ice cream cone, chocolate, prepackaged
	13121000	Ice cream sundae, NFS
	13121100	Ice cream sundae, fruit topping
	13121120	Banana split
	13121300	Ice cream sundae, hot fudge topping
	13121400	Ice cream sundae, caramel topping
	13126000	Ice cream, fried
	13130100	Light ice cream, NFS
	13130300	Light ice cream, vanilla
	13130310	Light ice cream, chocolate
	13130700	Soft serve, blended with candy or cookies, from fast food
	13135000	Light ice cream sandwich, vanilla
	13135010	Light ice cream sandwich, chocolate
	13140000	Light ice cream bar, vanilla
	13140100	Light ice cream bar, vanilla, chocolate coated
	13140115	Light ice cream bar, chocolate
	13140700	Creamsicle
	13140710	Creamsicle, light
	13140900	Fudgesicle
	13142100	Light ice cream cone, vanilla, prepackaged
	13142110	Light ice cream cone, chocolate, prepackaged
	13150000	Sherbet, all flavors
Milk products	13161600	Fudgesicle, light
	11100000	Milk, NFS
	11111000	Milk, whole
	11111100	Milk, low sodium, whole
	11111150	Milk, calcium fortified, whole
	11111160	Milk, calcium fortified, low fat (1%)
	11111170	Milk, calcium fortified, fat free (skim)
	11112110	Milk, reduced fat (2%)
	11112120	Milk, acidophilus, low fat (1%)
	11112130	Milk, acidophilus, reduced fat (2%)
	11112210	Milk, low fat (1%)
	11113000	Milk, fat free (skim)
	11114300	Milk, lactose free, low fat (1%)
	11114320	Milk, lactose free, fat free (skim)
	11114330	Milk, lactose free, reduced fat (2%)
	11114350	Milk, lactose free, whole
	11115000	Buttermilk, fat free (skim)

GRAS Determination for the use of β -casein produced by *E. coli* stF_24_139

Notifier: Fermify GmbH

Date: October 14, 2024

11115100	Buttermilk, low fat (1%)
11115200	Buttermilk, reduced fat (2%)
11115300	Buttermilk, whole
11115400	Kefir, NS as to fat content
11116000	Goat's milk, whole
11120000	Milk, dry, reconstituted, NS as to fat content
11121100	Milk, dry, reconstituted, whole
11121210	Milk, dry, reconstituted, low fat (1%)
11121300	Milk, dry, reconstituted, fat free (skim)
11210050	Milk, evaporated, NS as to fat content
11211050	Milk, evaporated, whole
11211400	Milk, evaporated, reduced fat (2%)
11212050	Milk, evaporated, fat free (skim)
11220000	Milk, condensed, sweetened
11400000	Yogurt, NFS
11400010	Yogurt, Greek, NS as to type of milk or flavor
11410000	Yogurt, NS as to type of milk or flavor
11411010	Yogurt, NS as to type of milk, plain
11411100	Yogurt, whole milk, plain
11411200	Yogurt, low fat milk, plain
11411300	Yogurt, nonfat milk, plain
11411390	Yogurt, Greek, NS as to type of milk, plain
11411400	Yogurt, Greek, whole milk, plain
11411410	Yogurt, Greek, low fat milk, plain
11411420	Yogurt, Greek, nonfat milk, plain
11430000	Yogurt, NS as to type of milk, fruit
11431000	Yogurt, whole milk, fruit
11432000	Yogurt, low fat milk, fruit
11433000	Yogurt, nonfat milk, fruit
11433990	Yogurt, Greek, NS as to type of milk, fruit
11434000	Yogurt, Greek, whole milk, fruit
11434010	Yogurt, Greek, low fat milk, fruit
11434020	Yogurt, Greek, nonfat milk, fruit
11434090	Yogurt, NS as to type of milk, flavors other than fruit
11434100	Yogurt, whole milk, flavors other than fruit
11434200	Yogurt, low fat milk, flavors other than fruit
11434300	Yogurt, nonfat milk, flavors other than fruit
11435000	Yogurt, Greek, NS as to type of milk, flavors other than fruit
11435010	Yogurt, Greek, whole milk, flavors other than fruit
11435020	Yogurt, Greek, low fat milk, flavors other than fruit
11435030	Yogurt, Greek, nonfat milk, flavors other than fruit
11435100	Yogurt, Greek, with oats
11436000	Yogurt, liquid
11446000	Yogurt parfait, low fat, with fruit
11511000	Chocolate milk, NFS
11511100	Chocolate milk, ready to drink, whole
11511200	Chocolate milk, ready to drink, reduced fat
11511300	Chocolate milk, ready to drink, fat free
11511400	Chocolate milk, ready to drink, low fat
11511550	Chocolate milk, ready to drink, reduced sugar, NS as to milk
11511600	Chocolate milk, ready to drink, low fat (Nesquik)
11511610	Chocolate milk, ready to drink, fat free (Nesquik)
11511700	Chocolate milk, ready to drink, low fat, no sugar added (Nesquik)
11512010	Hot chocolate / Cocoa, ready to drink
11512020	Hot chocolate / Cocoa, ready to drink, made with nonfat milk
11512100	Hot chocolate / Cocoa, ready to drink, with whipped cream
11512110	Hot chocolate / Cocoa, ready to drink, made with nonfat milk and whipped cream
11513000	Chocolate milk, made from dry mix, NS as to type of milk
11513100	Chocolate milk, made from dry mix with whole milk
11513150	Chocolate milk, made from dry mix with reduced fat milk
11513200	Chocolate milk, made from dry mix with low fat milk
11513300	Chocolate milk, made from dry mix with fat free milk
11513350	Chocolate milk, made from reduced sugar mix, NS as to type of milk
11513355	Chocolate milk, made from reduced sugar mix with whole milk
11513360	Chocolate milk, made from reduced sugar mix with reduced fat milk
11513365	Chocolate milk, made from reduced sugar mix with low fat milk
11513370	Chocolate milk, made from reduced sugar mix with fat free milk
11513380	Chocolate milk, made from dry mix, NS as to type of milk (Nesquik)
11513381	Chocolate milk, made from dry mix with whole milk (Nesquik)
11513382	Chocolate milk, made from dry mix with reduced fat milk (Nesquik)
11513383	Chocolate milk, made from dry mix with low fat milk (Nesquik)
11513384	Chocolate milk, made from dry mix with fat free milk (Nesquik)
11513390	Chocolate milk, made from no sugar added dry mix, NS as to type of milk (Nesquik)
11513391	Chocolate milk, made from no sugar added dry mix with whole milk (Nesquik)
11513392	Chocolate milk, made from no sugar added dry mix with reduced fat milk (Nesquik)
11513393	Chocolate milk, made from no sugar added dry mix with low fat milk (Nesquik)
11513394	Chocolate milk, made from no sugar added dry mix with fat free milk (Nesquik)
11513400	Chocolate milk, made from syrup, NS as to type of milk
11513500	Chocolate milk, made from syrup with whole milk
11513550	Chocolate milk, made from syrup with reduced fat milk
11513600	Chocolate milk, made from syrup with low fat milk
11513700	Chocolate milk, made from syrup with fat free milk
11513800	Chocolate milk, made from light syrup, NS as to type of milk
11513801	Chocolate milk, made from light syrup with whole milk
11513802	Chocolate milk, made from light syrup with reduced fat milk
11513803	Chocolate milk, made from light syrup with low fat milk
11513804	Chocolate milk, made from light syrup with fat free milk
11513850	Chocolate milk, made from sugar free syrup, NS as to type of milk
11513851	Chocolate milk, made from sugar free syrup with whole milk
11513852	Chocolate milk, made from sugar free syrup with reduced fat milk
11513853	Chocolate milk, made from sugar free syrup with low fat milk
11513854	Chocolate milk, made from sugar free syrup with fat free milk
11514100	Hot chocolate / Cocoa, made with dry mix and water
11514110	Hot chocolate / Cocoa, made with dry mix and whole milk
11514120	Hot chocolate / Cocoa, made with dry mix and reduced fat milk
11514130	Hot chocolate / Cocoa, made with dry mix and low fat milk
11514140	Hot chocolate / Cocoa, made with dry mix and fat free milk
11514310	Hot chocolate / Cocoa, made with no sugar added dry mix and water
11514320	Hot chocolate / Cocoa, made with no sugar added dry mix and whole milk
11514330	Hot chocolate / Cocoa, made with no sugar added dry mix and reduced fat milk

GRAS Determination for the use of β -casein produced by *E. coli* stF_24_139

Notifier: Fermify GmbH

Date: October 14, 2024

11514340	Hot chocolate / Cocoa, made with no sugar added dry mix and low fat milk
11514350	Hot chocolate / Cocoa, made with no sugar added dry mix and fat free milk
11519040	Strawberry milk, NFS
11519050	Strawberry milk, whole
11519105	Strawberry milk, reduced fat
11519200	Strawberry milk, low fat
11519205	Strawberry milk, fat free
11519210	Strawberry milk, reduced sugar
11526000	Milk, malted
11531000	Eggnog
11541110	Milk shake, home recipe, chocolate
11541120	Milk shake, home recipe, flavors other than chocolate
11541130	Milk shake, home recipe, chocolate, light
11541135	Milk shake, home recipe, flavors other than chocolate, light
11541400	Milk shake with malt
11542100	Milk shake, fast food, chocolate
11542200	Milk shake, fast food, flavors other than chocolate
11543000	Milk shake, bottled, chocolate
11543010	Milk shake, bottled, flavors other than chocolate
11560000	Chocolate milk drink
13120800	Ice cream soda, flavors other than chocolate
13120810	Ice cream soda, chocolate
42401100	Yogurt, coconut milk
11100000	Milk, NFS
11111000	Milk, whole
11111100	Milk, low sodium, whole
11111150	Milk, calcium fortified, whole
11111160	Milk, calcium fortified, low fat (1%)
11111170	Milk, calcium fortified, fat free (skim)
11112110	Milk, reduced fat (2%)
11112120	Milk, acidophilus, low fat (1%)
11112130	Milk, acidophilus, reduced fat (2%)
11112210	Milk, low fat (1%)
11113000	Milk, fat free (skim)
11114300	Milk, lactose free, low fat (1%)
11114320	Milk, lactose free, fat free (skim)
11114330	Milk, lactose free, reduced fat (2%)
11114350	Milk, lactose free, whole
11115000	Buttermilk, fat free (skim)
11115100	Buttermilk, low fat (1%)
11115200	Buttermilk, reduced fat (2%)
11115300	Buttermilk, whole
11115400	Kefir, NS as to fat content
11116000	Goat's milk, whole
11120000	Milk, dry, reconstituted, NS as to fat content
11121100	Milk, dry, reconstituted, whole
11121210	Milk, dry, reconstituted, low fat (1%)
11121300	Milk, dry, reconstituted, fat free (skim)
11210050	Milk, evaporated, NS as to fat content
11211050	Milk, evaporated, whole
11211400	Milk, evaporated, reduced fat (2%)
11212050	Milk, evaporated, fat free (skim)
11220000	Milk, condensed, sweetened
11400000	Yogurt, NFS
11400010	Yogurt, Greek, NS as to type of milk or flavor
11410000	Yogurt, NS as to type of milk or flavor
11411010	Yogurt, NS as to type of milk, plain
11411100	Yogurt, whole milk, plain
11411200	Yogurt, low fat milk, plain
11411300	Yogurt, nonfat milk, plain
11411390	Yogurt, Greek, NS as to type of milk, plain
11411400	Yogurt, Greek, whole milk, plain
11411410	Yogurt, Greek, low fat milk, plain
11411420	Yogurt, Greek, nonfat milk, plain
11430000	Yogurt, NS as to type of milk, fruit
11431000	Yogurt, whole milk, fruit
11432000	Yogurt, low fat milk, fruit
11433000	Yogurt, nonfat milk, fruit
11433990	Yogurt, Greek, NS as to type of milk, fruit
11434000	Yogurt, Greek, whole milk, fruit
11434010	Yogurt, Greek, low fat milk, fruit
11434020	Yogurt, Greek, nonfat milk, fruit
11434090	Yogurt, NS as to type of milk, flavors other than fruit
11434100	Yogurt, whole milk, flavors other than fruit
11434200	Yogurt, low fat milk, flavors other than fruit
11434300	Yogurt, nonfat milk, flavors other than fruit
11435000	Yogurt, Greek, NS as to type of milk, flavors other than fruit
11435010	Yogurt, Greek, whole milk, flavors other than fruit
11435020	Yogurt, Greek, low fat milk, flavors other than fruit
11435030	Yogurt, Greek, nonfat milk, flavors other than fruit
11435100	Yogurt, Greek, with oats
11436000	Yogurt, liquid
11446000	Yogurt parfait, low fat, with fruit
11511000	Chocolate milk, NFS
11511100	Chocolate milk, ready to drink, whole
11511200	Chocolate milk, ready to drink, reduced fat
11511300	Chocolate milk, ready to drink, fat free
11511400	Chocolate milk, ready to drink, low fat
11511550	Chocolate milk, ready to drink, reduced sugar, NS as to milk
11511600	Chocolate milk, ready to drink, low fat (Nesquik)
11511610	Chocolate milk, ready to drink, fat free (Nesquik)
11511700	Chocolate milk, ready to drink, low fat, no sugar added (Nesquik)
11512010	Hot chocolate / Cocoa, ready to drink
11512020	Hot chocolate / Cocoa, ready to drink, made with nonfat milk
11512100	Hot chocolate / Cocoa, ready to drink, with whipped cream
11512110	Hot chocolate / Cocoa, ready to drink, made with nonfat milk and whipped cream
11513000	Chocolate milk, made from dry mix, NS as to type of milk
11513100	Chocolate milk, made from dry mix with whole milk
11513150	Chocolate milk, made from dry mix with reduced fat milk
11513200	Chocolate milk, made from dry mix with low fat milk

GRAS Determination for the use of β -casein produced by *E. coli* stF_24_139

Notifier: Fermify GmbH

Date: October 14, 2024

	11513300	Chocolate milk, made from dry mix with fat free milk
	11513350	Chocolate milk, made from reduced sugar mix, NS as to type of milk
	11513355	Chocolate milk, made from reduced sugar mix with whole milk
	11513360	Chocolate milk, made from reduced sugar mix with reduced fat milk
	11513365	Chocolate milk, made from reduced sugar mix with low fat milk
	11513370	Chocolate milk, made from reduced sugar mix with fat free milk
	11513380	Chocolate milk, made from dry mix, NS as to type of milk (Nesquik)
	11513381	Chocolate milk, made from dry mix with whole milk (Nesquik)
	11513382	Chocolate milk, made from dry mix with reduced fat milk (Nesquik)
	11513383	Chocolate milk, made from dry mix with low fat milk (Nesquik)
	11513384	Chocolate milk, made from dry mix with fat free milk (Nesquik)
	11513390	Chocolate milk, made from no sugar added dry mix, NS as to type of milk (Nesquik)
	11513391	Chocolate milk, made from no sugar added dry mix with whole milk (Nesquik)
	11513392	Chocolate milk, made from no sugar added dry mix with reduced fat milk (Nesquik)
	11513393	Chocolate milk, made from no sugar added dry mix with low fat milk (Nesquik)
	11513394	Chocolate milk, made from no sugar added dry mix with fat free milk (Nesquik)
	11513400	Chocolate milk, made from syrup, NS as to type of milk
	11513500	Chocolate milk, made from syrup with whole milk
	11513550	Chocolate milk, made from syrup with reduced fat milk
	11513600	Chocolate milk, made from syrup with low fat milk
	11513700	Chocolate milk, made from syrup with fat free milk
	11513800	Chocolate milk, made from light syrup, NS as to type of milk
	11513801	Chocolate milk, made from light syrup with whole milk
	11513802	Chocolate milk, made from light syrup with reduced fat milk
	11513803	Chocolate milk, made from light syrup with low fat milk
	11513804	Chocolate milk, made from light syrup with fat free milk
	11513850	Chocolate milk, made from sugar free syrup, NS as to type of milk
	11513851	Chocolate milk, made from sugar free syrup with whole milk
	11513852	Chocolate milk, made from sugar free syrup with reduced fat milk
	11513853	Chocolate milk, made from sugar free syrup with low fat milk
	11513854	Chocolate milk, made from sugar free syrup with fat free milk
	11514100	Hot chocolate / Cocoa, made with dry mix and water
	11514110	Hot chocolate / Cocoa, made with dry mix and whole milk
	11514120	Hot chocolate / Cocoa, made with dry mix and reduced fat milk
	11514130	Hot chocolate / Cocoa, made with dry mix and low fat milk
	11514140	Hot chocolate / Cocoa, made with dry mix and fat free milk
	11514310	Hot chocolate / Cocoa, made with no sugar added dry mix and water
	11514320	Hot chocolate / Cocoa, made with no sugar added dry mix and whole milk
	11514330	Hot chocolate / Cocoa, made with no sugar added dry mix and reduced fat milk
	11514340	Hot chocolate / Cocoa, made with no sugar added dry mix and low fat milk
	11514350	Hot chocolate / Cocoa, made with no sugar added dry mix and fat free milk
	11519040	Strawberry milk, NFS
	11519050	Strawberry milk, whole
	11519105	Strawberry milk, reduced fat
	11519200	Strawberry milk, low fat
	11519205	Strawberry milk, fat free
	11519210	Strawberry milk, reduced sugar
	11526000	Milk, malted
	11531000	Eggnog
	11541110	Milk shake, home recipe, chocolate
	11541120	Milk shake, home recipe, flavors other than chocolate
	11541130	Milk shake, home recipe, chocolate, light
	11541135	Milk shake, home recipe, flavors other than chocolate, light
	11541400	Milk shake with malt
	11542100	Milk shake, fast food, chocolate
	11542200	Milk shake, fast food, flavors other than chocolate
	11543000	Milk shake, bottled, chocolate
	11543010	Milk shake, bottled, flavors other than chocolate
	11560000	Chocolate milk drink
	13120800	Ice cream soda, flavors other than chocolate
	13120810	Ice cream soda, chocolate
	42401100	Yogurt, coconut milk
Plant protein products	27564420	Frankfurter or hot dog sandwich, meatless, plain, on bun
	27564430	Frankfurter or hot dog sandwich, meatless, plain, on bread
	41440000	Textured vegetable protein, dry
	41810200	Bacon strip, meatless
	41810400	Breakfast link, patty, or slice, meatless
	41810600	Chicken, meatless, NFS
	41810610	Chicken, meatless, breaded, fried
	41811400	Frankfurter or hot dog, meatless
	41811600	Luncheon slice, meatless-beef, chicken, salami or turkey
	41811800	Meatball, meatless
	41811890	Vegetarian burger or patty, meatless, no bun
	41812000	Sandwich spread, meat substitute type
	41812450	Vegetarian chili, made with meat substitute
	41812600	Vegetarian, fillet
	59003000	Meat substitute, cereal- and vegetable protein-based, fried

DEPARTMENT OF HEALTH AND HUMAN SERVICES Food and Drug Administration GENERALLY RECOGNIZED AS SAFE (GRAS) NOTICE (Subpart E of Part 170)	Form Approved:OMB No. 0910-0342; Expiration Date: 08/31/2025 (See last page for OMB Statement)	
	FDA USE ONLY	
	GRN NUMBER 001228	DATE OF RECEIPT Oct 14, 2024
	ESTIMATED DAILY INTAKE	INTENDED USE FOR INTERNET
	NAME FOR INTERNET	
KEYWORDS		

Transmit completed form and attachments electronically via the Electronic Submission Gateway (*see Instructions*); OR Transmit completed form and attachments in paper format or on physical media to: Office of Food Additive Safety (*HFS-200*), Center for Food Safety and Applied Nutrition, Food and Drug Administration,5001 Campus Drive, College Park, MD 20740-3835.

SECTION A – INTRODUCTORY INFORMATION ABOUT THE SUBMISSION

1. Type of Submission (<i>Check one</i>)	
☒ New	☐ Amendment to GRN No. _____ ☐ Supplement to GRN No. _____
2. ☒ All electronic files included in this submission have been checked and found to be virus free. (<i>Check box to verify</i>)	
3. Most recent presubmission meeting (<i>if any</i>) with FDA on the subject substance (<i>yyyy/mm/dd</i>): <u>2024-04-07</u>	
4. For Amendments or Supplements: Is your amendment or supplement submitted in response to a communication from FDA? (<i>Check one</i>)	
☐ Yes	If yes, enter the date of communication (<i>yyyy/mm/dd</i>): _____
☐ No	

SECTION B – INFORMATION ABOUT THE NOTIFIER

1a. Notifier	Name of Contact Person Eva Sommer		Position or Title Co-Founder & CEO	
	Organization (<i>if applicable</i>) Fermify GmbH			
	Mailing Address (<i>number and street</i>) Richard Neutra Gasse 5			
City Vienna		State or Province Vienna	Zip Code/Postal Code 1210	Country Austria
Telephone Number +43 6 642 30 59 15		Fax Number	E-Mail Address eva@fermify.org	
1b. Agent or Attorney (<i>if applicable</i>)	Name of Contact Person Hannes Malfroy		Position or Title Regulatory & Novel Food Specialist	
	Organization (<i>if applicable</i>) Atova Regulatory Consulting			
	Mailing Address (<i>number and street</i>) Passeig de Gracia 50, 5			
City Barcelona		State or Province Barcelona	Zip Code/Postal Code 08019	Country Spain
Telephone Number +34 686 999 247		Fax Number	E-Mail Address hannes@atovaconsulting.com	

SECTION C – GENERAL ADMINISTRATIVE INFORMATION

1. Name of notified substance, using an appropriately descriptive term

β -casein produced by E. coli stF_24_139

2. Submission Format: *(Check appropriate box(es))*

☒ Electronic Submission Gateway

☐ Electronic files on physical media

☐ Paper

If applicable give number and type of physical media

3. For paper submissions only:

Number of volumes _____

Total number of pages _____

4. Does this submission incorporate any information in CFSAN's files? *(Check one)*

☐ Yes *(Proceed to Item 5)*

☒ No *(Proceed to Item 6)*

5. The submission incorporates information from a previous submission to FDA as indicated below *(Check all that apply)*

☐ a) GRAS Notice No. GRN _____

☐ b) GRAS Affirmation Petition No. GRP _____

☐ c) Food Additive Petition No. FAP _____

☐ d) Food Master File No. FMF _____

☐ e) Other or Additional *(describe or enter information as above)* _____

6. Statutory basis for conclusions of GRAS status *(Check one)*

☒ Scientific procedures *(21 CFR 170.30(a) and (b))*

☐ Experience based on common use in food *(21 CFR 170.30(a) and (c))*

7. Does the submission (including information that you are incorporating) contain information that you view as trade secret or as confidential commercial or financial information? *(see 21 CFR 170.225(c)(8))*

☐ Yes *(Proceed to Item 8)*

☒ No *(Proceed to Section D)*

8. Have you designated information in your submission that you view as trade secret or as confidential commercial or financial information *(Check all that apply)*

☐ Yes, information is designated at the place where it occurs in the submission

☐ No

9. Have you attached a redacted copy of some or all of the submission? *(Check one)*

☐ Yes, a redacted copy of the complete submission

☐ Yes, a redacted copy of part(s) of the submission

☐ No

SECTION D – INTENDED USE

1. Describe the intended conditions of use of the notified substance, including the foods in which the substance will be used, the levels of use in such foods, and the purposes for which the substance will be used, including, when appropriate, a description of a subpopulation expected to consume the notified substance.

β -casein produced by E. coli stF_24_139 is intended to be used as a food ingredient, at a maximum level of up to 15 g of dry matter of β -casein produced by E. coli stF_24_139 per 100 g in cheeses and cheese analogs and at a maximum level of 3.8 g of dry matter of β -casein produced by E. coli stF_24_139 per 100 g in non-alcoholic beverages, dairy product analogs, frozen dairy desserts, milk products and plant protein products.

2. Does the intended use of the notified substance include any use in product(s) subject to regulation by the Food Safety and Inspection Service (FSIS) of the U.S. Department of Agriculture?

(Check one)

☐ Yes

☒ No

3. If your submission contains trade secrets, do you authorize FDA to provide this information to the Food Safety and Inspection Service of the U.S. Department of Agriculture?

(Check one)

☐ Yes

☐ No, you ask us to exclude trade secrets from the information FDA will send to FSIS.

SECTION E – PARTS 2 -7 OF YOUR GRAS NOTICE

(check list to help ensure your submission is complete – PART 1 is addressed in other sections of this form)

- ☒ PART 2 of a GRAS notice: Identity, method of manufacture, specifications, and physical or technical effect (170.230).
- ☒ PART 3 of a GRAS notice: Dietary exposure (170.235).
- ☒ PART 4 of a GRAS notice: Self-limiting levels of use (170.240).
- ☒ PART 5 of a GRAS notice: Experience based on common use in foods before 1958 (170.245).
- ☒ PART 6 of a GRAS notice: Narrative (170.250).
- ☒ PART 7 of a GRAS notice: List of supporting data and information in your GRAS notice (170.255)

Other Information

Did you include any other information that you want FDA to consider in evaluating your GRAS notice?

☐ Yes ☒ No

Did you include this other information in the list of attachments?

☐ Yes ☐ No

SECTION F – SIGNATURE AND CERTIFICATION STATEMENTS

1. The undersigned is informing FDA that Fermify GmbH

(name of notifier)

has concluded that the intended use(s) of β-casein produced by E. coli stF_24_139

(name of notified substance)

described on this form, as discussed in the attached notice, is (are) not subject to the premarket approval requirements of the Federal Food, Drug, and Cosmetic Act based on your conclusion that the substance is generally recognized as safe recognized as safe under the conditions of its intended use in accordance with § 170.30.

2. Fermify GmbH agrees to make the data and information that are the basis for the
(name of notifier) conclusion of GRAS status available to FDA if FDA asks to see them;
agrees to allow FDA to review and copy these data and information during customary business hours at the following location if FDA asks to do so; agrees to send these data and information to FDA if FDA asks to do so.

Richard Neutra Gasse 5, 1210 Vienna, Austria

(address of notifier or other location)

The notifying party certifies that this GRAS notice is a complete, representative, and balanced submission that includes unfavorable, as well as favorable information, pertinent to the evaluation of the safety and GRAS status of the use of the substance. The notifying party certifies that the information provided herein is accurate and complete to the best of his/her knowledge. Any knowing and willful misinterpretation is subject to criminal penalty pursuant to 18 U.S.C. 1001.

3. Signature of Responsible Official,
Agent, or Attorney

Hannes Malfroy Digitally signed by Hannes Malfroy
Date: 2024.10.14 21:59:36 +02'00'

Printed Name and Title

Hannes Malfroy, Regulatory & Novel Food Specialist

Date (mm/dd/yyyy)

10/13/2024

SECTION G – LIST OF ATTACHMENTS

List your attached files or documents containing your submission, forms, amendments or supplements, and other pertinent information. Clearly identify the attachment with appropriate descriptive file names (or titles for paper documents), preferably as suggested in the guidance associated with this form. Number your attachments consecutively. When submitting paper documents, enter the inclusive page numbers of each portion of the document below.

Attachment Number	Attachment Name	Folder Location (select from menu) (Page Number(s) for paper Copy Only)
	Form3667.pdf	Administrative
	Confidential_GRAS_dossier_Fermify_Beta-casein_produced_by_E_coli_stF_24_139_signed.pdf	GRAS Notice
	COSM_3667_22030_FermifyGmbH.pdf	Administrative
	GRAS_Cover_letter_signed.pdf	Administrative
	Confidential_GRAS_dossier_Fermify_Beta-casein_produced_by_E_coli_stF_24_139_signed.pdf	Administrative

OMB Statement: Public reporting burden for this collection of information is estimated to average 170 hours per response, including the time for reviewing instructions, searching existing data sources, gathering and maintaining the data needed, and completing and reviewing the collection of information. Send comments regarding this burden estimate or any other aspect of this collection of information, including suggestions for reducing this burden to: Department of Health and Human Services, Food and Drug Administration, Office of Chief Information Officer, PRASStaff@fda.hhs.gov. (Please do NOT return the form to this address.). An agency may not conduct or sponsor, and a person is not required to respond to, a collection of information unless it displays a currently valid OMB control number.